Extremal dimensions and critical isomorphism growth of metabelian three-Engel Lie algebras

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Abstract

We determine sharp dimensions and near-minimum isomorphism populations for metabelian three-Engel Lie algebras in characteristic two and their binary quadratic orbit realizations. Two minimum dimension formulas and their complete equality classifications measure the cost of compatibility with the original polar bracket. At the unrestricted metabelian minimum the type count grows linearly with class, while the first admissible dimension surplus already supports doubly exponential growth. We prove fixed-surplus laws, recover exact principal orbits, and identify a sharp normalized-surplus threshold $\sqrt{2}$ for concentration at minimum derived dimension. An explicit logarithmic population profile and a discrete staircase describe the allocation between quotient and derived surplus. The proofs combine a multigraded commutator module, intrinsic recovery of coordinate actions, and uniform cocycle bounds. The metabelian correspondence is proved independently on all linear morphisms. The population measure is uniform on unlabelled isomorphism types; ordinary counts are stated over every finite characteristic-two field with at least four elements.

Keywords. Metabelian Lie algebra; three-Engel identity; extremal dimension; isomorphism growth; critical threshold; quadratic map.

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1 Introduction

How does an isomorphism population change immediately above its smallest possible dimension? We answer this question for metabelian three-Engel Lie algebras in characteristic two and for their binary quadratic orbit realizations. The minimum is rigid, but the first admissible surplus already supports a doubly exponential population. A larger surplus produces an explicit transition in the typical derived dimension.

The binary equation is

$$f \circ f = f + \mathrm{id}\,. \tag{1}$$

For quadratic f, its polar is a Lie bracket. We impose metabelianity and prove a complete normal form and an ordinary $\mathbb{F}_4$-Lie correspondence, retaining every linear intertwiner. This proof uses only the elementary polar identities and the metabelian assumption; the global structural theory of [17] is not an input. The ordinary formulation also permits counting over every finite characteristic-two field $\mathbb{F}_q$ with $q \geq 4$.

1.1 From exact minima to the first surplus

Write $\mu(c)$ for the least binary dimension at polar class c, and $\nu(c)$ for the corresponding compatible metabelian minimum. For $c \geq 5$ the sharp formulas are

$$\mu(c) = 2^c + 2c - 6, \qquad \nu(c) = 2^c + 2c - 2.$$

We classify all equality objects, their automorphisms, and the small class endpoints. The four-dimension difference measures the extra cost of extending the scalar structure to the original polar bracket. The auxiliary ordinary Lie structure exists throughout; compatibility is an additional condition on the original polar algebra.

At the minimum, the metabelian type count grows linearly in c. At dimension $\mu(c) + 2$, if A_c counts all binary-linear conjugacy classes, then

$$\log_4 A_c = 2^{c-1} + O(c^2).$$

Theorem 5.1 constructs indecomposable families and bounds the entire layer. The fixed-surplus law, exact principal orbits and full coordinate recovery determine how the additional dimensions enter the population.

1.2 A sharp population transition

For ordinary objects over $\mathbb{F}_q$, put $d = c - 1$ and $n = 2^d$; d is the class-dependent core scale, not the intrinsic generator rank of every object. Let r be total dimension surplus over the minimum, and let $\pi_{d,r}(q)$ be the proportion of unlabelled isomorphism types with minimum possible derived dimension. Theorem 6.2 gives

$$\limsup r/\sqrt{n} < \sqrt{2} \implies \pi_{d,r}(q) \to 1, \qquad \liminf r/\sqrt{n} > \sqrt{2} \implies \pi_{d,r}(q) \to 0.$$

We determine the macroscopic logarithmic population profile and dimension allocation, then resolve the competing integer defects through a discrete staircase. The exact boundary weights in shrinking bands remain a separate question. The measure throughout these population theorems is uniform on isomorphism types, as specified in the corresponding statements; it is not a uniform ideal measure.

The proof mechanism joins three tasks. A multigraded commutator module gives sharp terminal quotients. Intrinsic recovery then identifies every allowed change of coordinates in the near-minimum layers. Finally action normalization and cocycle estimates compare competing allocations of the same dimension surplus. In particular one additional central derived direction couples to the quotient actions strongly enough to produce the sharp constant.

1.3 Sources and scope

The commutator-module presentation is classical Koszul theory [14, 11]; partial Engel linearization is used in [16]. Cohn's construction [5] supplies a classical regular compatible model and the zero-parameter slice identified below. The equality classification keeps the additional parameters and all their coordinate changes. Generic algebra enumeration in [2] uses a different fixed-variety regime. Here the minimum depends on class and the populations occupy layers immediately above it.

The companion [17] develops general quadratic orbit structure, and [18] studies ordinary central-depth laws. The ideal-growth companion [12] fixes generator rank and allows unbounded restricted-power height. Its ideal/probability measure and growth limit differ from the type layers studied here. These papers develop separate parts of the common research manuscript [19].

Section 3 proves the independent correspondence. Sections 4 and 8 give the sharp minima and equality cases. The intervening population sections organize fixed surplus, the critical scale and fine defects; the appendices provide the complete orbit and uniform-bound arguments. All brackets are alternating and left-normed. Dimensions are binary for orbit solutions and explicitly field-valued in the ordinary counts.

2 The polar Lie algebra

Proposition 2.1. *Every solution of* (1) *satisfies*

$$f^3 = \mathrm{id}, \qquad f(0) = 0, \qquad f(x + f(x)) = x.$$

Its only fixed point is zero. Every nonzero orbit spans a binary plane, and these planes partition $V \setminus \{0\}$. *In particular* $\dim V$ *is even.*

Proof. Evaluate (1) at $f(x)$. It gives $f^3(x) = f^2(x) + f(x) = x$, so f is bijective. If $a = f(0)$, then $f(a) = a$; applying f^3 at zero gives $a = 0$. A fixed point in (1) must be zero. The remaining identity follows from $x + f(x) = f^2(x)$. The nonzero orbits have length three and sum zero, and $3 \mid 2^{\dim V} - 1$ implies the parity assertion. □

Theorem 2.2. *Let* f *be a quadratic solution. Its polar form* $[x, y] = f(x + y) + f(x) + f(y)$ *is a Lie bracket. It satisfies*

$$[x, f(x)] = 0, \tag{2}$$

$$f([x, y]) = [f(x), f(y)] + [x, y] + [f(x) + f(y), [x, y]], \tag{3}$$

$$[[a, b], [c, d]] + [[a, c], [b, d]] + [[a, d], [b, c]] = 0. \tag{4}$$

Proof. The polar is alternating and bilinear because f is quadratic and $f(0) = 0$. Expand $f(x + f(x)) = x$ by polarization. The terms $f(x) + f^2(x)$ sum to x, leaving (2). Take the third additive difference at zero of $x \mapsto [x, f(x)]$. The terms involving individual values of f cancel, leaving

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

This is Jacobi.

Next polarize $f^2 = f + \mathrm{id}$. Since $f(x + y) = f(x) + f(y) + [x, y]$, the expansion of the left side gives (3).

For (4), introduce Boolean variables $t_1, \ldots, t_4$ and vectors $x_1, \ldots, x_4$. The exact expansion is

$$f\Big(\sum_i t_i x_i\Big) = \sum_i t_i f(x_i) + \sum_{i<j} t_i t_j [x_i, x_j].$$

In applying f again, the coefficient of $t_1t_2t_3t_4$ can only come from a polar cross term between two disjoint pair summands. There are three such pairings, giving the left side of (4). The composition has degree at most two, so this coefficient vanishes. All coefficient comparisons take place in the Boolean function algebra, with $t_i^2 = t_i$. □

The two polarization steps have distinct roles. Binary degree two gives an alternating bilinear form; the commuting identity (2) forces its Jacobi identity. The fourth polarization then supplies (4), an additional restriction on the resulting Lie algebra. The polynomial in (4) is the associative standard polynomial of degree four expressed as a Lie element in characteristic two [8, Theorem 9]. It supplies an additional identity beyond Jacobi.

Proposition 2.3. *All terms of the derived, lower central, and upper central series of the polar Lie algebra $L = (V, [\ ,\ ])$ are f-invariant. The center is f-invariant and f is linear on it. Every nonzero factor of any of these series has even binary dimension.*

Proof. Let I be an f-invariant ideal. In (3), take $x \in I$. Each term on the right belongs to $[I, L]$, which is an ideal. Quadratic expansion extends invariance from bracket generators to their span: the extra polar terms also lie in that ideal. The same argument with $x, y \in I$ proves invariance of $[I, I]$, using that it is an ideal of L. Induction starts with $I = L$.

Polarization of (2) gives

$$[x, f(y)] + [y, f(x)] + [x, [x, y]] + [y, [x, y]] = 0. \tag{5}$$

If $x \in Z(L)$, this says $[y, f(x)] = 0$ for every y. Thus $Z(L)$ is invariant; its polar is zero, so f is linear there. Pass successively to quotients by invariant central ideals to obtain the upper central assertion. Every series factor inherits a solution of (1), so Proposition 2.1 proves its parity. □

Lemma 2.4. *A nonabelian nilpotent polar Lie algebra has binary abelianization dimension at least four.*

Proof. The abelianization is a nonzero even-dimensional space with the linear action induced by f. If its dimension were two, any nonzero coset $x + L'$ and its image $f(x) + L'$ would form a binary basis. The subalgebra H generated by $x, f(x)$ is abelian by (2). Moreover $L = H + L'$. Inductively, $L = H + \gamma_j(L)$ implies $L = H + \gamma_{j+1}(L)$, because $[H + \gamma_j, H + \gamma_j] \subseteq H + \gamma_{j+1}$. Nilpotency then gives $L = H$, a contradiction. □

3 A complete metabelian correspondence

Assume in this section that the polar Lie algebra is metabelian. Then $W = L'$ is abelian, $f|_W$ is linear, and the induced map on $U = V/W$ is linear. Both operators satisfy $T^2 + T = \mathrm{id}$. We give U and W the $\mathbb{F}_4$-structures in which these operators are multiplication by ω.

Lemma 3.1 (The operator pencil)**.** *In any binary-linear splitting of $V \to U$, write*

$$f(x, z) = (\omega x, \omega z + L_x z + q(x)). \tag{6}$$

The binary-linear pencil $x \mapsto L_x$ satisfies

$$L_x\omega = \omega L_x, \quad L_x^3 = 0, \quad L_{\omega x} = \omega L_x + L_x^2, \quad L_xL_y = L_yL_x, \quad L_x^2L_y = 0.$$

Proof. Put $J = \omega\,\mathrm{id}_W$ and $M_x = J + L_x$. Both M_x and $\mathrm{id} + M_x$ are invertible: they are the restrictions on coset differences of the permutations f and $f + \mathrm{id} = f^2$. Substitution in (1) gives

$$M_{\omega x}M_x = \mathrm{id} + M_x.$$

For $A = M_x$, the three matrices along the ω-orbit are $A, \mathrm{id}+A^{-1}, (\mathrm{id}+A)^{-1}$. Their sum is J, because the three L-terms sum to zero. Thus J is a rational function of A, and in particular commutes with it. Clearing the denominators gives

$$JA(\mathrm{id}+A) = A^3 + A + \mathrm{id}\,.$$

Since $J^2 + J = \mathrm{id}$ and $J^3 = \mathrm{id}$, this is $(A+J)^3 = 0$. It follows that L_x commutes with J and is cube zero. Expanding $(J + L_x)^{-1}$ now gives $L_{\omega x} = JL_x + L_x^2$. The additivity of the left side forces the L_x's to commute.

For $P = L_x, Q = L_y$, the identity $(P+Q)^3 = 0$ yields $P^2Q = PQ^2 =: T$. Apply the same identity to $P' = JP + P^2 = L_{\omega x}$ and Q. It gives $(J^2+J)T = P^2Q^2$, hence $T = P^2Q^2 = PT$. As P is nilpotent, this implies $T = 0$. □

Theorem 3.2 (Normal form and equivalence)**.** *Every metabelian quadratic solution is binary-linearly conjugate to*

$$f(x,z) = \big(\omega x, \omega z + P(x)z + Q(x) + P(x)Q(x)\big) \quad \text{on } \mathbb{F}_4^d \oplus W, \tag{7}$$

where

$$P(x) = \sum_i x_i P_i, \qquad Q(x) = \sum_{i<j} x_i x_j a_{ij}, \tag{8}$$

$$P_iP_j = P_jP_i, \qquad P_i^2 = 0, \tag{9}$$

$$P_ia_{jk} + P_ja_{ik} + P_ka_{ij} = 0 \quad (i<j<k), \tag{10}$$

$$W = \sum_i P_iW + \mathrm{span}_{\mathbb{F}_4}\{a_{ij} : i<j\}. \tag{11}$$

Conversely these conditions define a quadratic solution with intrinsic polar image W.

The category of these maps with all binary-linear intertwiners is equivalent to the category of finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras $\mathfrak{l}$ satisfying

$$(\mathrm{ad}\,u)^2|_{\mathfrak{l}'} = 0 \qquad (u \in \mathfrak{l}). \tag{12}$$

with all $\mathbb{F}_4$-Lie homomorphisms as morphisms. For corresponding objects, $C_{\mathrm{GL}(V,2)}(f) \cong \mathrm{Aut}_{\mathbb{F}_4}(\mathfrak{l})$. Their lower central dimensions agree after multiplication by two.

Proof. Lemma 3.1 provides the preliminary actions. We first normalize a section and separate the scalar weights of the cocycle (Steps 1–2). The decisive step is to combine binary degree two with intrinsic generation to force square zero (Step 3). We then verify sufficiency, construct the auxiliary Lie algebra and compare lower central series (Steps 4–5). Finally, tracking changes of coordinates upgrades the normal form to an equivalence on morphisms (Step 6).

Step 1: a normalized section. Choose an $\mathbb{F}_4$-basis e_i of U, with arbitrary lifts v_i in V. Define a binary-linear section by $e_i \mapsto v_i$, $\omega e_i \mapsto f(v_i)$. The images form a basis modulo W. Since $f(v_i + f(v_i)) = v_i$, the cocycle q in (6) vanishes on each coordinate $\mathbb{F}_4$-line.

Step 2: scalar weights. Initially put $P_i = L_{e_i}$, without yet asserting $P_i^2 = 0$. Lemma 3.1 gives

$$L_{\alpha e_i} = \alpha P_i + (\alpha + \alpha^2)P_i^2 \quad (\alpha \in \mathbb{F}_4).$$

Set $N(x) = \sum_i x_i(P_i + P_i^2)$. Then $L_x = N + N^2$, $N^3 = 0$, and $L_{\omega x} = \omega N + \omega^2 N^2$. Decompose $q = q_0 + q_1 + q_2$ into the three scalar weights $q_j(\omega x) = \omega^j q_j(x)$. Projection to each weight is a linear combination of $q(x), q(\omega x), q(\omega^2 x)$, so preserves binary degree and the vanishing on coordinate lines.

The cocycle equation is

$$q(\omega x) = \omega^2 q(x) + (\omega N + \omega^2 N^2)q(x).$$

Its weight components give

$$q_0 = Nq_2 + \omega N^2 q_1, \qquad q_1 = \omega N q_0 + \omega^2 N^2 q_2.$$

Using $N^3 = 0$ yields

$$q_0 = Nq_2, \qquad q_1 = N^2 q_2. \tag{13}$$

Step 3: binary degree forces square zero. The reduced $\mathbb{F}_4$-polynomial of a weight-two binary-quadratic map has the form $\sum_i b_i x_i^2 + \sum_{i<j} a_{ij} x_i x_j$. Line normalization removes the first sum, so $q_2 = Q$. Consequently

$$q_1 = N^2 Q = \sum_{k,i<j} x_k^2 x_i x_j P_k^2 a_{ij}.$$

This is where the degree bound becomes decisive. For $k = i$ or j, the exponent pattern is $(3, 1)$; otherwise it is $(2, 1, 1)$. Both patterns have binary degree three, whereas the weight projection q_1 has degree at most two. The cubed or squared coordinate identifies k, and the remaining coordinates identify $\{i, j\}$. Thus each displayed term has its own reduced monomial, forcing

$$P_k^2 a_{ij} = 0 \qquad \text{for all } k, i, j. \tag{14}$$

For completeness, the one-coordinate function basis $1, x, x^2, x^3$ has binary degrees $0, 1, 1, 2$. Its leading classes form a basis in the associated graded degree filtration. Tensoring over the coordinates proves that distinct reduced monomials of the displayed degree three cannot cancel to degree two. This justifies the comparison in the binary degree filtration.

We now extend this vanishing from the cocycle coefficients to the whole intrinsic derived ideal. Let $\mathfrak{m}W = \sum_i P_i W$. This is an $\mathbb{F}_4$-subspace because the P_i are $\mathbb{F}_4$-linear. By the preliminary identities, every P_k^2 annihilates $\mathfrak{m}W$. Modulo $\mathfrak{m}W$, the polar values in (6) are exactly the values of the polar of Q. For $i < j$ and $\alpha \in \mathbb{F}_4$,

$$\mathrm{pol}(Q)(e_i, \alpha e_j) = \alpha a_{ij}.$$

All polar values of Q lie in the $\mathbb{F}_4$-span of the a_{ij}, and the displayed values contain all their scalar multiples. Their binary span is therefore exactly that $\mathbb{F}_4$-span. Since W is the binary span of all polar values, this proves (11). Together with (14), it gives $P_k^2 = 0$ on all of W. Now $L_x = P(x)$ and $q = Q + PQ$. The degree-three terms of PQ with three distinct indices vanish precisely when (10) holds. The other terms have exponent patterns $(2, 1)$, of binary degree two. This proves necessity.

Step 4: sufficiency. Conditions (9) and (10) make $P(x)^2 = 0$ and make PQ binary-quadratic. Under $x \mapsto \omega x$, the functions Q and PQ have weights two and zero. Thus

$$(Q + PQ)(\omega x) = \omega^2 Q + PQ = \omega^2 (Q + PQ) + \omega P(Q + PQ).$$

Together with $(\omega \,\mathrm{id} + \omega P)(\omega \,\mathrm{id} + P) = \omega^2 \,\mathrm{id} + P$, this verifies $f^2 = f + \mathrm{id}$ by substitution. Mixed polar values generate $\mathfrak{m}W$; modulo that space, the values of the polar of Q generate the a_{ij}'s over $\mathbb{F}_4$. Equation (11) therefore makes the polar image exactly W.

Step 5: the associated Lie algebra. Put $\beta(x, y) = Q(x + y) + Q(x) + Q(y)$ and define

$$[(x, z), (y, t)]_0 = (0, P(x)t + P(y)z + \beta(x, y)). \tag{15}$$

Jacobi follows from commuting operators and (10). This is a metabelian $\mathbb{F}_4$-Lie algebra with derived ideal W, and it satisfies (12). Conversely, splitting the abelianization of any algebra with these properties gives exactly such operators and an alternating cocycle. Choose its cross-term refinement Q to recover (7).

For both the original polar bracket and (15), the action on W is $P(x)$. Hence, with $\mathfrak{m}$ the ideal generated by these actions,

$$\gamma_2 = W, \qquad \gamma_k = \mathfrak{m}^{k-2}W \quad (k \geq 3). \tag{16}$$

This proves the lower central comparison.

Step 6: morphisms. A binary-linear intertwiner maps the intrinsic space W into W'. Its diagonal blocks $A : U \to U'$ and $C : W \to W'$ commute with the intrinsic scalar actions, hence are $\mathbb{F}_4$-linear. Write its off-diagonal block as $t_s + t_a$, where $t_s(\omega x) = \omega t_s(x)$ and $t_a(\omega x) = \omega^2 t_a(x)$. Direct substitution in $Sf = f'S$ gives the two conditions

$$CP(x) = P'(Ax)C, \tag{17}$$

$$Q'(Ax) = CQ(x) + t_a(x) + P'(Ax)t_s(x). \tag{18}$$

Indeed the unseparated cocycle equation reduces to the second one by multiplication with $\mathrm{id} + P'(Ax)$, whose inverse is itself. Polarization over $\mathbb{F}_4$ eliminates t_a and gives

$$\beta'(Ax, Ay) = C\beta(x, y) + P'(Ax)t_s(y) + P'(Ay)t_s(x). \tag{19}$$

Equations (17) and (19) say exactly that $(x, z) \mapsto (Ax, Cz + t_s(x))$ is an $\mathbb{F}_4$-Lie homomorphism.

Conversely, for any such Lie homomorphism the function $Q'(Ax) + CQ(x) + P'(Ax)t_s(x)$ has zero polar and is an ordinary homogeneous quadratic. It is therefore a unique sum of squares $\sum b_i x_i^2$. Taking this as t_a gives a unique intertwiner. Scalar parts respect composition because A and C are $\mathbb{F}_4$-linear. Thus the correspondence is an equivalence on both objects and all morphisms, proving also the centralizer assertion. No inverse of A, C, or the intertwiner was used. Thus the argument includes maps with nontrivial kernel or cokernel. □

Corollary 3.3. *Every finite-dimensional metabelian polar algebra is nilpotent and satisfies* $(\operatorname{ad} v)^3 = 0$ *for every* v. *If* $w = \dim_{\mathbb{F}_4} W$, *its class is at most* $w + 1$.

Proof. The commuting square-zero operators generate a nilpotent associative ideal. They have a common nonzero kernel whenever $W \neq 0$; repeating on the quotient gives a strictly triangular flag, so all products of w operators vanish. Apply (16). Finally $\operatorname{ad} v$ maps V into W, and its square on W is zero. □

3.1 An intrinsic change of addition

Let $\mathcal{Q}$ have finite binary-quadratic solutions $f^2 = f + I$ with metabelian polar algebra as objects, and all binary-linear intertwiners $Tf = gT$ as morphisms. Let $\mathcal{E}$ have finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras satisfying $(\operatorname{ad} u)^2 \mathfrak{l}' = 0$, with all $\mathbb{F}_4$-Lie homomorphisms.

Proposition 3.4 (An involutive twist of binary Lie structures)**.** *Let* $(E, +, B)$ *be a metabelian binary Lie algebra with derived ideal* W, *satisfying* $(\operatorname{ad} u)^2 W = 0$ *for every* u. *Set* $P_u = \operatorname{ad} u|_W$ *and*

$$C(u, v) = (I + P_u + P_v)B(u, v), \qquad u \star v = u + v + C(u, v).$$

Then $(E, \star, C)$ *is another binary Lie algebra in the same category. Applying this operation twice recovers* $(+, B)$. *It is a canonical involutive functor on this precise Lie category, with the same underlying morphism functions.*

Proof. The operators P_u commute because W is abelian, and square to zero by hypothesis. For $a = P_x, b = P_y, c = P_z$, put $w = C(x, y) = (I + a + b)B(x, y)$. Direct expansion gives

$$C(x \star y, z) + C(x, z) + C(y, z) = bB(x, z) + aB(y, z) + (I + a + b + c)c(I + a + b)B(x, y).$$

The last operator is c, so this is zero by the original Jacobi identity. Thus C is additive for $\star$ in both arguments, before associativity is assumed. Both associations of three inputs now equal $x+y+z+C(x,y)+C(x,z)+C(y,z)$. The operation is therefore an abelian group operation, with identity zero and exponent two. On W, it is the old addition, and $C(x,w) = B(x,w)$ for $w \in W$. The Jacobi sum for C is the old Jacobi sum plus $\sum_{\text{cyc}} P_x(P_y + P_z)B(y,z)$; the latter vanishes by multiplying the old Jacobi identity by P_x, P_y, P_z and summing.

The new derived ideal is W. One inclusion follows from $C(E,E) \subseteq W$; for the other use

$$B(u,v) = C(u,v) + C(u \star v, C(u,v)).$$

The actions on W are unchanged, so the new square condition holds. If $S = u \star v$, $w = C(u,v)$, then $S \star w \star C(S,w) = u+v$: the last extra term is $P^2_{u+v}w = 0$. The second twisted bracket is $(I + P_u + P_v)^2 B(u,v) = B(u,v)$. This proves involutivity. Every original Lie homomorphism preserves the defining formulas; involutivity gives the converse as well. □

There is a choice-free realization of that equivalence. In the next formulas $+$ and B denote the original binary addition and polar bracket of a source object. Define on its underlying set

$$C(u,v) = B(u,v) + B(u+v, B(u,v)), \qquad u \oplus v = u + v + C(u,v), \qquad \omega \cdot u = f(u). \quad (20)$$

The other two $\mathbb{F}_4$ scalars are zero and the identity, and $\omega^2 u = f^2(u)$.

Theorem 3.5 (Canonical all-morphism equivalence)**.** *The operations* (20) *make the underlying set a vector space over* $\mathbb{F}_4$*, and* C *makes it an object of* $\mathcal{E}$*. On morphisms the correspondence uses the same set map. It is fully faithful and has a canonical inverse: for an object* $(L, \oplus, C)$ *of* $\mathcal{E}$*, define*

$$u + v = S \oplus C(u,v) \oplus C(S, C(u,v)), \qquad S = u \oplus v, \qquad f(u) = \omega u. \quad (21)$$

These are the binary vector addition and map of an object of $\mathcal{Q}$*. The two constructions are inverse on objects and on their underlying morphism functions.*

Proof. It suffices to establish the formulas in any normal-form coordinates; their right sides are already intrinsic. Put

$$H_f(x,z) = (x, (I + P(x))(z + Q(x))), \qquad H_f^{-1}(x,z) = (x, (I + P(x))z + Q(x)).$$

For the morphisms in Step 6 of Theorem 3.2, direct use of its cocycle equation gives the naturality identity

$$H_g T_f = T_L H_f. \quad (22)$$

Indeed the second coordinate inside $I + P'(Ax)$ becomes $C(z + Q(x)) + (I + P'(Ax))t_s(x)$, whose image is $C(I + P(x))(z + Q(x)) + t_s(x)$.

For completeness, write $u = (x,z)$, $v = (y,t)$, $a = P(x)$, $b = P(y)$, and $\beta = Q(x+y) + Q(x) + Q(y)$. A calculation using $a^2 = b^2 = 0$, $ab = ba$, gives both

$$H_f^{-1}(H_f u + H_f v) + u + v = [H_f u, H_f v]_0 = B(u,v) + (a+b)B(u,v).$$

Here H_f^{-1} is the identity on W, and $(a+b)B(u,v) = B(u+v, B(u,v))$. Thus H_f transports $\oplus, C, f$ to ordinary vector addition, the auxiliary Lie bracket, and multiplication by ω. This proves that (20) defines the claimed structure in every class, not merely in a low-class truncation.

To recover the old addition put $w = C(u,v) \in W$ and $S = u \oplus v$. On $V \times W$, the two bracket formulas agree and their actions square to zero. Hence $S \oplus w = S + w + C(S,w)$, and subsequently adding $C(S,w)$ by $\oplus$ cancels that last term. The result is $u + v$, proving (21).

For an arbitrary object of $\mathcal{E}$, choose a vector-space splitting only to verify the construction: its usual action and cocycle data give the existing normal-form solution, and the preceding calculation transports that solution to exactly (21). Thus this formula is associative, has exponent two, and gives the required quadratic solution; no splitting enters its definition. Applying the two formulas successively recovers the original operations.

Every source intertwiner preserves $+, f, B$, hence $\oplus, C$ and the $\mathbb{F}_4$ scalars. Conversely, a target Lie homomorphism preserves the right side of (21) and $f = \omega$, so it is an original binary-linear intertwiner. This proves full faithfulness and the asserted inverse functor. □

Exactness and central extensions. The canonical functor changes operations, not the underlying sets or morphism functions. Kernels and images therefore have the same point sets. Short exact sequences are preserved and reflected, with $\dim_{\mathbb{F}_2} = 2\dim_{\mathbb{F}_4}$ on corresponding objects. Even ranks are also immediate from the original orbit-parity argument and are not a separate new invariant.

The original derived ideal is the same set W as the new derived ideal. On W the additions coincide, and on $V \times W$ the brackets coincide; hence all lower central terms agree as sets. The original polar bracket can be recovered as

$$B(u, v) = C(u, v) \oplus C(u \oplus v, C(u, v)).$$

Thus the two centers coincide as sets. An original f-stable binary subspace is the same set as a new $\mathbb{F}_4$-Lie subalgebra; original f-stable polar Lie ideals correspond to new $\mathbb{F}_4$-Lie ideals. No claim is made about original ideals lacking f-invariance. These assertions follow in both directions directly from the two operation formulas. In particular, central short exact sequences whose middle object remains metabelian correspond to central Lie extensions. This does not cover central extensions that leave the metabelian category.

Scope. The formula uses the metabelian normal-form identities, especially commuting square-zero actions on the derived ideal. The general construction in [17] treats non-metabelian objects with an additional quartic term. The canonical change of addition preserves the original classification through its full morphism correspondence.

4 Sharp metabelian bounds and equality cases

The metabelian correspondence turns the dimension problem into a quotient problem for one multigraded commutator module. We first use that module to determine the least dimension at each polar class, then classify every equality case. Section 5 examines the much larger populations immediately above this minimum.

4.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = \mathbb{F}_4[t_1, \ldots, t_d]/(t_1^2, \ldots, t_d^2), \qquad \mathfrak{m} = (t_1, \ldots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{23}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [14, Theorem 6.2] and [11, Sections 2–3].

Every derived module in Theorem 3.2 is a quotient of W_d: send t_i to P_i and v_{ij} to a_{ij}. The intrinsic-generation condition makes this homomorphism surjective. Conversely, every quotient of W_d, with these images, gives the normal form (7).

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i = e_i$, $\deg v_{ij} = e_i + e_j$.

Proposition 4.1. *The coefficient-degree Hilbert polynomial of W_d is*

$$H_d(T) = \sum_{k=2}^{d} \binom{d}{k} \big((k-1)T^{k-2} + kT^{k-1}\big) + \binom{d}{2} T^2. \tag{24}$$

Consequently

$$\dim_{\mathbb{F}_4} W_d = (d-1)2^d + 1 - d + \binom{d}{2}.$$

The universal solution on $\mathbb{F}_4^d \oplus W_d$ has class $\max(4, d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k \geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (23) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i, j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i, j\}$, giving the final term of (24). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d = 2, 3$, and $d-1$ for $d \geq 4$. Equation (16) gives the class. □

In particular, binary degree two permits unbounded polar nilpotency class. For the exact minimum dimension we must retain a terminal commutator in as small a quotient of W_d as possible.

4.2 Terminal tensors and contraction modules

Lemma 4.2. *Let $\mathfrak{l}$ be a metabelian $\mathbb{F}_4$-Lie algebra satisfying* (12)*, of class $c \geq 3$. Some nonzero terminal commutator has a repeated input between its first bracket pair and an operator position. If $c \geq 5$, there is a subalgebra of the same class generated by exactly $c-1$ elements modulo its derived ideal.*

Proof. The last lower central ideal is spanned by

$$T(x_1, x_2; x_3, \ldots, x_c) = P(x_3) \cdots P(x_c)\beta(x_1, x_2).$$

Indeed $W = \operatorname{span} \beta + \mathfrak{m}W$, and multiplication by $\mathfrak{m}^{c-2}$ kills the second summand. The operator positions are symmetric, and a repeated input within them gives zero.

Suppose every value $T(x, y; x, z_4, \ldots, z_c)$ were zero. Polarizing the repeated input gives

$$T(x, y; z, \ldots) = T(z, y; x, \ldots).$$

Together with symmetry of the first pair and the operator positions, this makes T fully symmetric. Jacobi on the first three inputs then gives $3T = T = 0$, a contradiction.

Choose a nonzero repeated-input value. Its inputs span a subspace U_0 of the abelianization of dimension at most $c-1$. Choose a linear section on a basis of U_0 and use its linear lifts

for all the inputs. Changing a lift inside W does not alter the terminal commutator: the additional term belongs to $\mathfrak{m}^{c-1}W = 0$. These basis lifts generate a subalgebra H of class c, with abelianization exactly U_0. Indeed, any linear combination of the chosen generators whose image in U_0 vanishes is zero, and all remaining Lie words lie in H'. Its module is a quotient of W_{d_0}, where $d_0 = \dim U_0$. By Proposition 4.1, class $c \geq 5$ requires $d_0 \geq c-1$. Thus $d_0 = c - 1$. □

For $d \geq 4$, the top coefficient degree of W_d is $d - 1$. It has basis $T_1, \ldots, T_d$, where

$$T_j = \Big(\prod_{h \neq i} t_h\Big) v_{ij} \quad (i \neq j). \tag{25}$$

Triangle relations identify the choices of i.

Lemma 4.3. *The top component is the* $\mathrm{GL}(d,4)$*-module* $\det(U) \otimes U$*. For every nonzero functional* λ *on that component, extended by zero to the other degrees,*

$$\dim_{\mathbb{F}_4} R_d\lambda = 2^d - 2.$$

If a quotient M *of* W_d *retains a nonzero top component, then* $\dim_{\mathbb{F}_4} M \geq 2^d - 2$*, whether or not the quotient is graded.*

Proof. Under a diagonal basis change A, T_j scales by $\det(A)A_{jj}$. A transvection $e_j \mapsto e_j + \alpha e_k$ sends T_j to $T_j + \alpha T_k$, and fixes the other top basis vectors. To check the formula choose $i \notin \{j, k\}$ in (25); any extra product has a squared variable. For the remaining vectors an index outside the affected entries can be chosen since $d \geq 4$. Together with permutations these transformations prove the module assertion. They act transitively on nonzero dual vectors, including scalar multiples.

Normalize $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \neq j$. We claim

$$\mathrm{Ann}_{R_d}(\lambda) = \Big(\prod_{i \neq j} t_i\Big). \tag{26}$$

The annihilator is multigraded: contractions by distinct monomials have distinct resulting dual multidegrees. For a square-free monomial t_S not containing j, the remaining multidegree is of the one-doubled-index type. It survives unless S contains every other index. If $j \in S$, the remaining multidegree has no doubled index. It survives whenever an index is omitted, as witnessed by an edge joining that index to j; it vanishes when all indices are used. These are exactly the two monomials in the ideal on the right of (26). This proves the claim and the dimension formula.

For an arbitrary quotient M, pull back a functional nonzero on the top image to $\phi \in W_d^*$. Its highest homogeneous component ϕ_{d-1} is nonzero. Independent homogeneous contractions of ϕ_{d-1} lift to independent contractions of ϕ, by comparison of their leading dual degrees. Hence

$$\dim M \geq \dim R_d\phi \geq \dim R_d\phi_{d-1} = 2^d - 2.$$

The filtered contraction argument applies to every quotient $W_d \to M$, including those with nongraded kernel. □

4.3 Exact minima

Theorem 4.4. *Among metabelian quadratic solutions of polar class* $c \geq 2$*, the minimum binary dimension is*

$$\mu(2) = 6, \quad \mu(3) = 8, \quad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5).$$

Proof. Write $d = \dim_{\mathbb{F}_4}(V/W)$, $w = \dim_{\mathbb{F}_4} W$. Nonabelian objects have $d \geq 2$ by Lemma 2.4. A nonzero product $P_1 \cdots P_k v$, for commuting square-zero operators, yields 2^k independent vectors

$$v_S = \Big(\prod_{i \in S} P_i\Big) v \qquad (S \subseteq \{1, \ldots, k\}).$$

In a relation, choose a term with nonzero coefficient of smallest support and multiply by its complementary operators. Every other surviving term repeats an operator and vanishes, while the chosen term does not. Taking $k = c - 2$ gives $w \geq 2^{c-2}$. For $c = 2, 3, 4$ this gives the respective lower bounds $6, 8, 12$.

They are attained by cyclic quotients of $W_2 = R_2 v_{12}$: use $R_2/\mathfrak{m}$, $R_2/(t_2)$, and R_2, respectively, as the derived module in (7). The first two representatives are

$$\begin{aligned}(a, b, z) &\longmapsto (\omega a, \omega b, \omega z + ab),\\ (a, b, y, z) &\longmapsto (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).\end{aligned}$$

A class-four attaining model on $\mathbb{F}_4^6$ is

$$\begin{aligned}(a, b, c, d, e, h) \longmapsto \big(&\omega a, \omega b, \omega c + ab, \omega d + ac + a^2 b,\\ &\omega e + bc + ab^2, \omega h + ae + bd\big).\end{aligned} \tag{27}$$

It is the normal form with $a_{12} = c$ and commuting square-zero actions $P_1 c = d$, $P_2 c = e$, $P_1 e = P_2 d = h$, all other basis images zero. Thus its derived module has basis c, d, e, h and its lower-central dimensions are $(12, 8, 6, 2, 0)$.

Let $c \geq 5$. Proposition 4.1 gives $d \geq c - 1$. Lemma 4.2 supplies a same-class subalgebra with $d_0 = c - 1$. Its derived module retains the top of W_{c-1}, so Lemma 4.3 gives $w \geq 2^{c-1} - 2$. Therefore

$$\dim_{\mathbb{F}_2} V = 2(d + w) \geq 2^c + 2c - 6.$$

To attain the bound, set $d = c - 1 \geq 4$, choose a top functional λ as in Lemma 4.3, and put

$$M_d = W_d / \{w : \lambda(rw) = 0 \text{ for every } r \in R_d\}. \tag{28}$$

The denominator is an R_d-submodule. The quotient has dimension $2^d - 2$ and retains the terminal value, so its solution has class $d + 1 = c$ and binary dimension $2(d + 2^d - 2)$. □

The change of formula at class five comes from the top degree of the universal module. For two and three quotient generators,

$$H_2(T) = 1 + 2T + T^2, \qquad H_3(T) = 3 + 8T + 6T^2.$$

In these cases the degree-two double-endpoint terms contribute at the top, allowing class four even with two quotient generators. For $d \geq 4$, they lie below the top degree $d - 1$; the terminal component $\det(U) \otimes U$ then controls class $d + 1$ uniformly. This is why the contraction argument yields one formula from class five onward, while the cyclic quotients of W_2 give the three smaller values.

The extremal quotient M_d in (28) has coefficient Hilbert polynomial

$$(d - 1) + \sum_{k=1}^{d-1} \binom{d}{d - 1 - k} T^k.$$

For example, the class-five extremizer has derived Hilbert vector $(3, 6, 4, 1)$ and binary lower central dimensions $(36, 28, 22, 10, 2, 0)$. The contraction construction supplies an explicit finite module description at every class, without an orbit search.

4.4 All minimum-dimensional metabelian solutions

The bound $\mu(c)$ also admits an equality classification. Let E_c denote the number of binary-linear conjugacy classes of metabelian quadratic solutions of class c and dimension $\mu(c)$. The three small classes close directly in the cyclic module W_2.

Lemma 4.5 (The three small minima)**.** *There is exactly one metabelian extremizer in each of classes two, three, and four. Thus $E_2 = E_3 = E_4 = 1$.*

Proof. Equality in the separate bounds $d \geq 2$ and $w \geq 2^{c-2}$ used in Theorem 4.4 forces $d = 2$ and $w = 1, 2, 4$, respectively. Intrinsic generation makes W cyclic over $R_2 = k[t_1, t_2]/(t_1^2, t_2^2)$, $k = \mathbb{F}_4$, generated by a_{12}: iterate $W = ka_{12} + \mathfrak{m}W$ and use $\mathfrak{m}^3 = 0$. Hence $W = R_2/I$, with the cocycle represented by its cyclic generator. For $w = 1$ the ideal is $I = \mathfrak{m}$, and for $w = 4$ it is zero.

If $w = 2$, then $I \subseteq \mathfrak{m}$ has dimension two. Every nonzero ideal of R_2 contains its socle kt_1t_2: multiply a nonzero linear part by a suitable generator, or start with a socle element. Thus I also contains a nonzero linear form u, after subtracting a multiple of t_1t_2, and $I = (u)$. A change of basis in $\mathrm{GL}_2(k)$ takes (u) to (t_2). The accompanying nonzero determinant factor on a_{12} is absorbed by a rescaling of the cyclic module generator. These are precisely the three quotients displayed above, up to the isomorphisms of Theorem 3.2. □

For the uniform range, equality first fixes the derived module. We then classify its surviving cocycles under all changes of quotient basis and splitting; this last step distinguishes equality objects with identical lower central dimensions. Fix $c \geq 5$, and put $k = \mathbb{F}_4$, $d = c - 1$, $m = d - 1$, and

$$U = ke_0 \oplus H, \qquad H = \mathrm{span}_k\{e_1, \ldots, e_m\}, \qquad R = k[t_0, \ldots, t_m]/(t_0^2, \ldots, t_m^2).$$

Write $K = (t_1, \ldots, t_m)$, $\tau = t_0 \cdots t_m$, and $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. For $\kappa = (\kappa_i) \in H^*$ and $\lambda = (\lambda_{ij})_{0 \leq i < j \leq m} \in k^{\binom{d}{2}}$, define an alternating cocycle $\beta_{\kappa,\lambda} : \Lambda^2 U \to K$ by

$$\begin{aligned} \beta_{\kappa,\lambda}(e_0, e_i) &= t_i + \lambda_{0i}\tau_{0i} && (1 \leq i \leq m), \\ \beta_{\kappa,\lambda}(e_i, e_j) &= t_0(\kappa_i t_j + \kappa_j t_i) + \lambda_{ij}\tau_{ij} && (1 \leq i < j \leq m). \end{aligned} \tag{29}$$

Let $F_{\kappa,\lambda}$ be the map in (7) with

$$W = K, \qquad P(x)w = \Big(\sum_{i=0}^{m} x_i t_i\Big)w, \qquad Q(x) = \sum_{i<j} x_i x_j \beta_{\kappa,\lambda}(e_i, e_j).$$

Thus these representatives require only multiplication in a square-zero polynomial ring.

Theorem 4.6 (Unrestricted metabelian extremizers)**.** *Every metabelian quadratic solution of class $c \geq 5$ and binary dimension $\mu(c)$ is binary-linearly conjugate to some $F_{\kappa,\lambda}$. Its derived module is isomorphic to K.*

The exact equivalence relation on the displayed parameters is the following ordinary Lie-cohomology orbit relation:

$$[\beta_{\kappa',\lambda'}] = \big[a\,\sigma_B\beta_{\kappa,\lambda}(B^{-1}\cdot, B^{-1}\cdot)\big], \qquad B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}, \tag{30}$$

where $a \in k^\times$, $M \in \mathrm{GL}(H)$, $v \in H$, $\sigma_B(t_j) = \sum_i B_{ij}t_i$, and brackets denote classes in $H^2(U, K)$ for the action $e_i w = t_i w$. In particular, no nonlinear substitutions or Galois identifications are required.

Their number E_c is

$$E_c = \begin{cases} (7c - 13)/2, & c \geq 5 \text{ odd}, \\ 7(c-2)/2 + 21_{\,6|c}, & c \geq 6 \text{ even}. \end{cases} \tag{31}$$

Proof. Equality in the contraction bound forces the derived module to be the dual of $R/(t_1 \cdots t_m)$, hence to be K. An ordinary cohomology calculation gives a terminal star together with the κ- and λ-classes in (29); intrinsic generation is equivalent to the star being nonzero, and it has been normalized to one. Module intertwining maps give exactly (30). Theorem 3.2 then transfers the auxiliary-Lie classification to binary conjugacy. Appendix B proves the common-module assertion, the cohomology and morphism statements, and the complete orbit count. □

For $d \geq 5$, the orbit relation is especially explicit. If $A^{[2]}$ means entrywise squaring, its action is

$$\kappa' = a^2 M^{-T} \kappa, \qquad \lambda' = a \det(B)(\Lambda^2 B^{-T})^{[2]} \lambda. \tag{32}$$

After setting $\Omega = \lambda^{[2]}$ and writing $\Omega = e_0^* \wedge \ell + b$, the classification reduces to an alternating form b on H, two covectors on its radical, and one zero/nonzero flag. In the corank-two stratum an additional three-valued scalar invariant survives exactly when $6 \mid c$; this accounts for the correction in (31). At $c = 5$, two cohomological degrees coincide and the parameter representation is nonsplit. A symmetric-matrix description nevertheless gives all eleven classes, including this endpoint in the same count.

Models inside Cohn's construction. The regular Lie model and the zero-λ slice have a concrete classical source. Take characteristic two and the finite ground field $k = \mathbb{F}_4$ in Cohn's construction [5, Section 2, pp. 402–403]. Reindex its transcendental elements as $\xi_0, \xi_1, \ldots$, put $u = u(0)$, and let $D_i = \operatorname{ad} \xi_i|_M$. For $S \subseteq \{0, \ldots, d-1\}$, set

$$v_S = \Big(\prod_{i \in S} D_i\Big) u = \Big(\prod_{i \in S} \xi_i\Big) \sum_{T \subseteq S} u(\mathbf{1}_T).$$

Here $\mathbf{1}_T$ is the indicator sequence indexing Cohn's module basis. The displayed expansion is triangular in those basis vectors, so the v_S's are independent even over Cohn's larger coefficient field. The operators commute and square to zero, and $D_i v_S = v_{S \cup \{i\}}$ if $i \notin S$, and zero otherwise. Consequently the k-span of the ξ_i's and v_S's is the regular-module semidirect product $k^d \ltimes R_d$, of dimension $d + 2^d$ and class $d + 1$. Its polar realization is (62), or the compatible model F_0 after writing a ring element as its constant term plus an element of J.

The zero-λ models in (29) also occur as finite subalgebras. Choose

$$e_0 = \xi_0 + v_\varnothing, \qquad e_i = \xi_i + \kappa_i v_{\{0\}} \quad (1 \leq i < d).$$

Their derived module is spanned by the v_S's with $S \cap \{1, \ldots, d-1\} \neq \varnothing$, and is therefore K. Their action and brackets are multiplication by t_i, $\beta(e_0, e_i) = t_i$, and $\beta(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i)$. Thus this finite subalgebra has dimension $d + 2^d - 2$, class $d + 1$, and gives $F_{\kappa,0}$ through Theorem 3.2. These finite identifications are derived here from Cohn's construction; his paper establishes the infinite-dimensional example rather than an extremal dimension formula.

Cohn's full characteristic-two algebra satisfies

$$[[[x, y], x], y] = 0.$$

Indeed, for $x = a + u'$, $y = b + v'$, its bracket is $D_b u' + D_a v'$; the commuting operators D_a, D_b square to zero, so applying $D_a D_b$ kills it. In the present family,

$$[[[e_i, e_j], e_i], e_j] = t_i t_j \beta(e_i, e_j) = \lambda_{ij} \tau.$$

Hence the nonzero-λ models lie outside every Lie subquotient of Cohn's characteristic-two example. The sharp bounds and full orbit classifications determine how the classical slice and these additional models populate the extremal layers. The same finite-subalgebra identifications hold over any finite field of characteristic two; the binary-map interpretation uses $k = \mathbb{F}_4$.

5 Fixed-surplus populations

At the sharp minimum the metabelian types have linear growth in the polar class. The first two added binary dimensions already produce a doubly exponential population. This section identifies the mechanism and gives the fixed-surplus law and exact principal orbits.

For orientation, put $n = 2^{c-1}$, and let r be the surplus in k-dimension, corresponding to $2r$ binary dimensions at $q = 4$. The three population scales are

$r \geq 1$ fixed :	fixed-surplus law (Theorem 5.3);
$r/\sqrt{n} \to \alpha \in (0, \infty)$:	growth and allocation (Theorem 6.5);
$r/\sqrt{n} \to \infty$:	minimum-derived density tends to zero (Theorem 6.1).

The critical staircase then resolves the transition on the integer defect scale in Section 6.5.

Counting convention. Throughout this section, $d = c - 1$ is the class-dependent core parameter. It is distinct from the point rank $d(L)$ of an individual solution: in a surplus branch the quotient can have dimension $d + s$, where s is the quotient surplus. The corresponding proof appendices use the same local convention. At $q = 4$, ordinary $\mathbb{F}_4$-Lie isomorphism counts agree with binary-linear conjugacy counts through the metabelian correspondence. We specify the field whenever the count is stated over a general $\mathbb{F}_q$.

5.1 Two more binary dimensions

The linear class count at $\mu(c)$ suggests a stability question: does a small dimension surplus preserve the rigidity of the minimum? The next theorem gives a sharp answer. Already the first admissible increase in dimension supports a much larger family, while keeping the same derived module as a metabelian extremizer.

Theorem 5.1 (A sharp jump above the minimum)**.** *For each $c \geq 5$, there is a family of indecomposable, noncompatible metabelian quadratic solutions of polar class c and binary dimension*

$$\mu(c) + 2 = 2^c + 2c - 4.$$

Put $d = c-1$ and $D_{\mathrm{par}} = 2^d - d - 3$. If T_c counts the binary-linear conjugacy classes represented by this family, and A_c counts all metabelian classes of polar class c and binary dimension $\mu(c) + 2$, then

$$\begin{gathered} \frac{4^{D_{\mathrm{par}}} - 1}{3\, 4^d |\, \mathrm{GL}_d(4)|} \leq T_c \leq 4^{D_{\mathrm{par}}} - 1, \\ T_c \leq A_c \leq E_c\, 4^{\,2^d - 2} + 4^{(3d^2 - d)/2}. \end{gathered} \tag{33}$$

In particular

$$\log_4 A_c = 2^{c-1} + O(c^2) \qquad (c \longrightarrow \infty).$$

Thus doubly exponential class growth occurs at the first admissible dimension above the metabelian minimum.

Proof. We construct the family, identify an intrinsic line in its quotient, and use that line to bound how many parameters can become conjugate. The key point is parameter recovery under arbitrary binary conjugacy.

Construction. Use

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \qquad K = (t_1, \ldots, t_{d-1}), \qquad k = \mathbb{F}_4.$$

Write $z = t_1 \cdots t_{d-1}$, $\tau = t_0 z$, and $I = (z) = kz \oplus k\tau = \operatorname{Ann}_R K$. Let $\mathcal{P}_d$ be the space of sums of square-free monomials of degree at least two, excluding z and τ. It is a fixed representative space for $\operatorname{rad}(R/I)^2$, has dimension $D_{\mathrm{par}} = 2^d - d - 3$, and is contained in K.

For $p \in \mathcal{P}_d$, take $U = \operatorname{span}_k\{e_0, \dots, e_{d-1}, v\}$, with actions on K and cocycle

$$\begin{aligned} P(e_i) &= t_i \cdot, & P(v) &= p\cdot, \\ \beta(e_0, e_i) &= t_i \quad (i \geq 1), & \beta(e_0, v) &= p, \end{aligned}$$

all other quotient-basis pair values being zero. The actions commute and square to zero. The only additional triangle equation is $t_i p + p t_i = 0$, and the t_i, $i \geq 1$, generate K. These are therefore data for Theorem 3.2. Explicitly, put $b_p(X, V) = \sum_{i=1}^{d-1} X_i t_i + Vp$. The map is

$$G_p(a, X, V, w) = \big(\omega a, \omega X, \omega V, \omega w + (at_0 + b_p)w + ab_p + a^2 t_0 b_p\big), \quad w \in K. \tag{34}$$

Indeed $Q = ab_p$ and $PQ = a^2 t_0 b_p$, since $b_p^2 = 0$. All displayed terms have binary degree at most two. The quotient has k-dimension $d+1$, and $\dim_k K = 2^d - 2$, giving the stated binary dimension. As $p \in \mathfrak{m}^2$, the radical action ideal is still $\mathfrak{m} = (t_0, \dots, t_{d-1})$, and

$$\mathfrak{m}^{d-1} K = k\tau \neq 0, \qquad \mathfrak{m}^d K = 0.$$

Thus every model has polar class $d + 1 = c$.

Parameter recovery. The unital algebra generated by the actions on the derived ideal is $\mathcal{A} = R/I$, acting faithfully on K. Its radical $\mathfrak{r}$ is intrinsic. The map

$$U \longrightarrow \mathfrak{r}/\mathfrak{r}^2, \qquad u \longmapsto P(u) \pmod{\mathfrak{r}^2},$$

has kernel exactly kv. A binary conjugacy between G_p and G_q induces k-linear maps on U and K, since their intrinsic scalar actions are multiplication by ω. It must therefore induce a quotient map of the form

$$Ae_i = \sum_j B_{ji} e_j + \alpha_i v, \qquad Av = \delta v,$$

where $B \in \mathrm{GL}_d(k)$, $\alpha \in k^d$, and $\delta \in k^\times$. There are $3\,4^d |\,\mathrm{GL}_d(4)|$ possible such maps.

Let C be the map on the derived ideal. Mixed polar brackets give $CP_p(u)C^{-1} = P_q(Au)$. Hence conjugation by C induces an algebra isomorphism $\Phi : \mathcal{A} \to \mathcal{A}$ satisfying

$$\Phi(t_i) = \ell_i + \alpha_i q, \qquad \ell_i = \sum_j B_{ji} t_j, \qquad \Phi(p) = \delta q.$$

In this algebra every radical element squares to zero. Substituting the fixed polynomial representative of p therefore gives the exact identity

$$\delta q = p(\ell + \alpha q) = p(\ell) + qh, \qquad h = \sum_i \alpha_i (\partial_i p)(\ell) \in \mathfrak{r}.$$

The derivatives are taken on that representative in R, before evaluation in R/I; no derivation on the quotient is assumed. Since p has no linear part, h has zero constant term. Consequently $\delta + h$ is a unit, and

$$q = (\delta + h)^{-1} p(\ell) = (\delta^{-1} + \delta^{-2} h) p(\ell) \quad \text{in } R/I. \tag{35}$$

For a fixed quotient map A, this determines the canonical representative q uniquely. Thus a conjugacy class meets $\mathcal{P}_d$ in at most $3\,4^d |\,\mathrm{GL}_d(4)|$ parameters. This necessary condition

suffices for the counting bound; it does not require deciding which quotient maps lift to full conjugacies.

Indecomposability and counting. Restrict to $p \neq 0$. The action map $U \to \mathcal{A}$ is then injective: the t_i have independent linear parts and p is a nonzero element of $\mathfrak{r}^2$. A central vector of the original polar algebra must consequently lie in K. There its centralizer condition is $t_i w = 0$ for all i, so its center is exactly $k\tau$, of binary dimension two. In a nontrivial direct sum of solutions, each nilpotent polar summand has a nonzero center of even binary dimension (Proposition 2.3). Such a direct sum would have center dimension at least four. Hence every G_p, $p \neq 0$, is indecomposable. None is compatible, because $\mu(c) + 2 < \nu(c)$.

There are $4^{D_{\mathrm{par}}} - 1$ nonzero parameters, and the fiber bound proves the bounds on T_c in (33).

An upper bound for the entire dimension layer. The separate dimension bounds from Theorem 4.4 leave only two possibilities for the k-dimensions of quotient and derived ideal:

$$(d+1, 2^d - 2) \quad \text{or} \quad (d, 2^d - 1).$$

In the first case, Lemma 4.2 supplies a same-class subalgebra with d quotient generators. Its derived ideal already has dimension at least $2^d - 2$, so equals that of the full algebra. It is therefore one of the E_c metabelian extremizers. After choosing its representative, the extra quotient action belongs to $\mathrm{End}_R K = R/I$. Its constant term is zero, and its linear part can be removed by adding core quotient directions. At most $4^{D_{\mathrm{par}}}$ actions remain. For a fixed action p, the mixed cocycle values $a_i = \beta(e_i, v)$ satisfy

$$t_i a_j + t_j a_i = p\beta(e_i, e_j).$$

Their differences are ordinary one-cocycles. Changing the lift of v by $w \in K$ adds the one-coboundary $(t_i w)_i$. Lemma B.3 gives $\dim_k H^1(U_0, K) = d + 1$, where U_0 is the core quotient. This case therefore contributes at most $E_c 4^{D_{\mathrm{par}}+d+1} = E_c 4^{2^d-2}$ classes.

In the second case the derived module is a quotient of W_d retaining the top. Lemma B.2 gives at most $4^{\binom{d}{2}}$ possible module types and $\dim_k H^2(U, W) \leq d^2$. For each fixed module, changes of section identify cocycles differing by a two-coboundary. Thus at most $4^{\binom{d}{2}+d^2} = 4^{(3d^2-d)/2}$ classes occur in this case. These are upper bounds: no uniqueness of the chosen core or the codimension-one module presentation is required.

Combining the two cases proves the bound on A_c. Since $E_c = O(c)$ and $\log_4 |\mathrm{GL}_d(4)| = d^2 + O(1)$, the upper and lower bounds give the asserted logarithmic asymptotic. Finally all solution dimensions are even by Proposition 2.1, so two is the first possible positive surplus above $\mu(c)$. □

The two upper bounds locate the source of the jump. Adding one $\mathbb{F}_4$-dimension to the derived ideal allows at most $4^{(3d^2-d)/2}$ classes. Keeping that ideal at its minimum and adding one quotient direction already produces the doubly exponential contribution. The location of the extra dimension therefore matters as well as its size.

The contrast is already quantitative at class seven. The minimum dimension $\mu(7) = 136$ has exactly $E_7 = 18$ classes, whereas (33) gives at least 32 485 958 indecomposable noncompatible classes in dimension 138. This lower bound comes from the uniform recovery argument, rather than an enumeration of the models.

Lie-algebra interpretation. The associated auxiliary $\mathbb{F}_4$-Lie algebras have the same one-dimensional center $k\tau$, so they too are indecomposable. They admit no restricted square. Indeed, $(\operatorname{ad} e_0)^2$ is zero on K. If it were $\operatorname{ad} y$, injectivity of the quotient action would force $y \in K$. But

$$(\operatorname{ad} e_0)^2 e_1 = t_0 t_1, \qquad [y, e_1] = t_1 y,$$

and the latter has zero t_0t_1-coefficient because $y \in K$ has no t_0-term. Thus $(\operatorname{ad} e_0)^2$ is not inner, contradicting the restricted-square adjoint axiom. The correspondence therefore also gives the stated number of distinct indecomposable, nonrestrictable Lie algebras of class c and $\mathbb{F}_4$-dimension $2^{c-1} + c - 2$.

The construction uses the classical abelian-extension framework [3]. Its additional feature is that an intrinsic quotient line and (35) control all binary conjugacies despite the large group acting on the derived module. The upper bound also controls the full first-surplus layer. Explicit orbit representatives and an exact finite count for every c remain separate classification questions.

5.2 A law for every fixed surplus

The first-surplus transition belongs to a uniform counting law. It is convenient to formulate it for ordinary Lie algebras over a finite field $k = \mathbb{F}_q$ of characteristic two, with $q \geq 4$. Dimensions in this subsection are over k, unless explicitly called binary. A Lie algebra is *3-Engel* if $(\operatorname{ad} x)^3 = 0$ for every x. The following classical linearization identifies the relevant metabelian category; it is the metabelian consequence of [16, Section 1.2, identity (6)]. We include the short argument to specify the field-size hypothesis.

Lemma 5.2. *Let L be metabelian over a characteristic-two field with at least four elements, and put $W = L'$. Then L is 3-Engel if and only if $(\operatorname{ad} x)^2|_W = 0$ for every $x \in L$.*

Proof. The restricted operators $P_x = \operatorname{ad}(x)|_W$ commute, since $[L', L'] = 0$. If their squares vanish, then $(\operatorname{ad} x)^3 z = P_x^2[x, z] = 0$. Conversely, apply the 3-Engel identity to $x + ty$ and z. Commutativity and characteristic two give

$$0 = P_x^2[x, z] + tP_x^2[y, z] + t^2P_y^2[x, z] + t^3P_y^2[y, z].$$

This polynomial has degree at most three, below the field size. Its coefficient of t vanishes, so $P_x^2[y, z] = 0$. Brackets span W, proving the assertion. □

Write $\ell(c)$ for the minimum dimension of a metabelian 3-Engel algebra of class $c \geq 5$. Lemma C.1 gives $\ell(c) = 2^{c-1} + c - 3$. Let $B_{c,r}(q)$ count ordinary k-Lie isomorphism classes of class c and dimension $\ell(c) + r$ in this category.

Theorem 5.3 (The fixed-surplus law)**.** *Put $d = c - 1 \geq 4$ and $D_{\mathrm{par}} = 2^d - d - 3$. For $1 \leq r \leq D_{\mathrm{par}}$,*

$$B_{c,r}(q) \;\geq\; \frac{\displaystyle\prod_{j=0}^{r-1}(q^{D_{\mathrm{par}}} - q^j)}{q^{dr}\,|\operatorname{GL}_d(q)|\,|\operatorname{GL}_r(q)|}. \tag{36}$$

The lower bound is realized by indecomposable algebras with one-dimensional center and no restricted square. For each fixed q and $r \geq 1$,

$$\log_q B_{c,r}(q) = r\,2^{c-1} + O_{q,r}(c^2) \qquad (c \longrightarrow \infty). \tag{37}$$

Proof. Put

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \quad K = (t_1, \ldots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}) = \operatorname{Ann}_R K.$$

Put $\mathfrak{m} = (t_0, \ldots, t_{d-1})$, so $\dim K = 2^d - 2$. Take canonical representatives $\mathcal{P}_d$ of $\operatorname{rad}(R/I)^2$, excluding the two monomials in I; their dimension is D_{par}. Choose an ordered independent frame $\mathbf{p} = (p_1, \ldots, p_r)$ in $\mathcal{P}_d$. On $U \oplus K$, where $U = \langle e_0, \ldots, e_{d-1}, v_1, \ldots, v_r\rangle_k$, define commuting actions and an alternating cocycle by

$$P(e_i) = t_i\cdot, \quad P(v_a) = p_a\cdot, \qquad \beta(e_0, e_i) = t_i \ (i \geq 1), \quad \beta(e_0, v_a) = p_a,$$

with all other basis pair values zero. In particular, the Jacobi equation on e_0, v_a, v_b is $p_a p_b + p_b p_a = 0$. The cocycle values generate K, so the derived ideal is exactly K. All radical elements of R square to zero; hence the algebra is 3-Engel by Lemma 5.2. Its lower central ideals are $\gamma_j = \mathfrak{m}^{j-2}K$ for $j \geq 2$, with terminal term $k(t_0 \cdots t_{d-1})$. Its class is $d+1$, and its dimension is $d + r + 2^d - 2 = \ell(c) + r$.

The faithful generated action algebra is $\mathcal{A} = R/I$. The kernel of the intrinsic map $U \to \operatorname{rad}(\mathcal{A})/\operatorname{rad}(\mathcal{A})^2$ is $V_r = \langle v_1, \ldots, v_r \rangle_k$. Any isomorphism between two models therefore induces

$$Ae_i = \sum_j B_{ji} e_j + \sum_b \alpha_{bi} v_b, \qquad Av_a = \sum_b \Delta_{ba} v_b,$$

where $B \in \mathrm{GL}_d(k)$, $\Delta \in \mathrm{GL}_r(k)$. There are $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ possible quotient maps. For target frame $\mathbf{p}'$, conjugation on the derived ideal induces an algebra isomorphism satisfying

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}'), \qquad \ell_i = \sum_j B_{ji} t_j. \tag{38}$$

All polynomials here are evaluated on their fixed representatives.

For a fixed quotient map this equation has at most one target frame. Indeed, compare two solutions and let e be their difference. Every source polynomial has degree at least two, and all its arguments lie in $\operatorname{rad}(\mathcal{A})$. Exact telescoping of each monomial therefore gives $\Delta^T e = Me$, with every entry of M in that radical. Thus $e = Ne$, where $N = (\Delta^T)^{-1} M$. Since $\operatorname{rad}(\mathcal{A})^d = 0$, also $N^d = 0$, and iteration gives $e = 0$. The exact telescoping identity retains all products involving distinct target parameters. This uniqueness is an instance of the nilpotent implicit-function principle of Bahturin–Olshanskii [1, Sections 1.2–1.3]; the intrinsic quotient subspace and the equation forced by isomorphism specify its application here. Each isomorphism class consequently contains at most $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ ordered frames from this family, proving (36).

Frame independence makes $U \to \mathcal{A}$ injective. The center is therefore contained in K, where it is the one-dimensional socle $k(t_0 \cdots t_{d-1})$. This implies indecomposability. If $(\operatorname{ad} e_0)^2 = \operatorname{ad} y$, its zero action on K forces $y \in K$. But $(\operatorname{ad} e_0)^2 e_i = t_0 t_i$, whereas $t_i y$ has zero $t_0 t_i$-coefficient. Hence no restricted square exists.

For fixed q, r, the numerator of (36) has logarithm $rD_{\mathrm{par}} + O_{q,r}(1)$, while the denominator has logarithm $d^2 + dr + O_{q,r}(1)$. This proves the required lower estimate. Proposition C.3 gives the matching upper estimate, including all possible derived-module defects, and proves (37). □

Corollary 5.4 (Generic structure)**.** *With isomorphism classes counted uniformly, the proportion in $B_{c,r}(q)$ failing any of the following properties is at most $q^{-2^d + O_{q,r}(d^2)}$: the derived ideal has minimum dimension $2^d - 2$ and is the standard core module K; the action $L/L' \to \mathrm{End}_k(L')$ is injective; the center is one-dimensional; the algebra is indecomposable and has no restricted square.*

Proof. In the upper-bound proof, write the quotient and derived dimensions as $d + s$ and $2^d - 2 + h$, where $s + h = r$. The branches $h \geq 1$ together contribute at most $q^{(r-1)2^d + O_{q,r}(d^2)}$ classes. When $h = 0$, the derived module is K. Removing the linear parts of the extra actions gives an r-tuple in $\mathcal{P}_d$. Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}} + r(r-1)}$: choose $r-1$ spanning vectors and their coefficient matrix. The remaining cocycle choices cost only $q^{O_{q,r}(d^2)}$. Thus these exceptions have the same bound. Injectivity of the full quotient action is intrinsic, so different choices of core do not affect this division into cases.

For a full-rank tuple, the center and indecomposability follow as above. In a normalized extremal core, (92) gives $\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K}$. Hence $(\operatorname{ad} e_0)^2 e_i$ still has coefficient one

at $t_0 t_i$. The same inner-derivation obstruction excludes a restricted square, independently of the mixed cocycle choices. Dividing the exceptional bound by (36) proves the asserted proportion. □

Corollary 5.5 (Binary fixed-surplus law)**.** *Let $A_{c,r}$ count all metabelian quadratic solutions of polar class c and binary dimension $\mu(c) + 2r$, up to binary-linear conjugacy. Then $A_{c,r} = B_{c,r}(4)$, and, for each fixed $r \geq 1$,*

$$\log_4 A_{c,r} = r\, 2^{c-1} + O_r(c^2).$$

Almost all have binary polar center of dimension two and are indecomposable; their auxiliary $\mathbb{F}_4$-Lie algebras admit no restricted square.

Proof. Use Theorem 3.2, Lemma 5.2, and $\mu(c) = 2\ell(c)$. For the generic objects the faithful quotient action forces the original polar center into the derived ideal, where it is the same one-dimensional $\mathbb{F}_4$-socle. Its binary dimension is two, excluding a nontrivial direct sum by Proposition 2.3. The lower-bound family is noncompatible for every r: the compatibility criterion would require $t_i b_0 = t_0 t_i$ with $b_0 \in K$, whose $t_0 t_i$-coefficients are respectively zero and one. □

Corollary 5.6 (Compatibility at fixed surplus)**.** *For every fixed $r \geq 2$, the proportion of compatible objects among the $A_{c,r}$ classes is at most $4^{-2^d + O_r(d^2)}$. At $r = 2$, their dimension is precisely $\nu(c)$, and the compatible classes are precisely the extremizers counted by N_c. With $D_{\mathrm{par}} = 2^d - d - 3$,*

$$\frac{N_c}{A_{c,2}} \leq \frac{60(1+\eta_d)\, 4^{-2^d+3d+4}}{(1-4^{-D_{\mathrm{par}}})(1-4^{1-D_{\mathrm{par}}})} = 4^{-2^d+O(d)}. \tag{39}$$

Here η_d is the explicit bound from Corollary 8.13.

Proof. Lemma 8.2 gives $\dim_{\mathbb{F}_4} L' \geq 2^d - 1$, so compatible objects lie in the $h \geq 1$ upper-bound branches. This proves the first assertion. For the sharper $r = 2$ estimate, combine

$$N_c \leq \frac{4^{D_{\mathrm{par}}+1}}{3|\,\mathrm{GL}_d(4)|}(1+\eta_d)$$

with (36). The $\mathrm{GL}_d(4)$ factors cancel, and $|\,\mathrm{GL}_2(4)| = 180$ gives (39). □

5.3 Exact principal orbits at every fixed surplus

The minimum-derived stratum has a complete orbit model for any number of additional quotient directions. Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and put

$$d = c - 1 \geq 4, \quad C_d = \binom{d}{2}, \quad D_{\mathrm{par}} = 2^d - d - 3, \quad w_0 = 2^d - 2.$$

Use R, K, I from (91), and let $\mathcal{P}_d$ be the canonical monomial representatives of $\mathrm{rad}(R/I)^2$. On $U = \langle e_0, \ldots, e_{d-1}, v_1, \ldots, v_r \rangle_k$, let $P_{\mathbf{p}}(e_i) = t_i \cdot$ and $P_{\mathbf{p}}(v_a) = p_a \cdot$, for $\mathbf{p} \in \mathcal{P}_d^r$. Let η denote the degree-one star coefficient, and set

$$\mathcal{Y}_{d,r} = \coprod_{\mathbf{p} \in \mathcal{P}_d^r} \{[\beta] \in H^2(U, K; P_{\mathbf{p}}) : \eta = 1\}.$$

Each fiber has dimension

$$f_{d,r} = (r+1)d + C_d + r + \binom{r}{2} - 1.$$

Write $\mathcal{Y}_{d,r}^{\mathrm{fr}}$ for its restriction to linearly independent tuples $p_1, \dots, p_r$.

Let P_H be the parabolic preserving $H = \langle e_1, \dots, e_{d-1} \rangle_k$. The effective quotient group is

$$G_{d,r} = \left\{ A_g = \begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix} : B = \begin{pmatrix} a & 0 \\ b & M \end{pmatrix} \in P_H,\ \alpha \in k^{r \times d},\ \Delta \in \mathrm{GL}_r(k) \right\}, \tag{40}$$

with

$$|G_{d,r}| = (q-1) q^{rd+d-1} |\,\mathrm{GL}_{d-1}(q)|\,|\mathrm{GL}_r(q)|.$$

For $\ell_i = \sum_j B_{ji} t_j$, its action is

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}') \qquad \text{in } R/I, \tag{41}$$

$$[\beta'] = [a \sigma_g \beta(A_g^{-1} \cdot, A_g^{-1} \cdot)], \qquad \sigma_g(t_i) = \ell_i + \sum_a \alpha_{ai} p'_a. \tag{42}$$

The implicit system has a unique solution in $\mathcal{P}_d^r$.

Theorem 5.7 (Complete minimum-derived orbits)**.** *The isomorphism classes of metabelian 3-Engel k-Lie algebras of class c, dimension $\ell(c) + r$, and minimum derived dimension w_0, are exactly $\mathcal{Y}_{d,r}/G_{d,r}$. The faithful-action subfamily is exactly $\mathcal{Y}_{d,r}^{\mathrm{fr}}/G_{d,r}$. Equations* (41)–(42) *are sufficient as well as necessary for equivalence.*

Theorem 5.8 (Multiplicative enumeration at fixed surplus)**.** *For every fixed $q \geq 4$ of characteristic two and fixed $r \geq 1$,*

$$B_{c,r}(q) \sim \frac{\displaystyle\prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)\, q^{C_d + r + \binom{r}{2}}}{(q-1)|\,\mathrm{GL}_{d-1}(q)|\,|\mathrm{GL}_r(q)|} \qquad (c \longrightarrow \infty). \tag{43}$$

Almost all isomorphism classes have full automorphism-group order

$$|\,\mathrm{Aut}_k(L)| = q^{2w_0 + d + r - 1} = q^{2^c + c + r - 6}. \tag{44}$$

For $q = 4$, these are respectively the binary conjugacy count $A_{c,r}$ and its generic binary-linear centralizer order.

The proof retains all cocycle coefficients, including brackets between different added directions. Nontrivial linear parts are controlled by an exterior-rank estimate. For a pure matrix shear, specializing its active variables to zero gives a necessary affine multiplication constraint. This retains the higher cross-products in the implicit substitution and controls the summed Burnside error for every fixed r. Appendix D proves both statements. At $r = 1$, (43) reduces exactly to

$$B_{c,1}(q) \sim \frac{(q^{D_{\mathrm{par}}} - 1) q^{C_d + 1}}{(q-1)^2 |\,\mathrm{GL}_{d-1}(q)|} \sim \frac{q^{2^d - C_d - 5}}{(1 - q^{-1})^2 \prod_{j=1}^{\infty} (1 - q^{-j})}.$$

6 The critical transition

The fixed-surplus law persists while the surplus is small relative to the square root of the core dimension. At that scale the whole population changes its typical derived dimension. We first locate the transition, then determine its constant, allocation profile, and discrete staircase. Let $B_{c,r}^{\min}(q)$ count the classes in $B_{c,r}(q)$ whose derived ideal has the minimum possible dimension $2^{c-1} - 2$, and put

$$\pi_{d,r}(q) = \frac{B_{d+1,r}^{\min}(q)}{B_{d+1,r}(q)}, \qquad d = c - 1.$$

Both the numerator and denominator count unlabelled, k-linear Lie isomorphism classes. In particular this is not a parameter-weighted proportion.

Theorem 6.1 (A transition scale for minimum-derived objects)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and let $d \to \infty$. Uniformly over positive integers r,*

$$\log_q B_{d+1,r}(q) = \Theta\big(r2^d + r^3\big). \tag{45}$$

For integer sequences $r = r(d) \geq 1$,

$$r = o(2^{d/2}) \quad \Longrightarrow \quad \log_q B_{d+1,r}(q) \sim r2^d, \qquad \pi_{d,r}(q) \longrightarrow 1, \tag{46}$$

$$r/2^{d/2} \longrightarrow \infty \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{47}$$

In the second regime, $\log_q \pi_{d,r}(q) \leq -(2/27 - o(1))r^3$.

The small-surplus assertion follows by retaining the derived defect h in Proposition C.3; its loss of $h2^d$ dominates the remaining terms. For the opposite regime, two-step alternating-tensor families supply a cubic population. Their contribution can be counted through the characteristic quotient $L/\gamma_3(L)$, whose dimension discards the large minimum core. Appendix C.1 gives the precise bounds and proof.

Thus $2^{(c-1)/2}$ separates the two off-window behaviors of the minimum-derived population. Theorem 6.2 below sharpens this scale to an explicit critical constant. The finer allocation and staircase theorems then localize the remaining undetermined densities to shrinking boundary bands. The large-surplus lower family uses direct sums, but the conclusion concerns only the derived dimension; it makes no assertion that a typical object in that regime is decomposable. At $q = 4$, the same conclusions apply to binary conjugacy classes in dimension $\mu(c) + 2r$, with minimum binary derived dimension $2^c - 4$, by Corollary 5.5.

6.1 The critical constant for minimum derived dimension

The transition scale admits a sharp constant away from its boundary. Two effects determine it. Normalizing the subspace spanned by the extra actions removes a quadratic coordinate redundancy from the minimum-derived count. A single additional central derived direction, coupled to those same quotient directions, supplies enough alternating forms to outweigh that count on the other side. The latter family consists of indecomposable algebras.

Theorem 6.2 (The critical constant)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. For every integer sequence $r = r(d) \geq 1$, as $d \to \infty$,*

$$\limsup_{d\to\infty} \frac{r}{2^{d/2}} < \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \tag{48}$$

$$\liminf_{d\to\infty} \frac{r}{2^{d/2}} > \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{49}$$

In particular, if $r/2^{d/2} \to \alpha \neq \sqrt{2}$ with $0 \leq \alpha \leq \infty$, the limit is one for $\alpha < \sqrt{2}$ and zero for $\alpha > \sqrt{2}$. These are proportions of unlabelled k-linear isomorphism types.

Proof. Write $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, and use B_h for the branch with derived dimension $n - 2 + h$, so $\pi_{d,r}(q) = B_0/\sum_h B_h$. For the first implication there is an $\epsilon > 0$ such that eventually $r^2/2 \leq (1-\epsilon)n$. In particular $r \leq D_{\mathrm{par}}/2$, and

$$Z(d, r) = 2d^2 + 3dr + 2r + 6 = o(n).$$

Proposition E.3 gives, with $a = n - r^2/2 - Z(d, r) \geq (\epsilon - o(1))n$,

$$1 - \pi_{d,r}(q) \leq \frac{q^{-a}}{1 - q^{-a}} \longrightarrow 0.$$

For the second implication, first consider the indices with $r \le D_{\mathrm{par}}/2$. Here $r \ge 3$ eventually and Proposition E.4 yields

$$\pi_{d,r}(q) \le \frac{B_0}{B_1} \le q^{\,n-r^2/2+dr+d^2+r/2+\log_q(\frac{3}{2}(r+1))}.$$

The strict lower limit assumption gives an $\epsilon > 0$ with $n - r^2/2 \le -\epsilon r^2$ eventually. Since $r \gg d$, every remaining exponent term is $o(r^2)$; the last display tends to zero along these indices. On the remaining indices, $r > D_{\mathrm{par}}/2$, one has

$$\frac{r}{\sqrt{n}} > \frac{n-d-3}{2\sqrt{n}} \longrightarrow \infty.$$

The all-surplus upper-side assertion (47) of Theorem 6.1 therefore applies along this subsequence and also gives $\pi_{d,r}(q) \to 0$. Both subsequences have the same limit. This argument uses no monotonicity of $\pi_{d,r}(q)$ as the dimension changes. □

For $q = 4$, Theorem 3.2 transfers the same critical constant to binary-linear conjugacy classes of metabelian quadratic solutions in binary dimension $\mu(c) + 2r$, where $d = c - 1$. The counted minimum derived dimension is $2^c - 4$. The coupled lower family gives indecomposable quadratic solutions by Remark E.5. Theorems 6.5 and 6.7 refine this constant-scale statement to the macroscopic allocation and the discrete critical profile. The relative boundary weights inside the remaining shrinking bands are separate questions.

The following initial localization is sharpened in Corollary 6.8.

Corollary 6.3 (An initial outer localization of the transition)**.** *Fix a finite characteristic-two field of order $q \ge 4$, put $n = 2^d$, and let $r = r(d) \ge 1$ be integers. Then, as $d \to \infty$,*

$$\begin{aligned} r \le \sqrt{2n} - 3d - 3 \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \\ r \ge \sqrt{2n} - d + 1 \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \end{aligned}$$

The two thresholds are separated by $2d + 4$, giving an outer localization of order $d = c - 1$ around $\sqrt{2n}$. This does not assert that the actual transition has that width or determine its boundary density.

Proof. Set $t = \sqrt{2n}$. The exponent

$$a(d, r) = n - \tfrac{1}{2}r^2 - 2d^2 - 3dr - 2r - 6$$

in Proposition E.3 is decreasing in real $r \ge 0$. Its value at the lower endpoint is

$$a(d, t - 3d - 3) = t + \tfrac{5}{2}d^2 + 6d - \tfrac{9}{2} =: A_d.$$

For all sufficiently large d, the assumed lower-side range is contained in $1 \le r \le D_{\mathrm{par}}/2$. Hence

$$1 - \pi_{d,r}(q) \le \frac{q^{-A_d}}{1 - q^{-A_d}} \longrightarrow 0.$$

For the upper side, Corollary E.7 proves the stronger endpoint by combining a uniform count of the minimum-derived stratum with the full-central-cocycle lower family. That comparison is independent of the present lower-side argument. No monotonicity of $\pi_{d,r}(q)$ is assumed. □

6.2 Uniform enumeration through the transition scale

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. The minimum-derived stratum admits a uniform multiplicative count even when the surplus grows. Put $d = c - 1$, $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, $C_d = \binom{d}{2}$, and $w_0 = n - 2$. Write $B_0 = B^{\min}_{d+1,r}(q)$. The range in the following theorem is

$$2d \leq r \leq \frac{D_{\mathrm{par}}}{4d}.$$

It includes every sequence $r/\sqrt{n} \to \alpha$ with $0 < \alpha < \infty$.

Theorem 6.4 (Multiplicative principal enumeration at growing surplus)**.** *Uniformly for the integer range* (119)*, as $d \to \infty$,*

$$B_0 \sim \frac{\prod_{i=0}^{r-1}(q^{D_{\mathrm{par}}} - q^i)\, q^{C_d + r + \binom{r}{2}}}{(q-1)|\,\mathrm{GL}_{d-1}(q)||\,\mathrm{GL}_r(q)|}. \tag{50}$$

Almost all classes in this stratum have full automorphism order $q^{2w_0+d+r-1}$. These statements concern the whole minimum-derived stratum, not the complete dimension layer above the critical constant.

The proof reparametrizes a fixed pure shear by its induced ring automorphism σ and the action subspace Q:

$$\mathrm{span}\{\sigma(t_i) - t_i\} \subseteq Q \subseteq \mathrm{Fix}(\sigma).$$

A displacement-rank inequality and Gaussian incidence counts then control the complete stabilizer sum. This avoids a loss depending on the number of added directions in the fixed-surplus estimate. Appendix E.3 gives the proof and the sharper upper-side localization used in Corollary 6.3. The full dimension layer has the same multiplicative asymptotic on the subrange $2d \leq r \leq \sqrt{2n} - 3d - 3$, by the independent minimum-derived dominance bound. Above the critical constant, (50) counts only the minimum-derived stratum.

6.3 Exact growth and the typical surplus allocation

At a fixed ratio of surplus to the square-root core scale, an explicit profile determines both the growth of the entire population and its typical allocation. Continue to put $n = 2^d$, $d = c - 1$, and count ordinary k-linear Lie isomorphism types uniformly. For $\alpha > 0$, define

$$\gamma_\alpha = \begin{cases} \alpha, & 0 < \alpha \leq \sqrt{2}, \\ (\alpha + \sqrt{\alpha^2 + 6})/3, & \alpha > \sqrt{2}, \end{cases} \qquad \beta_\alpha = \alpha - \gamma_\alpha, \qquad \varphi(\alpha) = \gamma_\alpha + \frac{\beta_\alpha \gamma_\alpha^2}{2}. \tag{51}$$

Theorem 6.5 (Exact logarithmic growth and allocation)**.** *Fix a finite field $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$, and suppose $r/\sqrt{n} \to \alpha \in (0, \infty)$. Then*

$$\log_q B_{d+1,r}(q) = \big(\varphi(\alpha) + o(1)\big) n^{3/2}. \tag{52}$$

For every $\varepsilon > 0$, there is an $\eta > 0$ such that, for all sufficiently large d, all but a proportion $q^{-\eta n^{3/2}}$ of these types satisfy

$$\left| \frac{h}{\sqrt{n}} - \beta_\alpha \right| < \varepsilon, \qquad h = \dim_k L' - (n-2). \tag{53}$$

Thus for $\alpha > \sqrt{2}$, the typical derived share is

$$\frac{h}{r} \longrightarrow \frac{2 - \sqrt{1 + 6/\alpha^2}}{3}, \qquad \frac{s}{r} = 1 - \frac{h}{r}.$$

The core contributes the term γ, while alternating central brackets contribute $\beta\gamma^2/2$. Their competition gives the profile φ and its unique maximizing allocation. The main upper-bound issue is that arbitrary nonsplit derived modules may have additional actions. Restricting those actions to the core socle separates their visible part from a controlled kernel. A common rational flag bounds the visible tuples, and ordinary cohomology charges their rank against the available cocycles. This coupled bound covers every derived module, so the result counts the whole layer. Appendix G proves the theorem using the classical triangular square-zero count and ordinary Chevalley–Eilenberg tools.

At $q = 4$, the same statement counts metabelian quadratic solutions in binary dimension $\mu(c) + 2r$. At the critical ratio it implies $h = o(\sqrt{n})$ in type-probability; the finer question whether $h = 0$, and the density inside the remaining critical band, requires additional information.

There is also a finer comparison at the first positive derived defect. Writing B_h for the branch with derived dimension $n - 2 + h$, Corollary I.5 gives, uniformly for $a \le r/\sqrt{n} \le b$ with fixed $0 < a < b < \infty$,

$$\log_q B_1 = (r-1)n - 3(r-1) + O_{q,a,b}(d),$$
$$\log_q(B_1/B_0) = -n + \binom{r+d}{2} + O_{q,a,b}(d).$$

The proof classifies the three possible one-defect core modules and identifies which extra actions survive Jacobi. This locates the B_1-versus-B_0 comparison near $\sqrt{2n} - d + \frac{1}{2}$; controlling all defects on this finer scale is the subject of Section 6.5.

6.4 Explicit comparison bounds above the threshold

The exact allocation law identifies the limiting derived share. The next estimate gives explicit interior margins and an exponential error bound for each fixed finite supercritical ratio. Continue to put $d = c - 1$, $n = 2^d$, and $w_0 = n - 2$. In the layer of dimension $\ell(c) + r$, write

$$\dim_k L' = w_0 + h, \qquad \dim_k(L/L') = d + s, \qquad h + s = r.$$

Thus h and s measure the parts of the surplus spent on the derived ideal and on the quotient, respectively. Proportions below give equal weight to each k-linear isomorphism type.

Proposition 6.6 (Explicit interior margins)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \ge 4$, and a finite constant $\alpha > \sqrt{2}$. Suppose $r/2^{d/2} \to \alpha$ as $d \to \infty$. There are constants $\epsilon_h, \epsilon_s, \eta > 0$ such that the proportion of all types in $B_{d+1,r}(q)$ satisfying*

$$\epsilon_h 2^{d/2} < h < r - \epsilon_s 2^{d/2} \tag{54}$$

is at least $1 - q^{-\eta 2^{3d/2}}$ for all sufficiently large d. One may take

$$\delta = \alpha^2/2 - 1, \qquad \epsilon_h = \frac{\delta}{16\alpha}, \qquad \epsilon_s = \frac{\alpha}{4(1+6\alpha^2)}, \qquad \eta = \min\left\{\frac{\delta^2}{16\alpha}, \frac{\alpha}{4}\right\}.$$

In particular, both h and s have order r for almost all isomorphism types in this regime.

The comparison family couples any central derived defect h to the same faithful high-class core. Every member of that family is indecomposable. Comparing its count with the complete branch upper bounds gives the explicit constants above; the limiting allocation is determined by Theorem 6.5. At $q = 4$, the estimate concerns binary conjugacy classes in dimension $\mu(c) + 2r$, with binary derived dimension $2^c - 4 + 2h$. Appendix F gives the comparison construction and proof.

The constants depend on the fixed finite ratio $\alpha > \sqrt{2}$. The off-window regimes, including $r/2^{d/2} \to \infty$, are covered separately by Theorem 6.1.

6.5 The discrete critical transition

The macroscopic allocation law of Theorem 6.5 determines $h/\sqrt{n}$. On a finer scale, the entire population is governed by an integer defect profile. Put

$$T = \sqrt{2n}, \qquad R_d^* = T - d + \tfrac{1}{2}, \qquad x_{d,r} = r - R_d^*, \qquad M(x) = \max_{h \in \mathbb{Z}_{\geq 0}} \{hx - h(h-1)\}.$$

Write $B_h = B_{d+1,r}^{(h)}(q)$ for the number of types with derived dimension $n - 2 + h$; thus $B_0 = B_{d+1,r}^{\min}(q)$ depends on d, r, q. The centering by d matters here: bounded changes in $x_{d,r}$ can change the typical integer derived defect even though they disappear in $r/\sqrt{n}$.

Theorem 6.7 (The critical profile and the defect staircase)**.** *Fix $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$. For each fixed $A > 0$, uniformly for integer r with $|x_{d,r}| \leq A$,*

$$\log_q \frac{B_{d+1,r}(q)}{B_0} = TM(x_{d,r}) + O_{q,A}(d). \tag{55}$$

If $x_{d,r} \to x$, then all but a proportion $q^{-\kappa T}$ of the types have the following derived defects, for some $\kappa > 0$:

(i) *$h = 0$ if $x < 0$;*

(ii) *$h = j$ if $2j - 2 < x < 2j$, for an integer $j \geq 1$;*

(iii) *$h \in \{j, j+1\}$ if $x = 2j$, for an integer $j \geq 0$.*

Near an adjacent-branch boundary,

$$\log_q(B_{j+1}/B_j) = T(x_{d,r} - 2j) + O_{q,j}(d). \tag{56}$$

The relative weights inside the remaining shrinking boundary band are not determined by this estimate.

Thus the typical defects are successively $0, 1, 2, \ldots$, with changes at offsets $0, 2, 4, \ldots$. The score difference $f_x(h+1) - f_x(h) = x - 2h$ is elementary; the counting content is a uniform bound on every module and defect branch, including the growing-defect tail.

Corollary 6.8 (A shrinking outer band for minimum derived dimension)**.** *For every integer sequence $r = r(d) \geq 1$,*

$$\begin{aligned} r \leq R_d^* - 4d/T \quad \textit{eventually} \quad &\Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \\ r \geq R_d^* + 4d/T \quad \textit{eventually} \quad &\Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \end{aligned}$$

The two outer endpoints have separation $8d/T = o(1)$.

Appendix I proves both statements. The key structural step is an exact ordinary-cohomology obstruction: when the derived center has its maximum dimension $h + 1$, the core module is a diagonal central extension E_Q of K, and Jacobi removes precisely $h(d-1)$ nonpolynomial action parameters per extra direction. The linear stabilizer of $\ker Q$ then makes the first-defect logarithmic error $O_q(d)$. The socle-sensitive bounds discard the complementary modules and the growing-defect tail. Diagonal Frobenius terms are standard in characteristic-two extension algebra; compare [11, Section 2]. Their use here is in the complete nearcritical census and its resulting discrete distribution.

For odd d, $T = 2^{(d+1)/2}$ is an integer. At each fixed integer offset ℓ, $r = T - d + \ell$, almost every type has

$$h = \max\{0, \lceil \ell/2 \rceil\}.$$

In particular, the first change occurs between the two consecutive integer surpluses $T - d$ and $T - d + 1$. The boundary weights in the theorem concern limits of the real offsets $x_{d,r}$ along integer surplus sequences.

7 When the original polar bracket is $\mathbb{F}_4$-bilinear

Return to the normal form (7) and keep the intrinsic scalar actions on $U = V/W$ and W. Its auxiliary bracket (15) is always $\mathbb{F}_4$-bilinear. The following definition concerns the original polar bracket of f.

Definition 7.1. A metabelian solution is *compatible* if V admits an $\mathbb{F}_4$-vector space structure that restricts to the intrinsic structure on W, induces the intrinsic structure on U, and makes the original polar bracket $\mathbb{F}_4$-bilinear.

In the criterion below we use the symmetric convention $a_{ji} = a_{ij}$ and $a_{ii} = 0$.

Theorem 7.2. *For a normal form with data P_i, a_{ij}, the following are equivalent:*

(i) *the solution is compatible;*

(ii) *there are $b_1, \dots, b_d \in W$ satisfying*

$$P_i b_i = 0, \qquad P_i b_j = P_j a_{ij} \quad (i \neq j); \tag{57}$$

(iii) *the auxiliary $\mathbb{F}_4$-Lie algebra has a restricted square r with $r(W) = 0$ and $r(\mathfrak{l}) \subseteq W$.*

If a compatible scalar lift exists, the number of such scalar actions on the fixed binary space, with the prescribed actions on U, W, is 4^{dk}, where $k = \dim_{\mathbb{F}_4} \bigcap_i \ker P_i$.

Proof. Every binary-linear lift of multiplication by ω has the form

$$T(x, z) = (\omega x, \omega z + t(x)).$$

The relation $T^2 + T = \mathrm{id}$ is exactly $t(\omega x) = \omega^2 t(x)$. Thus $t(x) = b(x) = \sum_i x_i^2 b_i$. The involutory binary shear $S(x, z) = (x, z + b(x))$ conjugates T to standard scalar multiplication. It changes the cocycle of f to

$$Q + b + P(Q + b).$$

The first two terms have $\mathbb{F}_4$-bilinear polars. The last term has scalar weight zero. Its polar could be $\mathbb{F}_4$-bilinear only if that polar were zero, since an $\mathbb{F}_4$-bilinear polar has weight two. An additive zero-constant function of weight zero is zero: additivity and invariance under ω give its value at $\omega^2 x = x + \omega x$ equal to zero, and then invariance gives its value at x equal to zero. Compatibility is therefore equivalent to

$$P(x)(Q(x) + b(x)) = 0. \tag{58}$$

After the triangle cancellations, comparison of x_i^3 and $x_i x_j^2$ gives precisely (57).

We recall explicitly the restricted-square axioms in characteristic two: $r(\alpha u) = \alpha^2 r(u)$, $r(u+v) = r(u) + r(v) + [u, v]_0$, and $\operatorname{ad} r(u) = (\operatorname{ad} u)^2$. Any square with the range and vanishing conditions in (iii) must have the form

$$r(x, z) = (0, P(x)z + q(x)), \qquad q(x) = Q(x) + b(x). \tag{59}$$

Its first two axioms are automatic. For $u = (x, z), v = (y, t)$, the two adjoint expressions have final coordinates

$$P(x)P(y)z + P(x)\beta(x, y), \qquad P(x)P(y)z + P(y)q(x),$$

respectively. Condition (58) implies $P(x)\beta(x, y) = P(y)q(x)$: take the coefficient of s in $P(x + sy)q(x + sy) = 0$. Its degree in s is at most three, below $|\mathbb{F}_4|$, so the comparison is valid for functions over $\mathbb{F}_4$. This proves the third restricted axiom. Conversely that axiom, applied to u, gives $[u, r(u)]_0 = 0$, hence (58).

Finally two solutions of (57) differ by independent choices of d vectors in $\bigcap_i \ker P_i$. The displayed count follows. It counts scalar actions, not their conjugacy classes. □

Example 7.3. Write the eight-dimensional class-three model over $\mathbb{F}_4$ as

$$N(a,b,y,z) = (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).$$

On $W = \operatorname{span}_{\mathbb{F}_4}\{y,z\}$, its data are $P_1 z = y$, $P_1 y = 0$, $P_2 = 0$, and $a_{12} = z$. The equation with $(i,j) = (2,1)$ in (57) would require

$$0 = P_2 b_1 = P_1 a_{21} = y,$$

which is impossible. Its auxiliary bracket is $\mathbb{F}_4$-bilinear, but the original polar bracket admits no compatible scalar action. Thus the class-three metabelian minimum $\mu(3) = 8$ lies outside the compatible subclass, whose minimum is $\nu(3) = 12$. The general dimension cost is proved in the next section.

Remark 7.4. The restricted square in Theorem 7.2 is a square on the auxiliary algebra. The theorem imposes both its range and its restriction to the derived ideal. General correspondences between quadratic maps and restricted Lie structures, such as [13, Section 9], carry their own additional identities. Here the exact additional condition is the solvable linear system (57).

8 Sharp compatible bounds and extremizers

8.1 The abelianization bound

Lemma 8.1. *A nonabelian compatible solution of class c, with $d = \dim_{\mathbb{F}_4}(V/W)$, satisfies $c \le d$.*

Proof. Use a restricted square from Theorem 7.2. The identity $P(x)\beta(x,y) = P(y)r(x)$, with a lift of x understood, implies

$$P(x)P(y)\beta(x,y) = 0. \tag{60}$$

In the universal module this kills the submodule K spanned by $t_i t_j v_{ij}$, $i < j$. It has dimension $\binom{d}{2}$ and lies in coefficient degree two. Every multiplication by an endpoint kills it by a square, and multiplication by a third index kills it by a triangle relation.

The space K is $\mathrm{GL}(U)$-invariant: the expression $P(x)P(y)\beta(x,y)$ is additive in each variable and homogeneous of degree two in each. For additivity in x, the cross term is

$$P(y)\big(P(x)\beta(z,y) + P(z)\beta(x,y)\big) = P(y)^2\beta(x,z) = 0$$

by Jacobi. Its basis values therefore span precisely K.

Put $\overline{W}_d = W_d/K$. The multigrading in Proposition 4.1 shows that its highest coefficient degree is $d-1$, for every $d \ge 2$, and that its top has the basis T_j of (25). This top is $\det(U)\otimes U$ also for $d = 2, 3$: the same diagonal and transvection calculations apply, with any additional double-endpoint term now in K. For $d = 2$ this is immediate from $T_1 = t_1 v_{12}, T_2 = t_2 v_{12}$.

Every compatible derived module M is a quotient of $\overline{W}_d$, so $c \le d+1$. Suppose $c = d+1$. Choose a functional on M nonzero on its terminal component. A change of quotient basis normalizes its top pullback to $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \ne j$. Using $P_j a_{ij} = P_i b_j$, we have, for $i \ne j$,

$$\operatorname{im}(T_j) = \Big(\prod_{h\ne j} P_h\Big) b_j. \tag{61}$$

Write $b_j = \sum c_{kl} a_{kl} \pmod{\mathfrak{m}M}$, using intrinsic generation. The product on the right kills $\mathfrak{m}M$. If j is an endpoint of v_{kl}, then $(\prod_{h\ne j} t_h)v_{kl}$ is a top vector whose doubled endpoint is different from j. If j is not an endpoint, the vector has both endpoints doubled and is in K. Thus the normalized top functional is zero on the right side of (61), but one on the left. This contradiction proves $c \le d$. □

8.2 Counting preimages of an operator cube

Lemma 8.2. *A compatible solution of class $c \geq 2$ satisfies*

$$\dim_{\mathbb{F}_4} W \geq 2^{c-1} - 1.$$

Proof. The case $c = 2$ says $W \neq 0$. Let $c \geq 3$. By Lemma 4.2, choose a nonzero terminal value

$$P(x)P(z_1)\cdots P(z_{c-3})\beta(x, y).$$

Restrictedness gives $P(x)\beta(x, y) = P(y)b$, where $b = r(x) \in W$. Put $m = c - 2$, $u_1 = y, u_2 = z_1, \ldots, u_m = z_{c-3}$. Then $P(u_1)\cdots P(u_m)b \neq 0$. The 2^m vectors

$$b_S = \Big(\prod_{i\in S} P(u_i)\Big)b \qquad (S \subseteq \{1, \ldots, m\})$$

are independent by the complementary-operator argument used in Theorem 4.4. Let C be their span. Since $[x, r(x)]_0 = 0$, $P(x)$ kills b, hence all of C.

For each nonempty S, choose $i \in S$ and set

$$v_S = \Big(\prod_{j\in S\setminus\{i\}} P(u_j)\Big)\beta(x, u_i).$$

These vectors lie in the intrinsic derived ideal, and restrictedness gives $P(x)v_S = b_S$. Thus

$$\dim\ker(P(x)|_W) \geq 2^m, \qquad \operatorname{rank}(P(x)|_W) \geq 2^m - 1.$$

Rank-nullity yields $\dim W \geq 2^{m+1} - 1 = 2^{c-1} - 1$. □

Theorem 8.3. *For every $c \geq 2$, the minimum binary dimension among compatible solutions of polar class c is*

$$\nu(c) = 2^c + 2c - 2.$$

Proof. Lemmas 8.1 and 8.2 give the lower bound by addition. For attainment put $s = c - 1$, $R_s = \mathbb{F}_4[t_1, \ldots, t_s]/(t_i^2)$, and use

$$f(x, z) = \left(\omega x, \omega z + \Big(\sum_i x_i t_i\Big)z\right) \quad \text{on } \mathbb{F}_4^s \oplus R_s. \tag{62}$$

The square of $\sum x_i t_i$ is zero, so direct substitution gives (1). The polar bracket is $\mathbb{F}_4$-bilinear and has intrinsic derived ideal $\mathfrak{m}R_s$. The scalar actions induced by f on that ideal and on the abelianization are the displayed ones, so the map is compatible. Its lower central ideals are $\gamma_k = \mathfrak{m}^{k-1}R_s$ for $k \geq 2$, giving class $s + 1 = c$. The derived and quotient dimensions are $2^s - 1$ and $s + 1$. The binary dimension is therefore $2(s + 2^s) = 2^c + 2c - 2$. □

The regular Lie model in (62) is a finite subalgebra of Cohn's characteristic-two construction [5]; Section 4.4 gives the identification. The preceding bounds establish its precise extremal dimension.

For $c \geq 5$, the extra four binary dimensions have an exact structural meaning. The sharp bounds in the two subclasses are

	d	w
metabelian	$c - 1$	$2^{c-1} - 2$
compatible metabelian	c	$2^{c-1} - 1$

where each entry is the corresponding lower bound, attained simultaneously. Compatibility therefore requires one additional $\mathbb{F}_4$-dimension in the quotient and one in the derived ideal, giving four additional binary dimensions. The restricted square explains the strengthened bounds: it constrains terminal commutators and supplies independent preimages for the nonempty faces of the operator cube. The elementary semidirect construction attains this cost.

8.3 A common module and a full parameter family

The equality cases of the two bounds have a rigid module structure, although the resulting maps admit many inequivalent parameters. Throughout this part put $k = \mathbb{F}_4$, $s = c - 1 \geq 2$, and

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \quad J = (t_1, \dots, t_s), \quad \tau = t_1 \cdots t_s, \quad Z = k\tau. \tag{63}$$

The socle $Z = \operatorname{Ann}_R(J)$ is one-dimensional. We use the parameter space

$$\mathcal{V}_s = J^2/Z = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad D_s = \dim_k \mathcal{V}_s = 2^s - s - 2. \tag{64}$$

In characteristic two the exterior algebra is the displayed commutative square-zero algebra. Each parameter is represented by its unique sum of terms of degrees two through $s-1$. These are degrees of fixed ring coefficients; the maps below still have binary degree at most two.

Proposition 8.4 (Canonical extremal models)**.** *For $p \in \mathcal{V}_s$, put*

$$a_p(X, V) = \sum_{i=1}^{s} X_i t_i + Vp \quad (X \in k^s,\ V \in k).$$

Then

$$F_p(X, V, z) = \big(\omega X, \omega V, \omega z + a_p(X, V)(z + V)\big), \qquad z \in J, \tag{65}$$

is a compatible solution of class $c = s + 1$ and binary dimension $\nu(c)$. Its intrinsic derived ideal is J, and

$$\gamma_j = J^{j-1} \quad (j \geq 2), \qquad \dim_k \gamma_j = \sum_{h=j-1}^{s} \binom{s}{h}.$$

Proof. Every element of J squares to zero. Write $r_p = F_p - \omega\,\mathrm{id}$. At the image under F_p, the expressions a_p and $z + V$ become ωa_p and $(\omega + a_p)(z + V)$, respectively. Hence $r_p(F_p(y)) = \omega^2 r_p(y)$, proving $F_p^2 = F_p + \mathrm{id}$. The displayed expression is quadratic over k, and thus has binary degree at most two. Its original polar bracket is k-bilinear:

$$[(X, V, z), (Y, T, w)] = (0, 0, a_p(X, V)(w + T) + a_p(Y, T)(z + V)).$$

With quotient basis $e_1, \dots, e_s, v$, the action and cocycle are

$$P(e_i) = t_i \cdot, \quad P(v) = p \cdot, \quad \beta(e_i, e_j) = 0, \quad \beta(e_i, v) = t_i. \tag{66}$$

The square values of r_p are $b_i = 0$, $b_v = p$. The cocycle values generate J, and the core actions generate its augmentation filtration, proving all assertions about the derived ideal and lower central series. The dimension is $2((s + 1) + (2^s - 1)) = 2^c + 2c - 2$. □

Lemma 8.5 (The common extremal module)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ has derived module isomorphic to J under $s = c - 1$ independent quotient directions acting by $t_1, \dots, t_s$. There is one remaining quotient direction, whose action, after a change of basis, is multiplication by a parameter in $\mathcal{V}_s$. In particular all these extremizers have the lower central dimensions in Proposition 8.4.*

Proof. Equality in the two separate bounds forces $d = c$ and $\dim_k W = 2^{c-1} - 1$. In the proof of Lemma 8.2, put $m = c - 2$, $T = P(x)$, $A = k[t_1, \dots, t_m]/(t_i^2)$, and $C = Ab$. The complementary-operator argument identifies C with the regular module A. Every nonempty face lies in $\operatorname{im} T$, whereas $C \subseteq \ker T$. Equality of dimensions now gives

$$\ker T = C \cong A, \qquad \operatorname{im} T = I_A b, \quad I_A = (t_1, \dots, t_m).$$

Here and below kernels and images are taken on W.

For completeness, the required splitting follows directly from the Frobenius pairing. Let ℓ_A take the top coefficient in A, and extend $ab \mapsto \ell_A(a)$ from C to a linear functional ℓ on W. Nondegeneracy of $(a, a') \mapsto \ell_A(aa')$ defines an A-linear retraction $\rho : W \to A$ by $\ell_A(a\rho(w)) = \ell(aw)$ for every $a \in A$. Thus $W = C \oplus N$, and T identifies N with $I_A b$. In these coordinates $W = A \oplus I_A$, $T(a, q) = (q, 0)$. The map $(a, q) \mapsto t_0 a + q$ identifies this with the augmentation ideal of $A[t_0]/t_0^2$, giving the claimed J.

The directions $x, u_1, \dots, u_m$ are independent: applying an action relation to b first kills all coefficients of the u_i, and $T \neq 0$ kills the last one. There is therefore exactly one remaining direction v. The top-coefficient pairing makes R self-injective: $R \cong \mathrm{Hom}_k(R, k)$, and $\mathrm{Hom}_R(-, \mathrm{Hom}_k(R, k)) \cong \mathrm{Hom}_k(-, k)$ is exact. Every R-endomorphism of J extends to multiplication on R, so $\mathrm{End}_R(J) = R/Z$. Accordingly $P(v)$ is multiplication by some $p \in R/Z$. Its square is zero, forcing its constant coefficient to vanish. Subtracting its linear part from v leaves $p \in J^2/Z$, as required. □

Theorem 8.6 (All compatible extremizers)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ is binary-linearly conjugate to a model F_p in* (65)*.*

Proof. Theorem 7.2 allows us first to write the solution as $\omega\,\mathrm{id} + r$, with a compatible restricted square r. Use the common module from Lemma 8.5. The remaining data are core cocycle values $a_{ij} = \beta(e_i, e_j)$, core square values $b_i = r(e_i)$, mixed values $A_i = \beta(e_i, v)$, and $b_v = r(v)$. We normalize them in three steps. All changes of section take values in the intrinsic ideal J.

Core cohomology. Lemma A.1 computes the relevant low-degree complex explicitly. A section change puts the core data in the form

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i \quad (\kappa_i \in k). \tag{67}$$

The restricted identities give $t_i A_i = p b_i$, so $A_i = \kappa_i p + t_i z_i$, with $z_i \in R$. Mixed Jacobi gives $t_i t_j (z_i + z_j) = 0$; hence all z_i have the same constant η. If $\eta = 0$, the mixed values have no linear terms, and the linear core values span at most the proper hyperplane $\ker(\sum \kappa_i t_i^*)$ of J/J^2 (or zero if $\kappa = 0$). This contradicts intrinsic generation of J. Replace v by $\eta^{-1} v$, scaling p, A_i by η^{-1} and b_v by η^{-2}, to make $\eta = 1$. Also $t_i b_v = p A_i \in J^3$ for all i; since $s \geq 2$, b_v has no linear part.

Removing the core class. Replace e_i by $e_i + \kappa_i v$. The new square values $b_i + \kappa_i^2 b_v + \kappa_i A_i$ have no linear terms. The new actions are multiplication by $t_i + \kappa_i p$. The substitution $\phi(t_i) = t_i + \kappa_i p$ is an automorphism of R, because its linear part is the identity. Reidentify J by ϕ^{-1}; in particular the parameter must be changed simultaneously to

$$p_{\mathrm{new}} = \phi^{-1}(p) \pmod{Z}. \tag{68}$$

The core square values still have no linear terms. By the same cohomology calculation their class is zero, so a J-valued section change kills every b_i and a_{ij}.

Removing the mixed freedom. Now (A_i) is a one-cocycle. Its linear part is still (t_i), so Lemma A.1 gives $A_i = t_i + t_i w$ for a common $w \in J$. Replacing the section of v by $v + w$ makes $A_i = t_i$. Restrictedness then gives $t_i(b_v + p_{\mathrm{new}}) = 0$ for every i, whence $b_v = p_{\mathrm{new}} + \zeta\tau$. The last term is removed by the binary-linear central shear $z \mapsto z + \zeta\tau V^2$. We have reached precisely (66), with $b_i = 0$, $b_v = p_{\mathrm{new}}$, proving the theorem.

All ring substitutions in this argument reidentify the finite-dimensional k-space J; they are linear maps on that space. No nonlinear change of the input vector is used. □

8.4 The exact parameter orbits

Theorem 8.7 (Binary conjugacy of extremizers)**.** *Let $s \geq 2$. The maps F_p, F_q are binary-linearly conjugate if and only if*

$$\sigma_B(p) = \delta q \pmod{Z} \quad \text{for some } B \in \mathrm{GL}_s(k),\ \delta \in k^\times, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j. \tag{69}$$

Thus the compatible extremizers of class $c = s+1$ are in bijection with the orbits of $\mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times$ on $\mathcal{V}_s$, acting by $p \mapsto \delta^{-1}\sigma_B(p)$.

Proof. Let $SF_p = F_q S$. The intrinsic ideal J is preserved, so $S(u,z) = (Au, Cz + h(u))$. The diagonal blocks commute with the intrinsic action of ω, and are therefore k-linear; h is initially only binary-linear. The mixed polar relations show that C conjugates the unital action algebras, both equal to $H = R/Z$. The kernel of $U \to \mathrm{rad}(H)/\mathrm{rad}(H)^2$ is kv. Hence

$$Av = \delta v', \qquad Ae_i = \sum_j B_{ji} e'_j + \alpha_i v', \quad B \in \mathrm{GL}_s(k),\ \delta \neq 0.$$

Compare the zero core brackets modulo J^2. All terms from h lie in J^2, and the remaining equation is $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Independence of the columns of B and $s \geq 2$ force $\alpha_i = 0$ for every i. The induced algebra automorphism therefore sends each t_i to its linear substitution $B(t_i)$. The action of v gives (69). This proves necessity without assuming the off-diagonal block is k-linear.

For the chosen representatives, the condition is equality in R, because linear substitutions preserve degree. The map

$$S(X, V, z) = (BX, \delta V, \delta\sigma_B(z))$$

then intertwines the two models directly. For arbitrary representatives with $\theta = \sigma_B(p) + \delta q \in Z$, the same conclusion holds after adding $\delta\theta V^2$ to the last coordinate. Thus passing modulo the socle loses no invariant. □

The final equivalence group consists of linear variable substitutions and one common scalar. Arbitrary nonlinear automorphisms of R do not preserve this normal form with its mixed cocycle; nor is an extra Galois action allowed, since the diagonal blocks commute with the fixed intrinsic ω.

Corollary 8.8 (Extremal diversity)**.** *Let N_c be the number of compatible extremal conjugacy classes of polar class $c \geq 3$, and put $s = c-1$. Then*

$$N_c = \frac{1}{3|\,\mathrm{GL}_s(4)|} \sum_{B \in \mathrm{GL}_s(4)} \sum_{\delta \in k^\times} 4^{\sum_{j=2}^{s-1} \dim_k \ker(\Lambda^j B - \delta I)}, \tag{70}$$

$$N_c \geq \left\lceil \frac{4^{D_s}}{3|\,\mathrm{GL}_s(4)|} \right\rceil > \frac{1}{3} 4^{D_s - s^2}, \qquad N_c \leq 4^{D_s}. \tag{71}$$

In particular $\log_4 N_c = 2^{c-1} + O(c^2)$.

Proof. Apply Burnside's lemma to Theorem 8.7. The parameter set has 4^{D_s} points and every orbit has size at most $3|\,\mathrm{GL}_s(4)|$. Finally $|\,\mathrm{GL}_s(4)| = \prod_{j=0}^{s-1}(4^s - 4^j) < 4^{s^2}$. □

Corollary 8.9. *For every $c \geq 4$, at least $c-2$ compatible extremizers are given by $p = 0$ and the monomials $p = t_1 \cdots t_j$, $2 \leq j \leq s-1 = c-2$.*

Proof. The parameter action preserves degree and distinguishes zero from nonzero parameters. The listed models are therefore inequivalent. □

The growth in Corollary 8.8 is double exponential in the polar class c. In the extremal binary dimension $n = \nu(c) = 2^{s+1}+2s$, the bound is instead exponential, $N_c > 2^{n-2s^2-4s-4}/3$. For example, class seven and dimension 140 give at least 532249921743 classes by the displayed formula; no enumeration of that many models is involved.

Classical tools and the extremal interface. The algebra R is the group algebra of an elementary abelian two-group, and J is its augmentation ideal, or first syzygy of the trivial module. These are classical objects; see Dade's treatment of the syzygy modules and their endo-trivial properties [6, Part II, Section 7 and Theorem 10.1]. The low-degree complex used above is the ordinary periodic tensor resolution for this algebra. Its interpretation as restricted cohomology agrees with the Ext convention in [10, Section 2]. The extremal conclusions here require the equality-forced common module, elimination of every remaining cocycle and square parameter, and the necessity of the precise binary conjugacy criterion. They retain the mixed interactions that are absent from a classification of the module J alone.

8.5 Automorphisms of every extremal model

The general lifting problem for automorphisms of an abelian Lie extension is governed by its module action and extension class; see [3, Theorems 1.1–1.2]. For the extremal models these conditions can be solved explicitly, including the unique binary correction needed to commute with the map.

Theorem 8.10 (Automorphisms of compatible extremizers)**.** *Put $k = \mathbb{F}_4$, and let $c \geq 3$, $s = c - 1$,*

$$R_s = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J_s = (t_1, \dots, t_s), \qquad \tau_s = t_1 \cdots t_s.$$

Choose $p \in J_s^2/k\tau_s$, represented by its unique sum of terms of degrees $2, \dots, s-1$. For the canonical extremizer

$$F_p(X, V, w) = \big(\omega X, \omega V, \omega w + P_p(X, V)w + VP_p(X, V)\big), \qquad P_p(X, V) = \sum_i X_i t_i + Vp,$$

where $X \in k^s$, $V \in k$, $w \in J_s$, and $\omega^2 + \omega = 1$, define

$$\mathcal{S}_p = \{(B, \delta) \in \mathrm{GL}_s(k) \times k^\times : \sigma_B(p) = \delta p \text{ in } J_s^2/k\tau_s\}, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j.$$

The binary-linear centralizer fits into a split exact sequence

$$1 \longrightarrow K_s \longrightarrow C_{\mathrm{GL}(2(s+2^s),2)}(F_p) \longrightarrow \mathcal{S}_p \longrightarrow 1.$$

The kernel K_s is independent of p up to group isomorphism. It has $2^{s+1} + s - 2$ free k-parameters, with the group law described in Appendix A.1; in particular,

$$\big|C_{\mathrm{GL}(2(s+2^s),2)}(F_p)\big| = 4^{\,2^{s+1}+s-2}|\mathcal{S}_p| = 4^{\,2^c+c-3}|\mathcal{S}_p|.$$

The splitting sends (B, δ) to $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$.

Proof. The intrinsic derived ideal identifies the action algebra with $R_s/k\tau_s$. Its radical modulo its square identifies the core quotient directions and the remaining line. Comparing core brackets modulo J_s^2 forces the induced ring automorphism to be the linear substitution σ_B, giving the displayed map to $\mathcal{S}_p$. The stated splitting is verified by substitution. Appendix A.1 solves its kernel explicitly and proves that each auxiliary k-Lie automorphism has exactly one binary-linear lift commuting with F_p. No additional choices of the antiscalar block enter the order. □

For $p = 0$, one has $\mathcal{S}_0 = \mathrm{GL}_s(4) \times \mathbb{F}_4^\times$. For $p \neq 0$, projection onto B identifies $\mathcal{S}_p$ with the projective stabilizer of kp in $\mathrm{GL}_s(4)$; there is no further factor of three in this latter description.

Remark 8.11 (The class-two boundary). For the six-dimensional central model $F(X, V, z) = (\omega X, \omega V, \omega z + XV)$, the action algebra has no radical directions. The preceding sequence is replaced by

$$C_{\mathrm{GL}(6,2)}(F) \cong (k^2, +) \rtimes \mathrm{GL}_2(k), \qquad |C_{\mathrm{GL}(6,2)}(F)| = 2880.$$

This follows from the automorphisms of the three-dimensional Heisenberg algebra and the same unique binary lifting argument.

8.6 Asymptotic enumeration and generic symmetry

Lemma 8.12 (A uniform fixed-space bound)**.** *Let $k = \mathbb{F}_4$, $s \geq 2$, and put*

$$\mathcal{E}_s = \Lambda(k^s), \qquad \mathcal{V}_s = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad \mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times.$$

Let (B, δ) act on $\mathcal{V}_s$ by $p \mapsto \delta^{-1}(\Lambda B)p$. For every $(B, \delta) \neq (I, 1)$,

$$\operatorname{codim}_{\mathcal{V}_s} \operatorname{Fix}(B, \delta) \geq r_s := 2^{s-2} - s - 2.$$

Proof. Put $T = \Lambda B$, $A = T - \delta I$, and let L_v denote left exterior multiplication by $v \in k^s$. The identity

$$AL_v - L_{Bv}A = \delta L_{Bv - v}$$

follows from $TL_v = L_{Bv}T$. If $B \neq I$, choose $Bv \neq v$. Multiplication by a nonzero exterior generator has rank 2^{s-1}, so subadditivity of rank gives

$$2^{s-1} \leq \operatorname{rank}(AL_v) + \operatorname{rank}(L_{Bv}A) \leq 2 \operatorname{rank} A.$$

If $B = I$ and $\delta \neq 1$, then A is invertible. Finally, deleting degrees $0, 1, s$ loses at most $s + 2$ dimensions of rank. Since fixed points are the kernel of $A|_{\mathcal{V}_s}$, the claimed bound follows. □

Corollary 8.13 (Asymptotic enumeration and generic automorphisms)**.** *Write*

$$D_s = 2^s - s - 2, \qquad g_s = 3|\,\mathrm{GL}_s(4)|, \qquad \eta_s = 3\,4^{\,s^2+s+2-2^{s-2}},$$

and let $N_{s+1} = |\mathcal{V}_s/\mathcal{G}_s|$. Then

$$N_{s+1} = \frac{4^{D_s}}{g_s}(1 + \epsilon_s), \qquad 0 \leq \epsilon_s \leq (g_s - 1)4^{-r_s} \leq \eta_s.$$

For a uniformly chosen parameter $p \in \mathcal{V}_s$,

$$\Pr\big(\mathrm{Stab}_{\mathcal{G}_s}(p) \neq 1\big) \leq \epsilon_s.$$

The proportion of parameter orbits having nontrivial stabilizer is at most $2\epsilon_s/(1 + \epsilon_s)$. Consequently

$$N_{s+1} \sim C_4\, 4^{\,2^s - s^2 - s - 2}, \qquad C_4 = \frac{1}{3\prod_{j=1}^{\infty}(1 - 4^{-j})}.$$

Under the canonical classification, these are the compatible extremizers of polar class $c = s+1$. Thus asymptotically almost all, whether counted by parameters or by conjugacy classes, have full binary-linear centralizer K_s from Theorem 8.10. For every $s \geq 9$, such extremizers exist, since $\eta_s \leq 3\,4^{-36} < 1$.

Proof. Burnside's lemma gives the stated formula with

$$\epsilon_s = \sum_{g \in \mathcal{G}_s \setminus \{1\}} 4^{-\operatorname{codim} \operatorname{Fix}(g)}.$$

The fixed-space bound and $g_s < 3\,4^{s^2}$ give the error estimate. The union bound gives the assertion for parameters. Let β_s be the proportion of parameters with nontrivial stabilizer. Counting stabilizer orders shows that the proportion of nonfree orbits is $(\beta_s + \epsilon_s)/(1 + \epsilon_s)$, which is at most the claimed bound because $\beta_s \le \epsilon_s$. Finally,

$$|\operatorname{GL}_s(4)| = 4^{s^2} \prod_{j=1}^{s} (1 - 4^{-j}),$$

the infinite product has positive limit, and $\eta_s \to 0$. At $s = 9$ its exponent is -36, and it decreases thereafter. The centralizer assertion now follows from Theorem 8.10. □

It follows in particular that, for every $c \ge 10$, the minimum binary-linear centralizer order among compatible extremizers is exactly 4^{2^c+c-3}. It is attained by asymptotically almost all such classes. This is a minimum of the symmetry group size within the already minimal-dimensional family.

8.7 The first three extremal classifications

The canonical parameters give complete classifications at three low-class compatible minima.

Corollary 8.14. *The numbers of binary-linear conjugacy classes of compatible extremizers in polar classes* $3, 4, 5$ *are* $1, 2, 7$*, respectively. Their binary dimensions are* $12, 22, 40$*. For class five, write* $p = \alpha + \beta$*, with* $\alpha \in \Lambda^2 k^4$*,* $\beta \in \Lambda^3 k^4$*, and use the notation* $t_{ij} = t_i t_j$*,* $t_{ijk} = t_i t_j t_k$*. The seven parameter representatives and their distinguishing conditions are listed in Table 1.*

Table 1: The seven compatible class-five extremizers. The conditions refer to the quadratic and cubic parts of the parameter, not to the binary degree of the map, which remains at most two.

Parameter representative	Distinguishing condition
0	$\alpha = \beta = 0$
t_{123}	$\alpha = 0,\ \beta \neq 0$
t_{12}	$\operatorname{rank} \alpha = 2,\ \beta = 0$
$t_{12} + t_{123}$	$\operatorname{rank} \alpha = 2,\ 0 \neq \beta \in \alpha \wedge k^4$
$t_{12} + t_{134}$	$\operatorname{rank} \alpha = 2,\ \beta \notin \alpha \wedge k^4$
$t_{12} + t_{34}$	$\operatorname{rank} \alpha = 4,\ \beta = 0$
$t_{12} + t_{34} + t_{134}$	$\operatorname{rank} \alpha = 4,\ \beta \neq 0$

Proof. Put $s = c - 1$. The canonical orbit correspondence reduces the assertion to the action of $\operatorname{GL}_s(k) \times k^*$ on J^2/Z, where the second factor scales the whole parameter. For $s = 2$ this parameter space is zero. For $s = 3$ it is $\Lambda^2 k^3$; the usual alternating-bivector normal form gives the two ranks zero and two.

For $s = 4$, the homogeneous decomposition is $J^2/Z = \Lambda^2 k^4 \oplus \Lambda^3 k^4$. If $\alpha = 0$, the cubic part is either zero or a nonzero decomposable three-vector, giving the first two rows. Suppose $\operatorname{rank} \alpha = 2$, and normalize $\alpha = t_{12}$. Write $k^4 = A \oplus B$, with $A = \langle t_1, t_2 \rangle$, $B = \langle t_3, t_4 \rangle$. The

stabilizer of the line $k\alpha$ consists of block matrices $\left(\begin{smallmatrix} M & N \\ 0 & P \end{smallmatrix}\right)$; the common parameter scale is then $\det M$. There is a decomposition

$$\Lambda^3 k^4 = (\Lambda^2 A \wedge B) \oplus (A \wedge \Lambda^2 B).$$

If the second component is zero, the first is either zero or nonzero, and $\mathrm{GL}(B)$ is transitive in the latter case. If the second component is nonzero, the block N removes the first component, and $\mathrm{SL}(A)$ is transitive on the remaining nonzero vector in A. These are exactly the three rank-two rows.

Finally normalize the rank-four part to $\alpha = t_{12}+t_{34}$. The map $u \mapsto \alpha \wedge u$ is an isomorphism $k^4 \to \Lambda^3 k^4$, as is seen directly on the displayed basis. It intertwines the symplectic stabilizer of α with its action on k^4. A nonzero vector extends to a symplectic basis, so that group is transitive on nonzero vectors. Thus the cubic part gives precisely the last two rows. All conditions in the table are invariant, so the listed orbits are pairwise distinct and exhaustive. □

9 Further questions

The sharp threshold specifies the limit away from $\sqrt{2}$, and the staircase identifies competing integer defects. Multiplicative counts at its boundaries would determine finer centering and limiting weights. Beyond the stated principal range, high-order displacements and nontrivial linear parts of the coordinate action require further control. These questions concern the uniform type measure fixed here.

Compatibility changes both minimum dimension and the size of the extremal population. It would be useful to isolate other natural scalar-extension constraints for which a comparable dimension cost and population law can be proved from an intrinsic module criterion.

A Extremal rigidity and automorphism calculations

This appendix gives the low-degree elimination and full kernel calculations used in the compatible equality cases. Put $k = \mathbb{F}_4$ and retain R, J, Z, τ from (63).

Lemma A.1. *Let $k = \mathbb{F}_4$,*

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J = (t_1, \dots, t_s), \qquad \tau = t_1 \cdots t_s, \quad s \geq 2.$$

Give $k = R/J$ its augmentation-module structure, and write $H^n(M) = \mathrm{Ext}_R^n(k, M)$. Then

$$H^1(J) \cong k, \qquad H^2(J) \cong k^s.$$

These isomorphisms have the following explicit descriptions.

(i) *If $a_i \in J$ satisfy*

$$t_i a_i = 0, \qquad t_i a_j + t_j a_i = 0 \quad (i \neq j), \tag{72}$$

then there are a unique $\eta \in k$ and some $w \in J$ such that

$$a_i = \eta t_i + t_i w \qquad (1 \leq i \leq s).$$

Thus the class of $(t_i)_i$ generates $H^1(J)$.

(ii) *Let $b_i, a_{ij} \in J$, with $a_{ij} = a_{ji}$ and $a_{ii} = 0$, satisfy*

$$t_i b_i = 0, \tag{73}$$

$$t_i a_{ij} = t_j b_i, \qquad t_j a_{ij} = t_i b_j \quad (i \neq j), \tag{74}$$

$$t_i a_{jk} + t_j a_{ik} + t_k a_{ij} = 0 \quad (i < j < k). \tag{75}$$

Then there are unique $\kappa_1, \dots, \kappa_s \in k$ and some $w_1, \dots, w_s \in J$ such that

$$b_i = \kappa_i t_i + t_i w_i, \tag{76}$$

$$a_{ij} = \kappa_i t_j + \kappa_j t_i + t_i w_j + t_j w_i. \tag{77}$$

In particular κ_i is read directly from $b_i \equiv \kappa_i t_i \pmod{J^2}$. The class corresponding to $\kappa = (\kappa_i) \in k^s$ has representative

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i.$$

Proof. For $A_i = k[t_i]/(t_i^2)$, the complex

$$\cdots \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \longrightarrow k \longrightarrow 0$$

is an augmented free resolution: at each positive degree both kernel and image are (t_i). Tensor the s resolutions over k and take the total complex. Exact complexes of vector spaces split, so this tensor complex has homology k in degree zero and zero in positive degrees. It is a free R-resolution $P_\bullet \to k$, explicitly

$$P_n = \bigoplus_{|\alpha|=n} RE_\alpha, \qquad dE_\alpha = \sum_{\alpha_i > 0} t_i E_{\alpha - \varepsilon_i}. \tag{78}$$

These are tensor-periodic differentials, with no coefficient α_i. Their square is zero because $t_i^2 = 0$ and the mixed terms occur twice in characteristic two.

Put $C^n(M) = \operatorname{Hom}_R(P_n, M)$. A zero-cochain is $w \in M$, a one-cochain is a tuple (w_i), and a two-cochain has components b_i at $E_{2\varepsilon_i}$ and a_{ij} at $E_{\varepsilon_i + \varepsilon_j}$. The first differentials are

$$(\delta^0 w)_i = t_i w, \qquad (\delta^1(w_i))_{ii} = t_i w_i, \qquad (\delta^1(w_i))_{ij} = t_i w_j + t_j w_i. \tag{79}$$

Consequently the one-cocycle equations are (72). Evaluating the next differential at $E_{3\varepsilon_i}$, $E_{2\varepsilon_i + \varepsilon_j}$, $E_{\varepsilon_i + 2\varepsilon_j}$, and $E_{\varepsilon_i + \varepsilon_j + \varepsilon_k}$ gives precisely (73)–(75).

Let $\lambda : R \to k$ take the coefficient of τ. The pairing $(r, a) \mapsto \lambda(ra)$ is nondegenerate: the square-free monomial t_S pairs with t_{S^c}. It identifies R with $\operatorname{Hom}_k(R, k)$ as an R-module, and hence identifies the complexes

$$\operatorname{Hom}_R(P_\bullet, R) \cong \operatorname{Hom}_k(P_\bullet, k).$$

The latter is exact in positive cohomological degrees, since the underlying augmented vector-space resolution is exact. Thus $H^n(R) = 0$ for $n > 0$.

Apply $C^\bullet$ to $0 \to J \to R \to k \to 0$. Each P_n is free, so the result is a short exact sequence of cochain complexes. All differentials of $C^\bullet(k)$ vanish, giving $H^0(k) = k$ and $H^1(k) = k^s$. Moreover $H^0(R) = \operatorname{Ann}_R(J) = k\tau$, and its augmentation map to $H^0(k)$ is zero. The long exact sequence therefore gives the connecting isomorphisms

$$H^0(k) \xrightarrow{\sim} H^1(J), \qquad H^1(k) \xrightarrow{\sim} H^2(J).$$

Lifting $1 \in H^0(k)$ to $1 \in R$ gives the one-cocycle $(t_i)_i$. Lifting $\kappa \in H^1(k)$ to the constant tuple $(\kappa_i) \in C^1(R)$ and applying (79) gives the displayed two-cocycle representative. Subtracting these representatives leaves coboundaries with coefficients in J, proving both descriptions. Such a two-coboundary has $b_i = t_i w_i \in J^2$, so its homogeneous degree-one part vanishes; this also proves the stated identification and uniqueness of κ_i. □

For the extension data used in the extremal normal form, replacing the lift of e_i by $e_i + w_i$, with $w_i \in J$, changes the square value and bracket cocycle by

$$b_i \longmapsto b_i + t_i w_i, \qquad a_{ij} \longmapsto a_{ij} + t_i w_j + t_j w_i.$$

These are exactly the legal section coboundaries in the lemma. The constants $\kappa_i \in R$ in the connecting-map calculation are cochain lifts through $R \to k$; they are not section shifts in $W = J$. Thus the lemma distinguishes the removable section terms from the degree-one class that must be eliminated by the separate change of quotient basis in the normal-form proof.

A.1 The automorphism kernel of a compatible extremizer

Retain the notation of Theorem 8.10, and abbreviate $R = R_s$, $J = J_s$, $\tau = \tau_s$, and $\tau_i = \prod_{j \neq i} t_j$. Thus $\mathrm{Soc}(R) = \mathrm{Ann}_R(J) = k\tau$. An element of $R/k\tau$ is always represented with zero τ-coefficient. The auxiliary algebra $E(p)$, on $U \oplus J$ with $U = \langle e_1, \dots, e_s, v \rangle_k$, has

$$[e_i, e_j] = 0, \quad [e_i, v] = t_i, \quad [e_i, w] = t_i w, \quad [v, w] = pw, \quad [J, J] = 0.$$

This is also the original k-bilinear polar bracket of F_p.

The quotient and its splitting. The derived ideal is J, and its unital action algebra is $H = R/k\tau$. An automorphism induces maps $A : U \to U$, $C : J \to J$, and a section change $h : U \to J$. The kernel of

$$U \longrightarrow \mathrm{rad}(H)/\mathrm{rad}(H)^2$$

is precisely kv. Consequently $Av = \delta v$ and $Ae_i = \sum_j B_{ji} e_j + \alpha_i v$, with B invertible. The source core brackets vanish. The section-change terms lie in J^2, so their images modulo J^2 give $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Since $s \geq 2$, independence of the columns of B forces all $\alpha_i = 0$. It follows that the induced automorphism of H is σ_B, and that $\sigma_B(p) = \delta p$ modulo $k\tau$. The specified representatives make this an equality in R.

The standard Frobenius pairing of R, given by the highest coefficient of a product, yields $\mathrm{End}_R(J) = R/k\tau$. Thus $C = u\sigma_B$ for a unit $u \in H$. The mixed brackets modulo J^2 give $u(0) = \delta$. The maps $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$ therefore provide a homomorphic section of the quotient map.

All kernel elements. In the kernel, A is the identity and C is multiplication by $1 + a$, for a unique $a \in J/k\tau$. Write $h_i = h(e_i)$ and $h_v = h(v)$. Bracket preservation is equivalent to

$$t_i h_j + t_j h_i = 0 \qquad (i \neq j), \tag{80}$$

$$t_i(a + h_v) + p h_i = 0 \qquad (1 \leq i \leq s). \tag{81}$$

There is no additional equation $t_i h_i = 0$: the chosen restricted square is not a marked part of the automorphism problem.

Multiplying (80) by t_i shows that $t_i h_i = \lambda_i \tau$ for a unique $\lambda_i \in k$. Then $g_i = h_i + \lambda_i \tau_i$ satisfies $t_i g_i = 0$ and $t_i g_j + t_j g_i = 0$. These relations imply $g_i = t_i z$ for a common $z \in R/k\tau$. Here is a direct coefficient proof. The first relation allows only monomials containing t_i in g_i. For every proper subset $A \subsetneq \{1, \dots, s\}$, define the coefficient of t_A in z to be the coefficient of $t_{A \cup \{i\}}$ in g_i, choosing any $i \notin A$. The second relation makes the answer independent of the choice. Only the highest coefficient remains free, and it is removed modulo $k\tau$.

Since $p\tau_i = 0$, equation (81) now gives

$$\begin{gathered} h_i = t_i z + \lambda_i \tau_i, \qquad h_v = a + pz + \zeta\tau, \\ a \in J/k\tau,\ z \in R/k\tau,\ \lambda \in k^s,\ \zeta \in k. \end{gathered} \tag{82}$$

These parameters are unique, and every choice satisfies the two kernel equations. Since $a^2 = 0$, multiplication by $1 + a$ is invertible. There are $(2^s - 2) + (2^s - 1) + s + 1 = 2^{s+1} + s - 2$ free parameters.

The group law. Let $\mathrm{pr} : R \to R$ remove the highest term, and let $[t_A]r$ denote a monomial coefficient. For left parameters (a, z, λ, ζ) and right parameters $(a', z', \lambda', \zeta')$, the composite has coordinates

$$\begin{aligned} a'' &= \mathrm{pr}(a + a' + aa'), \\ z'' &= \mathrm{pr}\left(z + z' + az' + \sum_i \lambda_i'[t_i]a\,\tau_i\right), \\ \lambda'' &= \lambda + \lambda', \\ \zeta'' &= \zeta + \zeta' + [\tau](aa'). \end{aligned}$$

Indeed block composition is $(u, h)(u', h') = (uu', h + uh')$, and $a\tau_i = [t_i]a\,\tau$. The formula contains no p, proving that the abstract kernel is independent of the parameter. It has exponent four: a square has $a = \lambda = \zeta = 0$, so its square is the identity; taking $a = t_1$, $z = 1$, and $\lambda = \zeta = 0$ gives an element whose square is nonidentity.

For completeness, conjugation by the section corresponding to (B, δ) is explicitly described in block coordinates by

$$u \longmapsto \sigma_B(u), \qquad h \longmapsto \delta\sigma_B \circ h \circ A^{-1}, \quad A = \mathrm{diag}(B, \delta).$$

Thus the kernel and the stabilizer action specify the full semidirect product, not only its order.

The unique binary lift. Any binary-linear map commuting with F_p preserves the intrinsic derived ideal. Its diagonal blocks commute with multiplication by ω, so they are k-linear. Write its off-diagonal block as $h_s + h_a$, with h_s linear over k and $h_a(\alpha\xi) = \alpha^2 h_a(\xi)$. Comparing the polar identity at ξ and $\omega\xi$ shows that h_a takes values in the common action kernel $k\tau$. It therefore contributes nothing to the polar bracket, and the k-linear part is an automorphism of $E(p)$.

Conversely, let S_0 be such an auxiliary automorphism and put $r_p = F_p - \omega\,\mathrm{id}$. The difference

$$D(\xi) = r_p(S_0(\xi, 0)) - S_0(r_p(\xi, 0))$$

is Frobenius semilinear: its polar vanishes, and the two restricted squares have the same adjoint square. Its values belong both to J and to the Lie center, hence to $k\tau$. It vanishes on the derived ideal. Since $\omega^2 + \omega = 1$, adding the central off-diagonal map D to S_0 gives a map commuting with F_p. The lift is unique; the difference of two lifts would satisfy $(\mathrm{id} + P_p(A\xi))h_a(\xi) = 0$, and the first factor is invertible. Taking the scalar part respects composition, so this is a group isomorphism.

For the kernel parameters above, the actual binary transformation is

$$\begin{aligned} (X, V, w) \longmapsto \Big(X, V, (1 + a)w + \sum_i X_i(t_i z + \lambda_i\tau_i) \\ + V(a + pz + \zeta\tau) + \tau\sum_i \lambda_i X_i^2\Big). \end{aligned}$$

The last term is forced by λ; it is not an additional family of choices. Indeed $D(e_i) = \lambda_i\tau$ and $D(v) = 0$.

The class-two boundary. For $s = 1$, the auxiliary algebra is the three-dimensional Heisenberg algebra. Its automorphisms are $(u, z) \mapsto (Bu, \det(B)z + h(u))$, with $B \in \mathrm{GL}_2(k)$ and $h \in \mathrm{Hom}_k(k^2, k)$. For $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the corresponding binary lift with $h = 0$ is

$$(X, V, z) \longmapsto (aX + bV, cX + dV, \det(B)z + acX^2 + bdV^2).$$

The kernel is $(k^2, +)$, giving order $4^2|\,\mathrm{GL}_2(4)| = 2880$, as asserted in Remark 8.11.

A.2 Pure quadratic parameters and an intrinsic action rank

Pure quadratic parameters form a readable all-class subfamily of the canonical extremizers. Their ordinary polynomial degree refers to the fixed ring coefficients, and does not increase the binary degree of the associated map.

Proposition A.2. *Let $s = c - 1 \geq 3$. Among the canonical models with $p \in \Lambda^2 k^s$, there are exactly $\lfloor s/2 \rfloor + 1$ binary-linear conjugacy classes, represented by*

$$p_0 = 0, \qquad p_r = t_1t_2 + t_3t_4 + \cdots + t_{2r-1}t_{2r} \quad (1 \leq r \leq \lfloor s/2 \rfloor).$$

For $r \geq 1$, multiplication by p_r on the intrinsic derived module J has k-rank

$$\rho_{s,r} = 2^{s-1} - 2^{s-r-1} - 1. \tag{83}$$

The corresponding binary rank is $2\rho_{s,r}$.

Proof. Linear substitutions preserve the homogeneous degree. On $\Lambda^2 k^s$, the standard alternating-bivector normal form is determined by its even rank $2r$, giving the displayed representatives. Common nonzero scalar multiplication creates no further identifications between different ranks. The canonical orbit correspondence therefore gives the stated complete classification of this subfamily.

For the rank calculation, regard multiplication by p_r as a square-zero differential on R. Each paired factor $k[a, b]/(a^2, b^2)$, with differential given by multiplication by ab, is the direct sum of the acyclic block $k1 \to k(ab)$ and the zero differential on $ka \oplus kb$. Tensoring the r paired factors gives homology dimension 2^r. The remaining $s - 2r$ square-zero variables have zero differential and multiply that dimension by 2^{s-2r}. Thus

$$\dim_k H(R, p_r\cdot) = 2^{s-r}.$$

Since a square-zero operator has homology dimension equal to its ambient dimension minus twice its rank, multiplication by p_r on R has rank $2^{s-1} - 2^{s-r-1}$. Moreover $p_rR = kp_r \oplus p_rJ$: the first summand is homogeneous of degree two, whereas the second has degree at least three. Subtracting this one dimension proves (83). □

In the canonical model, kv is the intrinsic kernel of the quotient action modulo the square of the radical of its generated operator algebra. Consequently the rank of $P(v)$, up to rescaling v, is an intrinsic invariant. Formula (83) makes the alternating rank of a pure quadratic parameter visible directly in the extremal map. For example, at $s = 4$, parameters of ranks two and four give binary action ranks six and ten.

B The unrestricted metabelian equality classification

We prove Theorem 4.6. Throughout, $k = \mathbb{F}_4$, $c \geq 5$, $d = c - 1$, $m = d - 1$,

$$U = ke_0 \oplus H, \quad H = \mathrm{span}_k\{e_1, \ldots, e_m\}, \quad R = k[t_0, \ldots, t_m]/(t_0^2, \ldots, t_m^2), \quad K = (t_1, \ldots, t_m).$$

Put $z = t_1 \cdots t_m$, $I = (z) = \operatorname{span}_k\{z, t_0 z\}$, and $\tau = t_0 z$. In particular I is two-dimensional, whereas $\dim_k K = 2^d - 2$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of the abelian Lie algebra U, acting by $e_i w = t_i w$. The Koszul complex, Frobenius pairing, contraction homotopies and alternating-form reductions used here are standard linear-algebraic tools; see also [4, Theorem 1 and p. 279] for the classical Koszul background. We give the required calculations explicitly.

B.1 Equality forces a common derived module

Lemma B.1. *Every minimum-dimensional metabelian solution of class $c \geq 5$ has $d = c - 1$ quotient generators and derived R-module*

$$W \simeq D(R/I) \simeq K, \qquad D = \operatorname{Hom}_k(-, k).$$

Moreover $\operatorname{Ann}_R K = I$ *and* $\operatorname{End}_R K \simeq R/I$.

Proof. Equality in Theorem 4.4 gives $\dim_k W = 2^d - 2$ and $d = c - 1$. The quotient $W_d \to W$ retains a nonzero top component. Pull back a functional on W nonzero there to $\phi \in W_d^*$. By Lemma 4.3, a linear change of generators normalizes its highest homogeneous component to a top functional λ_0 with $\operatorname{Ann}_R \lambda_0 = I$. Write $\phi = \lambda_0 + \psi$, where the dual coefficient degrees of ψ are at most $d-2$. Since z has degree $d-1$, it annihilates ψ, and it annihilates λ_0 as well. Hence $I\phi = 0$, and $\dim_k R\phi \leq 2^d - 2$. The filtered contraction argument of the same lemma gives the reverse inequality. Thus $\operatorname{Ann}_R \phi = I$. As $R\phi \subseteq W^*$ already has dimension $\dim_k W$, $W^* = R\phi \simeq R/I$.

The pairing $(r, s) \mapsto [\tau](rs)$ is nondegenerate on R. It identifies $D(R/I)$ with $\operatorname{Ann}_R I = K$: a polynomial annihilates z precisely when its constant and t_0 coefficients vanish. The same observation gives $\operatorname{Ann}_R K = I$. Finally, duality identifies $\operatorname{End}_R K$ with the opposite of $\operatorname{End}_R(R/I) = R/I$, which is commutative. □

Lemma B.2 (One-dimensional derived-module defect)**.** *Let $d \geq 4$. Suppose a quotient W of the universal module W_d retains a nonzero top component and has* $\dim_k W = 2^d - 1$. *After a linear change of quotient generators, there is an exact sequence of R-modules*

$$0 \longrightarrow k \longrightarrow W \longrightarrow K \longrightarrow 0.$$

After this normalization, the number of possible R-module isomorphism types is at most $4^{\binom{d}{2}}$, and their ordinary cohomology satisfies $\dim_k H^2(U, W) \leq d^2$.

Proof. Pull back a functional nonzero on the top to $\phi \in W_d^*$, and normalize its highest component as in Lemma B.1. The same degree argument gives $I\phi = 0$, while the filtered contraction bound gives $\dim_k R\phi = 2^d - 2$. Thus $R\phi \simeq R/I$ is a codimension-one submodule of W^*. Its quotient is the trivial module k, since each t_i acts nilpotently. Dualizing gives the displayed sequence.

Such dual extensions are classified by $\operatorname{Ext}^1_R(k, R/I)$. Finite duality identifies this with $D \operatorname{Tor}^R_1(k, K)$. To compute the latter, put $S = k[t_1, \ldots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \ldots, t_{d-1})$. Then $K = k[t_0]/t_0^2 \otimes_k J_S$. The tensor-periodic minimal resolution of k over S, as in (78), has free rank $\binom{d+j-2}{j}$ in degree j. Deleting its final copy of S gives a resolution of J_S; extending scalars gives a minimal R-resolution of K. Its degree-one free rank is $\binom{d}{2}$. Hence

$$\dim_k \operatorname{Ext}^1_R(k, R/I) = \binom{d}{2},$$

giving the asserted bound on module types. Finally the ordinary CE long exact sequence of $0 \to k \to W \to K \to 0$ and Lemma B.3 give

$$\dim_k H^2(U, W) \le \dim_k H^2(U, k) + \dim_k H^2(U, K) = \binom{d}{2} + \left(d + \binom{d}{2}\right) = d^2.$$

Here module Ext_R and ordinary CE cohomology have different complexes; each is used with its stated meaning. □

B.2 Ordinary cocycles and intrinsic generation

Let $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. Write $\beta_\eta(e_0, e_i) = t_i$, with all its other basis values zero. For $\kappa \in H^*$, let

$$\beta_\kappa(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i) \quad (i, j \ge 1), \qquad \beta_\kappa(e_0, e_i) = 0.$$

For $\lambda = (\lambda_{ij})$, set $\beta_\lambda(e_i, e_j) = \lambda_{ij}\tau_{ij}$. Thus $\beta_{\kappa,\lambda} = \beta_\eta + \beta_\kappa + \beta_\lambda$ is exactly (29).

Lemma B.3. *Every class in $H^2(U, K)$ has a unique expression*

$$\eta[\beta_\eta] + [\beta_\kappa] + [\beta_\lambda], \qquad \eta \in k, \quad \kappa \in H^*, \quad \lambda \in k^{\binom{d}{2}}.$$

Consequently $\dim_k H^2(U, K) = d + \binom{d}{2}$. *Also* $\dim_k H^1(U, K) = d + 1$. *For a closed cocycle β, the condition*

$$K = \mathfrak{m}K + \mathrm{span}_k\{\beta(x, y) : x, y \in U\}, \qquad \mathfrak{m} = (t_0, \dots, t_m), \tag{†}$$

holds if and only if $\eta \neq 0$. In that case the associated solution has class $d + 1$.

Proof. The ordinary CE complex for the R-module R is the tensor product of the one-variable complexes $k[t_i]/t_i^2 \xrightarrow{t_i} k[t_i]/t_i^2$. Their zeroth cohomology is kt_i, and their first cohomology has representative 1. Hence $H^n(U, R)$ has basis

$$\Big(\prod_{i \notin S} t_i\Big) e_S^*, \qquad |S| = n.$$

In particular its coefficient degree is $d - n$.

Put $A = R/K = k[t_0]/t_0^2$. The H-action on A is zero, so $H^1(U, A)$ has representatives

$$h(e_0) = 1,\ h(H) = 0; \qquad h(e_i) = \kappa_i t_0,\ h(e_0) = 0.$$

Its dimension is $1 + m = d$. Its elements have coefficient degree zero or one; the same is true of $H^2(U, A)$. Since $d \ge 4$, the maps $H^1(U, R) \to H^1(U, A)$ and $H^2(U, R) \to H^2(U, A)$ vanish by coefficient degree. The long exact sequence therefore gives

$$0 \longrightarrow H^1(U, A) \longrightarrow H^2(U, K) \longrightarrow H^2(U, R) \longrightarrow 0. \tag{84}$$

In degree zero, $H^0(U, R) = k\tau \to H^0(U, A) = kt_0$ is zero. Together with the vanishing map on H^1 above, this gives $0 \to k \to H^1(U, K) \to H^1(U, R) \to 0$, proving its dimension $d + 1$. The connecting map gives β_η and β_κ. The displayed $H^2(U, R)$-basis lifts to the closed K-valued cocycles β_λ, proving the basis claim. Equivalently, the coefficient of τ_{ij} in component (i, j) extracts λ_{ij}, because a coboundary in that component is divisible by t_i or t_j.

The quotient $K/\mathfrak{m}K$ has basis $t_1, \dots, t_m$. The only degree-one closed cocycles are multiples of β_η; there are no degree-one coboundaries, since K has no constant component. Thus the images of the cocycle values in this quotient are precisely those of $\eta\beta_\eta$. They span it exactly when $\eta \neq 0$. This proves (†); one may also lift the spanning statement by Nakayama's lemma. Finally, $\mathfrak{m}^{d-1}K = k\tau \neq 0$ and $\mathfrak{m}^d K = 0$. Equation (16) gives class $d + 1$. □

Multiplication by a nonzero scalar normalizes η to one. Lemmas B.1 and B.3, followed by Theorem 3.2, prove the asserted coverage by $F_{\kappa,\lambda}$.

B.3 All morphisms and the normalized parabolic

Lemma B.4. *On the slice $\eta = 1$, auxiliary-Lie isomorphism is exactly* (30).

Proof. A Lie isomorphism preserves the intrinsic derived module K. Let $B \in \mathrm{GL}(U)$ be its induced quotient map and let C be its restriction to K. Intertwining the actions gives $C(rw) = \sigma_B(r)C(w)$, and hence $\sigma_B(I) = I$. The degree-m line of I is kz, and $\{u \in U : uz = 0\} = H$. Thus $B(H) = H$, so

$$B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}.$$

Conversely every such B preserves I and K. By Lemma B.1, all intertwining C have the form $u\sigma_B$, with $u \in (R/I)^\times$. Changes of section add exactly ordinary CE coboundaries.

Positive-degree multiplication is null-homotopic on this CE complex. Indeed, if ι_i is contraction with e_i, then $\delta\iota_i + \iota_i\delta = t_i$. Writing a positive-degree element as $\sum_i t_i r_i$ gives the homotopy $\sum_i r_i\iota_i$. Consequently the unit u acts on cohomology by its constant term u_0 alone. The star transforms as $\eta' = u_0\eta/a$, as is seen either in degree one or on its lift $h(e_0) = 1$ in A. The normalization $\eta = \eta' = 1$ therefore forces $u_0 = a$. This proves necessity of (30). For sufficiency, use $C = a\sigma_B$; equality of cohomology classes supplies the required change of section.

There are no further quotient substitutions: B is the actual linear map induced on U. For the original quadratic solutions, Theorem 3.2 identifies these Lie isomorphisms with binary conjugacies, including their uniquely determined off-diagonal binary-linear parts. In particular their diagonal blocks are k-linear, not Frobenius-semilinear. □

B.4 The graded action when $d \geq 5$

The coefficient degrees of the three parts of Lemma B.3 are $1, 2, d-2$. For $d \geq 5$ these are distinct, so the displayed splitting is equivariant. On A, the substitution is $t_0 \mapsto at_0$. Pulling back the lift $h(e_i) = \kappa_i t_0$ gives $\kappa' = a^2 M^{-T}\kappa$; its extra e_0-component is a multiple of $t_0 e_0^*$, which is a boundary on A. For the other part, the complementary-minor identity gives

$$[\tau_{ij}]\sigma_B(\tau_{kl}) = \det(B)(\Lambda^2 B^{-T})_{ij,kl}. \tag{‡}$$

The pullback on the two cochain inputs contributes the same minor. Extracting the τ_{ij}-coefficient in component (i,j) therefore proves (32). For example, take $B = \mathrm{diag}(b_0, \ldots, b_m)$, so $a = b_0$. The complementary monomial contributes

$$\sigma_B(\tau_{ij}) = \frac{\det B}{b_i b_j}\tau_{ij},$$

and the two pulled-back inputs contribute a further $(b_i b_j)^{-1}$. Consequently

$$\lambda'_{ij} = a\det(B)(b_i b_j)^{-2}\lambda_{ij}.$$

The same inverse minor occurs once from each source, explaining the entrywise square in (32). Identity (‡) also proves the action on the quotient $H^2(U, R)$ when $d = 4$; only its splitting fails at that endpoint.

Put $\Omega = \lambda^{[2]}$. This is a bijective reparametrization of coefficients, not an additional Galois identification of models. Since squaring twice is the identity on k, and $a^4 = a$,

$$\Omega' = \chi\Lambda^2 B^{-T}\Omega, \qquad \chi = a(\det M)^2.$$

Writing $\Omega = e_0^* \wedge \ell + b$, with $\ell \in H^*$, $b \in \Lambda^2 H^*$, and $w = M^{-1}v$, gives the complete action

$$\begin{aligned} \kappa' &= a^2 M^{-T}\kappa, \\ b' &= \chi M^{-T} b M^{-1}, \\ \ell' &= (\det M)^2 M^{-T}(\ell + \iota_w b). \end{aligned} \tag{85}$$

B.5 Complete rank and radical-pair reduction

Fix $\operatorname{rank} b = 2r$, put $Z = \operatorname{rad} b$, $n = \dim Z = m - 2r$, and choose $H = S \oplus Z$, with b standard symplectic on S. The parameter v removes exactly $\operatorname{im} b$ from ℓ. Its residual is thus $\ell_Z = \ell|_Z$. Write $\kappa = (\kappa_S, \kappa_Z)$. For $r > 0$, the stabilizer of the chosen b has

$$M = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}, \qquad A^T b A = \rho b, \quad D \in \mathrm{GL}(Z), \quad C \text{ arbitrary.}$$

The Pfaffian identity gives $\det A = \rho^r$. The condition $\chi = \rho$ is equivalent to

$$a = \rho^{1-2r} (\det D)^{-2}. \tag{86}$$

Every $\rho \in k^\times$ and every $D \in \mathrm{GL}(Z)$ occur. When $r = 0$, the same formulas hold with S absent and the freely selectable parameter $\rho = a(\det D)^2$. Because $x^3 = 1$ for $x \in k^\times$, the radical-pair action is

$$\kappa'_Z = \rho^{2-r} (\det D)^{-1} D^{-T} \kappa_Z, \qquad \ell'_Z = \rho^{-r} (\det D)^{-1} D^{-T} \ell_Z. \tag{87}$$

If $\kappa_Z \neq 0$, the free block C removes κ_S. If $\kappa_Z = 0$, it cannot change κ_S, and the symplectic group is transitive on its nonzero values. Thus there is exactly one zero/nonzero flag when $r > 0$ and $\kappa_Z = 0$, and no such flag otherwise.

The radical pair has five possible incidence types:

$$(0,0), \quad (0,\neq 0), \quad (\neq 0, 0), \quad \text{nonzero proportional}, \quad \text{independent}. \tag{$*$}$$

The last requires $n \geq 2$. In the proportional case the nonzero ratio is multiplied by ρ, so gives no further invariant. For $n \geq 2$, $\mathrm{SL}_n(k)$ is transitive on nonzero covectors. For $n \geq 3$, it is also transitive on ordered independent two-frames: adjust the determinant of a frame extension on an unused coordinate. These observations prove transitivity for all types in $(*)$, except possibly the last at $n = 2$. For $n = 1$, putting $D = (d_0)$ in (87) gives a freely selectable factor d_0 on a nonzero covector, so the same conclusion holds.

For $n = 2$ and an independent pair, let $\Delta = \det[\kappa_Z, \ell_Z]$. Equation (87) gives

$$\Delta' = \rho^{2-2r} \Delta, \tag{88}$$

since $(\det D)^3 = 1$. If $r \not\equiv 1 \pmod 3$, this character covers $k^\times$. If $r \equiv 1 \pmod 3$, the three nonzero values of Δ are separate invariants. These are complete: choose ρ using (88) when permitted, and then use $\mathrm{SL}_2(k)$, which is transitive on ordered bases of a fixed determinant.

This proves the following exhaustive count at each rank:

Rank condition	Radical condition	Number of orbits
$r = 0$	$n = m \geq 4$	5
$r > 0$	$n = 0$	2
$r > 0$	$n = 1$	6
$r > 0$	$n \geq 2$, except the next row	7
$r > 0,\ r \equiv 1 \pmod 3$	$n = 2$	9

Indeed, when $n \geq 1$ the first two types in $(*)$ each acquire the extra κ_S-flag for $r > 0$; the other types do not. For $n = 0$ only the first type remains, with its two flags. For $r = 0$ all five types occur and there is no flag. Summation gives

$$E_c = \begin{cases} 7h + 2\mathbf{1}_{h \equiv 2 \ (3)}, & m = 2h \geq 4, \\ 7h + 4, & m = 2h + 1 \geq 5. \end{cases}$$

Since $m = c - 2$, these are exactly (31) for $c \geq 6$.

B.6 The scalar invariant at $6 \mid c$

The three-valued case has a coordinate-free relative-invariant description. Suppose $m = 2r+2$ and $\operatorname{rank} b = 2r$. Let $\gamma_r(b)$ be the divided exterior power: its coefficient on a $2r$-subset is the Pfaffian of the corresponding submatrix. This polynomial definition uses no factorial division. It is natural under linear maps and homogeneous of degree r. Since $\gamma_r(b)$ spans the top exterior power of $\operatorname{im} b$, $(\iota_w b) \wedge \gamma_r(b) = 0$. Consequently the coefficient J of

$$\kappa \wedge \ell \wedge \gamma_r(b) \in \Lambda^m H^*$$

in a fixed volume form satisfies

$$J' = a^{r+2} (\det M)^{2r+1} J. \tag{89}$$

For $r \equiv 1 \pmod 3$ this character is trivial on $k^\times$; equivalently $c = 2r + 4$ is divisible by six. For example,

$$b = \sum_{j=1}^{r} e^*_{2j-1} \wedge e^*_{2j}, \quad \kappa = e^*_{2r+1}, \quad \ell = t e^*_{2r+2}, \qquad t \in \{1, \omega, \omega^2\},$$

gives three inequivalent parameters with identical rank and incidence data. The divided power is essential in characteristic two: ordinary b^r is zero for $r \geq 2$.

B.7 The nonsplit endpoint $d = 4$

Now $m = 3$. Define a symmetric matrix S and vector q by

$$S = \begin{pmatrix} \lambda_{23} & \kappa_3 & \kappa_2 \\ \kappa_3 & \lambda_{13} & \kappa_1 \\ \kappa_2 & \kappa_1 & \lambda_{12} \end{pmatrix}, \qquad q = \begin{pmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{03} \end{pmatrix}, \qquad D(S) = \begin{pmatrix} 0 & S_{33} & S_{22} \\ S_{33} & 0 & S_{11} \\ S_{22} & S_{11} & 0 \end{pmatrix}.$$

For $B = \left(\begin{smallmatrix} a & 0 \\ v & M \end{smallmatrix}\right)$ and $w = M^{-1}v$, the full action is

$$S' = a^2 (\det M)^{-1} M S M^T, \qquad q' = \det M (M^{-T})^{[2]} \big(q + D(S) w^{[2]}\big). \tag{90}$$

Here is a direct derivation. The coefficients of t_0 in the H-pair cocycles, with columns ordered as $(23), (13), (12)$, form a linear map $H^* \to H$; Jacobi is exactly symmetry of its matrix S. These coefficients cannot be altered by a K-valued degree-one coboundary. The change of volume on H, the factor $t_0 \mapsto a t_0$, and the normalizing scalar a give the first formula. Factoring $B = \operatorname{diag}(a, M) \left(\begin{smallmatrix} 1 & 0 \\ w & I \end{smallmatrix}\right)$, the unipotent factor leaves these t_0-coefficients unchanged. On q, the complementary-minor calculation gives exactly $D(S) w^{[2]}$; the diagonal factor gives $\det M (M^{-T})^{[2]}$. Thus no splitting of the colliding degrees has been assumed.

A single shear shows the collision explicitly. Take $a = 1$, $v = 0$, and $M = I + E_{21}$, so $e_1 \mapsto e_1 + e_2$. Start with $\lambda_{23} = 1$ and every other λ- and κ-coefficient zero. Then $S = E_{11}$, and

$$M S M^T = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\lambda'_{23} = \lambda'_{13} = \kappa'_3 = 1$, with all other coefficients zero. A pure λ-class has acquired a κ-component, so the split action for $d \geq 5$ cannot be used at this endpoint.

In fact the degree-two part of (84) is nonsplit already on the Levi $\mathrm{GL}_3(k)$. For $a = 1, v = 0$, the S-space is $\det(H)^{-1} \otimes \Gamma^2 H$, the symmetric tensors. Its alternating, or zero-diagonal, subspace is the κ-space; its diagonal quotient is the corresponding Frobenius-natural twist.

A diagonal-torus-equivariant splitting would have to lift each quotient basis vector to its diagonal symmetric matrix, since the weights x_i^2 and x_ix_j are distinct characters of $(k^\times)^3$, also after the common determinant twist. But a transvection $e_i \mapsto e_i + e_j$ sends that diagonal matrix to the two diagonal matrices plus a nonzero off-diagonal symmetric matrix. This contradicts equivariance. Projecting to S shows that the separate q-summand cannot split the full sequence either.

It remains to classify (90). Over the perfect field k, symmetric 3×3 matrices have five congruence types: zero, alternating rank two, and nonalternating ranks one, two and three. One may obtain this list by Gram–Schmidt; if an alternating plane remains after a nonsingular line, the identity $\langle 1\rangle \perp \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \simeq I_3$ diagonalizes the final nonalternating case. These are also the S-orbits in (90): any ordinary congruence M is realized by choosing $a^2 = \det M$. For each type the remaining complete invariants are

S	Remaining invariant	Orbits
0	$q = 0$ or $q \neq 0$	2
Alternating, rank 2	$q = 0;\ q \neq 0,\ Sq^{[2]} = 0;\ Sq^{[2]} \neq 0$	3
Nonalternating, rank $1, 2$, or 3	$\operatorname{diag}(S)^T q = 0$ or $\neq 0$	2 each

For $S = 0$, $\mathrm{SL}_3(k)$ is transitive on nonzero q. For alternating $S = J_2 \oplus 0$, its stabilizer has

$$M = \begin{pmatrix} A & b \\ 0 & a^2 \end{pmatrix}, \qquad A \in \mathrm{GL}_2(k), \quad b \in k^2.$$

If the first two coordinates of q are nonzero as a vector, the free block b removes the third coordinate and $\mathrm{SL}_2(k)$ is transitive on the remaining vector. If they vanish, $\det A$ scales the third coordinate freely. This proves the three cases in the table. Finally, if $u = \operatorname{diag} S \neq 0$, then $\operatorname{im} D(S) = \ker u^T$, so translations identify exactly the q with equal $u^T q$. Formula (90) gives $u'^T q' = a^2 u^T q$. The simultaneous choice $M = a^{-1} I_3$ fixes every S and multiplies q by a^2. All nonzero residues are therefore equivalent, proving the two cases for each nonalternating rank. There are exactly $2 + 3 + 2 + 2 + 2 = 11$ classes. This is E_5, and completes the proof of Theorem 4.6.

C Fixed surplus over finite characteristic-two fields

We prove the upper bound in Theorem 5.3. Throughout this appendix, k is a finite field of characteristic two, $|k| = q \geq 4$, and all Lie isomorphisms are k-linear. By Lemma 5.2, the metabelian 3-Engel identity is equivalent to

$$(\operatorname{ad} x)^2 L' = 0 \qquad (x \in L).$$

Put

$$d = c - 1 \geq 4, \qquad C_d = \binom{d}{2}, \qquad w_0 = 2^d - 2,$$

and use the ring and modules

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \quad \mathfrak{m} = (t_0, \ldots, t_{d-1}), \quad K = (t_1, \ldots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}). \tag{91}$$

The ideal I has dimension two. We write $D = \mathrm{Hom}_k(-, k)$, and ordinary cohomology is the Chevalley–Eilenberg cohomology of the indicated abelian quotient. Module Ext_R is used separately.

Lemma C.1 (The minimum and its common module over k)**.** *Among metabelian 3-Engel k-Lie algebras of class $c \geq 5$, the minimum k-dimension is*

$$\ell(c) = 2^{c-1} + c - 3 = d + w_0.$$

Every equality case has quotient dimension d. After a linear change of quotient generators, its derived R-module is K. For the core quotient $U_0 = k^d$, acting by $e_i w = t_i w$,

$$\operatorname{Ann}_R K = I, \qquad \operatorname{End}_R K \simeq R/I,$$
$$\dim_k H^1(U_0, K) = d + 1, \qquad \dim_k H^2(U_0, K) = d + C_d.$$

The degree-one free rank in a minimal R-resolution of K is C_d, and

$$\dim_k \operatorname{Ext}^1_R(k, K) = 1.$$

In every minimum core the degree-one cocycle can be normalized to

$$\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K} \quad (i \geq 1), \qquad \beta(e_i, e_j) \in \mathfrak{m}K \quad (i, j \geq 1). \tag{92}$$

In particular, the number of minimum-dimensional k-Lie isomorphism classes is at most q^{d+C_d}.

Proof. The universal commutator module has the same square-free presentation and triangle relations as in Section 4, with k in place of $\mathbb{F}_4$. The multigrading calculation in Proposition 4.1, the repeated-terminal argument in Lemma 4.2, and the contraction calculation in Lemma 4.3 use only characteristic two. The top module is $\det(U_0) \otimes U_0$, and $\mathrm{SL}_d(k)$ is transitive on its nonzero dual vectors. A normalized top functional has annihilator I and contraction dimension $2^d - 2$. Thus every class-c object has quotient dimension at least d and derived dimension at least w_0. The normalized contraction quotient attains both bounds.

For equality, the proof of Lemma B.1 applies over k: a top functional generates the whole dual derived module as R/I. The top-coefficient Frobenius pairing on R gives $D(R/I) \simeq K$, $\operatorname{Ann}_R K = I$, and $\operatorname{End}_R K \simeq R/I$. The ordinary Koszul calculation in Lemma B.3 likewise has no field-size dependence. It gives the displayed H^1 and H^2 dimensions. For clarity, its degree-one normalization can also be read directly from Jacobi. The degree-one coefficients lie in $\operatorname{span}_k\{t_1, \ldots, t_{d-1}\}$. Jacobi at $(0, i, j)$ first makes the degree-one part of $\beta(e_i, e_j)$ zero: its t_0-multiple cannot cancel against the other two terms. Comparing the remaining monomials $t_i t_l$, using $d - 1 \geq 3$, then gives $\beta(e_0, e_i)_{[1]} = \eta t_i$ with one common scalar η. Generation of $K/\mathfrak{m}K$ forces $\eta \neq 0$, and rescaling K normalizes it to one. Two-coboundaries have coefficients in $\mathfrak{m}K$, so no degree-one term is lost in this comparison. This proves (92). For a fixed derived module, cocycles modulo changes of section form $H^2(U_0, K)$; these classes cover all minimum-dimensional objects. This proves the last count.

For the module-resolution assertions, set $S = k[t_1, \ldots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \ldots, t_{d-1})$. The tensor-periodic minimal resolution of k over S, as in (78), has degree-j free rank $\binom{d+j-2}{j}$. Its shifted tail resolves J_S, and extension of scalars resolves $K = R \otimes_S J_S$. Its degree-one free rank is therefore C_d. Finally finite duality gives

$$\operatorname{Ext}^1_R(k, K) \simeq D \operatorname{Tor}^R_1(k, R/I).$$

The cyclic presentation of R/I has the principal minimal relation ideal I, so $\operatorname{Tor}^R_1(k, R/I) \simeq I/\mathfrak{m}I$ is one-dimensional. □

Lemma C.2 (Bounded derived defect)**.** *Let L be a metabelian 3-Engel k-Lie algebra of class c, with derived dimension $w_0 + h$ and quotient dimension at least d. There is a class-c core with d quotient generators. After choosing normalized core quotient coordinates, the full derived*

ideal $W = L'$, as an R-module, has a filtration with one factor K and h factors k. The possible normalized R-module types are covered by at most

$$M_h(d) = (h+1)q^{hC_d+dh^2+h} \tag{93}$$

representatives. Every such W satisfies

$$\dim_k \mathrm{End}_R(W) \le w_0 + hd + h^2, \qquad \dim_k \mathrm{Soc}_R(W) \le h+1, \tag{94}$$

$$\dim_k H^1(U_0, W) \le (h+1)d + 1, \qquad \dim_k H^2(U_0, W) \le d + (h+1)C_d. \tag{95}$$

Here the core quotient U_0 acts through R.

Proof. Lemma 4.2 supplies the core. Its generators may be chosen as linear lifts of a basis of a d-dimensional quotient subspace. Changing a lift inside W leaves a terminal commutator unchanged, since $\mathfrak{m}^{c-1}W = 0$. Thus the chosen core H still has class c, and its abelianization is that quotient subspace. Write $W_0 = H'$. Its dimension is $w_0 + t$ for some $0 \le t \le h$, and it is a quotient of the universal module W_d retaining its top component.

Choose a functional on W_0 nonzero on that top component, and pull it back to $\phi \in W_d^*$. Normalize its highest component to have annihilator I. Write this component as λ, so the remaining dual degrees of $\phi - \lambda$ are at most $d-2$. Multiplication by $t_1 \cdots t_{d-1}$ kills those lower components and kills λ. Therefore $I\phi = 0$. The filtered contraction bound and this annihilator bound give $\dim_k R\phi = w_0$, hence $R\phi \simeq R/I$. Its quotient in W_0^* has dimension t. Dualizing gives

$$0 \longrightarrow F_t \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N_a \longrightarrow 0, \qquad a = h - t, \tag{96}$$

where $\dim_k F_t = t$ and $\dim_k N_a = a$. The local algebra R has residue field k; refining the two small modules by composition series gives the claimed filtration. In particular the full W is handled through these two extensions; only W_0 is required to be a quotient of W_d.

A j-dimensional R-module can be specified by d matrices of size j, so there are at most q^{dj^2} possible module types. For fixed F_t, the degree-one free rank in Lemma C.1 gives

$$\dim_k \mathrm{Ext}^1_R(K, F_t) \le tC_d.$$

The first extension in (96) is thus covered by $q^{dt^2+tC_d}$ choices. Since $\dim_k \mathrm{Ext}^1_R(k,k) = d$, composition series and the long exact sequence give

$$\dim_k \mathrm{Ext}^1_R(k, F_t) \le td, \qquad \dim_k \mathrm{Ext}^1_R(k, W_0) \le td + 1.$$

Applying the long exact sequence in its first argument to a composition series of N_a then gives

$$\dim_k \mathrm{Ext}^1_R(N_a, W_0) \le a(td+1).$$

Thus the second extension costs at most $q^{da^2+a(td+1)}$ choices. Summing over $t + a = h$, the total is bounded by

$$\sum_{t=0}^{h} q^{d(t^2+a^2)+tC_d+a(td+1)} \le (h+1)q^{hC_d+dh^2+h}.$$

These extension representatives cover all normalized module types. Every unmarked Lie isomorphism class admits a normalized presentation, so choices of normalization introduce no additional factor in this upper bound.

For the remaining bounds, dimensions of Hom_R are subadditive across a short exact sequence in either argument. The filtration factors are one K and h copies of k, and

$$\dim_k \mathrm{Hom}_R(K,k) = d-1, \quad \dim_k \mathrm{Hom}_R(k,K) = 1, \quad \dim_k \mathrm{End}_R(K) = w_0.$$

Applying subadditivity in both arguments proves

$$\dim_k \mathrm{End}_R(W) \le w_0 + h(d-1) + h + h^2 = w_0 + hd + h^2.$$

The same filtration gives $\dim_k \mathrm{Soc}_R(W) = \dim_k \mathrm{Hom}_R(k, W) \le h + 1$. Ordinary CE cohomology is dimension-subadditive across its long exact sequence. Combining Lemma C.1 with $\dim H^1(U_0, k) = d$ and $\dim H^2(U_0, k) = C_d$ proves (95). □

Proposition C.3 (The fixed-surplus upper bound)**.** *Let $B_{c,r}(q)$ count metabelian 3-Engel k-Lie isomorphism classes of class c and dimension $\ell(c) + r$. For $r \ge 1$, put*

$$Q_r(d) = (2r+1)C_d + (r^2 + r + 1)d + r^3 + r + \binom{r}{2}(r+1).$$

Then

$$B_{c,r}(q) \le (r+1)^2 q^{rw_0 + Q_r(d)}. \tag{97}$$

More precisely, the branch with derived dimension $w_0 + h$, where $0 \le h \le r$ and $s = r - h$, contributes at most

$$(h+1)q^{sw_0 + (2h+1)C_d + d(h^2+2sh+s+1) + sh^2 + r + \binom{s}{2}(h+1)}. \tag{98}$$

In particular, for fixed q, r,

$$\log_q B_{c,r}(q) \le r2^{c-1} + O_{q,r}(c^2).$$

Proof. The separate minimum bounds in Lemma C.1 force quotient and derived dimensions

$$(d + s, w_0 + h), \qquad s + h = r, \quad s, h \ge 0.$$

Choose and normalize a d-generator core, then complete its quotient basis by $v_1, \dots, v_s$. For the fixed core R-module W, write $P_i = t_i \cdot$ and $T_a = \mathrm{ad}\, v_a|_W$. Metabelianity makes every T_a commute with all P_i, so $T_a \in \mathrm{End}_R(W)$. The action tuple therefore has at most

$$q^{s(w_0 + hd + h^2)}$$

choices. The commuting and square-zero conditions select a subset of these tuples.

The core bracket cocycle $\beta_{ij} = [e_i, e_j]$ is closed for the ordinary CE complex on U_0. Changing the core section by vectors of W adds a two-coboundary. By (95), the core cocycle can consequently be represented by at most $q^{d+(h+1)C_d}$ choices. Such section changes leave W, the P_i, and the T_a unchanged, so the module and action census continues to cover the resulting representatives.

Next put $A_{ia} = [e_i, v_a]$. The core/core/extra Jacobi equations are

$$P_i A_{ja} + P_j A_{ia} = T_a \beta_{ij}.$$

For fixed actions and core cocycle, two solutions for a given a differ by an ordinary one-cocycle. Replacing v_a by $v_a + w_a$, $w_a \in W$, adds the one-coboundary $(P_i w_a)_i$. Thus all mixed tuples can be represented by at most

$$q^{s((h+1)d+1)}$$

choices. These changes may also replace $[v_a, v_b]$ by $[v_a, v_b] + T_a w_b + T_b w_a$; those remaining brackets are counted after the mixed tuples have been fixed.

Write $Z_{ab} = [v_a, v_b]$. The core/extra/extra Jacobi equations give

$$P_i Z_{ab} = T_a A_{ib} + T_b A_{ia}.$$

Once the preceding data are fixed, two solutions for Z_{ab} differ by a vector in $\bigcap_i \ker P_i = \mathrm{Soc}_R(W)$. Therefore all extra pairs together have at most $q^{\binom{s}{2}(h+1)}$ choices. The remaining Jacobi equations and the generation and class conditions select a subset of the displayed data.

Multiplying these counts by (93) gives exponent

$$\begin{aligned} &hC_d + dh^2 + h + s(w_0 + hd + h^2) \\ &\qquad + d + (h+1)C_d + s((h+1)d + 1) + \binom{s}{2}(h+1) \\ &= sw_0 + (2h+1)C_d + d(h^2 + 2sh + s + 1) + sh^2 + r + \binom{s}{2}(h+1), \end{aligned}$$

which proves (98). Every Lie algebra under consideration admits one of these presentations; multiple choices of a core only enlarge this upper count. Finally

$$h^2 + 2sh + s + 1 \le r^2 + r + 1, \quad sh^2 \le r^3, \quad \binom{s}{2}(h+1) \le \binom{r}{2}(r+1),$$

and $\sum_{h=0}^{r}(h+1) \le (r+1)^2$. Summing the branches proves (97) and its logarithmic consequence. □

C.1 Proof of the transition scale

Write $n = 2^d$, $C_d = \binom{d}{2}$, $w_0 = n - 2$, and $D_{\mathrm{par}} = n - d - 3$. We use B_h for the branch with derived dimension $w_0 + h$, so $B_0 = B^{\min}_{d+1,r}(q)$ and $B_{d+1,r}(q) = \sum_{h=0}^{r} B_h$.

A uniform frame estimate. For $1 \le r \le D_{\mathrm{par}}$, Theorem 5.3 gives

$$\log_q B_{d+1,r}(q) \ge rn - d^2 - 2dr - r^2 - 3r - 1. \tag{99}$$

Indeed, after extracting $q^{rD_{\mathrm{par}}}$ from the frame numerator, its remaining product is at least

$$\prod_{j\ge 1}(1 - q^{-j}) \ge 1 - \sum_{j\ge 1} q^{-j} \ge \frac{2}{3}.$$

The denominator is at most $q^{dr+d^2+r^2}$, proving the estimate.

The minimum-derived branch below the transition scale. For $h \ge 1$, put $s = r - h$. The exponent in (98), apart from sw_0, is at most

$$h\big(3C_d + 3dr + 2r^2 + r\big).$$

Here $2h + 1 \le 3h$, $h^2 + 2sh + s + 1 \le 3rh$, $sh^2 \le r^2 h$, and $\binom{s}{2}(h+1) \le r^2 h$. Since $\log_q(h+1) \le h/2$, comparison with (99) gives, with

$$T = 3C_d + d^2 + 5dr + 3r^2 + 2r + 2,$$

the bound

$$\frac{B_h}{B_{d+1,r}(q)} \le q^{-h(w_0 - T)}. \tag{100}$$

When $a = w_0 - T > 0$, summing a geometric series yields

$$1 - \pi_{d,r}(q) \le \frac{q^{-a}}{1 - q^{-a}}. \tag{101}$$

If $r = o(\sqrt{n})$, then eventually $r \le D_{\mathrm{par}}$ and $T = o(n)$, so this tends to zero. Moreover the coarse upper bound (97) has an error $O(rd^2 + r^2 d + r^3)$. Together with (99), it gives $\log_q B_{d+1,r}(q) \sim rn$ in the same range.

A cubic lower family and a small characteristic quotient. Fix one minimum class-$d+1$ algebra L_0 from Lemma C.1. Its derived module is K, and

$$\dim_k\big(L_0/\gamma_3(L_0)\big) = d + \dim_k(K/\mathfrak{m}K) = 2d-1.$$

For $u = \lfloor 2r/3 \rfloor$ and $w = r-u$, each alternating map $b : \Lambda^2 k^u \to k^w$ defines a two-step algebra N_b on $k^u \oplus k^w$, with central second summand. No surjectivity condition is needed. All $q^{w\binom{u}{2}}$ labelled brackets give algebras $L_b = L_0 \oplus N_b$ counted by $B_{d+1,r}(q)$. Since $\gamma_3(N_b) = 0$,

$$L_b/\gamma_3(L_b) \simeq \big(L_0/\gamma_3(L_0)\big) \oplus N_b$$

has dimension $m = r+2d-1$. Distinct tensors give distinct brackets on this fixed labelled quotient space. Each quotient isomorphism class has at most $|\operatorname{GL}_m(q)|$ such labelled structures. Isomorphic L_b's have isomorphic characteristic quotients. Consequently,

$$B_{d+1,r}(q) \geq \frac{q^{w\binom{u}{2}}}{|\operatorname{GL}_{r+2d-1}(q)|}, \qquad \log_q B_{d+1,r}(q) \geq \frac{2}{27}r^3 - O\big((r+d)^2\big). \tag{102}$$

This argument uses the classical alternating-tensor count for two-step Lie brackets; see also the coordinate-explicit enumeration in [15]. It neither transfers a group-counting theorem to characteristic two nor assumes cancellation of direct summands.

The minimum-derived branch above the transition scale. Taking $h = 0$ in (98) gives, for every $r \geq 1$,

$$B_0 \leq q^{rw_0 + C_d + d(r+1) + r + \binom{r}{2}}. \tag{103}$$

If $r/\sqrt{n} \to \infty$, then $r \gg d$ and $n = o(r^2)$. Dividing (103) by (102) gives

$$\log_q \pi_{d,r}(q) \leq rn - \frac{2}{27}r^3 + O(r^2 + rd + d^2) = -\big(2/27 - o(1)\big)r^3.$$

This proves (47).

The uniform order of growth. The explicit upper bound in Proposition C.3 is $O(rn+r^3)$ on the logarithmic scale, since $d^2 = O(n)$ and $r^2 d \leq (r^3 + rd^2)/2$. For $r \leq \sqrt{n}$, (99) is at least a fixed positive multiple of rn, uniformly in r, once d is large. For $r \geq \sqrt{n}$, (102) is at least a fixed positive multiple of r^3, again uniformly in r. The larger of rn and r^3 is at least half their sum. This proves (45) and completes the proof of Theorem 6.1.

D Exact principal orbits and multiplicative enumeration

Let $k = \mathbb{F}_q$, $q \geq 4$, have characteristic two, and use $d, C_d, D_{\mathrm{par}}, w_0, R, K, I$ as in Section 5.3. Put $\tau = t_0 \cdots t_{d-1}$, $\tau_j = \prod_{i \neq j} t_i$, and $V_r = \langle v_1, \ldots, v_r \rangle_k$. The binary-map interpretation is used only at $q = 4$.

D.1 All ordinary cocycles

For each $p_a \in \mathcal{P}_d$, choose $a_{0a} = 0$, $a_{ia} \in \mathfrak{m}$, with

$$p_a = \sum_{i=1}^{d-1} t_i a_{ia}. \tag{104}$$

Assigning each monomial to its smallest positive index fixes this choice. Let $b = \eta\beta_\eta + \beta_\kappa + \beta_\lambda$ be the core representative from Lemma B.3. Use the core one-cocycles

$$\zeta_*(e_0) = 0, \quad \zeta_*(e_i) = t_0 t_i \ (i \geq 1), \qquad \zeta_j(e_i) = \mathbf{1}_{i=j}\tau_j \quad (0 \leq j < d).$$

Their classes form a basis of $H^1(U_0, K)$, where $U_0 = k^d$.

Lemma D.1. *For any tuple* $\mathbf{p} \in \mathcal{P}_d^r$,

$$\begin{aligned} H^2(U_0 \oplus V_r, K) &\simeq H^2(U_0, K) \oplus H^1(U_0, K) \otimes V_r^* \oplus H^0(U_0, K) \otimes \Lambda^2 V_r^*, \\ \dim H^2(U_0 \oplus V_r, K) &= d + C_d + r(d+1) + \binom{r}{2}, \\ \dim H^1(U_0 \oplus V_r, K) &= d + 1 + r. \end{aligned}$$

Explicit representatives are, with $\zeta^a \in \operatorname{span}\{\zeta_*, \zeta_0, \ldots, \zeta_{d-1}\}$ *and* $\theta_{ab} \in k$,

$$\begin{aligned} \beta(e_i, e_j) &= b(e_i, e_j), \\ \beta(e_i, v_a) &= \sum_j a_{ja} b(e_i, e_j) + \zeta^a(e_i), \\ \beta(v_a, v_b) &= \sum_{i<j} (a_{ia}a_{jb} + a_{ib}a_{ja}) b(e_i, e_j) + \sum_i (a_{ia}\zeta^b(e_i) + a_{ib}\zeta^a(e_i)) + \theta_{ab}\tau. \end{aligned} \tag{105}$$

These representatives are unique modulo coboundaries. Intrinsic generation is equivalent to $\eta \neq 0$.

Proof. In the ordinary Koszul cochain complex, replace e_i^* by $e_i^* + \sum_a a_{ia} v_a^*$, using (104). This triangular invertible R-linear change converts the differential $\sum_i t_i e_i^* \wedge + \sum_a p_a v_a^* \wedge$ to the core differential, with zero differential on $\Lambda(V_r^*)$. It is an operation on cochains, not a k-linear change of Lie quotient generators. Expanding the resulting representatives gives (105). The core cohomology bases and $H^0(U_0, K) = k\tau$ prove the dimensions and uniqueness. Every mixed coefficient has degree at least two, so modulo $\mathfrak{m}K$ only the core star remains. It spans $K/\mathfrak{m}K$ exactly when $\eta \neq 0$; apply Nakayama. □

Scalar multiplication of K normalizes $\eta = 1$. The minimum derived-module result covers every object with $h = 0$. Each extra action lies in $\operatorname{End}_R K = R/I$; square zero forces its constant term to vanish, and subtracting its linear part from the corresponding quotient generator gives a unique tuple in $\mathcal{P}_d^r$. Thus the normalized fibers cover the whole minimum-derived stratum.

D.2 The complete effective action

We use ordinary abelian-extension equivalence and section changes as in [3, Theorems 1.1–1.2]. The intrinsic kernel of $U \to \operatorname{rad}(R/I)/\operatorname{rad}(R/I)^2$ is V_r, whether or not the tuple is independent. Every isomorphism has quotient matrix $A_g = \left(\begin{smallmatrix} B & 0 \\ \alpha & \Delta \end{smallmatrix}\right)$, and action intertwining gives (41). The nilpotent implicit-function principle [1, Sections 1.2–1.3] also applies: iteration from zero increases the radical valuation of a difference, hence gives a unique target tuple. Its associated substitution has invertible linear part B.

Preserving $I = (z)$, $z = t_1 \cdots t_{d-1}$, in degree $d-1$ forces $B(H) = H$. Conversely, this condition suffices. The substitution σ_g is an automorphism of R and preserves K, since all p_a' lie in $\mathfrak{m}^2 \subset K$. In $\sigma_g(z)$, terms with at least two additional parameter factors have degree at least $d+1$ and vanish. The remaining terms give

$$\sigma_g(z) = (\det M)z + \xi\tau,$$

so $\sigma_g(I) = I$. No product between distinct parameters has been discarded where its degree could survive.

All module intertwiners are $u\sigma_g|_K$, $u \in (R/I)^\times$. Positive-degree multiplication is null-homotopic, by $\delta\iota_i + \iota_i\delta = t_i$. Consequently only the constant term u_0 acts on ordinary cohomology. The degree-one star transforms by u_0/a; its normalization forces $u_0 = a$. Section changes add precisely two-coboundaries. This proves that (42) is necessary and sufficient.

The formulas give a genuine group action. Under composition, two canonical R-lifts may differ by terms in I. Their maps on K intertwine the same automorphism of R/I, so their ratio is a module unit. Their leading maps agree, making its constant term one; its cohomological action is trivial. Thus the induced fiber maps compose as the actual quotient matrices, without assuming that the two R-lifts agree pointwise on K. This proves Theorem 5.7.

D.3 A summed fixed-point bound

Write

$$F_{d,r} = \prod_{j=0}^{r-1} (1 - q^{j - D_{\mathrm{par}}}),$$

which tends to one for fixed q, r. The frame base has size $q^{rD_{\mathrm{par}}} F_{d,r}$. All its cohomology fibers have the same cardinality, so a fixed-base bound also bounds the fixed proportion in $\mathcal{Y}^{\mathrm{fr}}_{d,r}$.

Suppose $(B, \Delta) \neq (I, I)$. The fixed equation is triangular by coefficient degree: in degree j its linear part is $\Lambda^j B \otimes I_r - I \otimes \Delta^T$, and all nonlinear terms depend on lower degrees. On the full exterior algebra tensored with k^r, put $T = \Lambda B \otimes I_r - I \otimes \Delta^T$. Then

$$T(L_v \otimes I_r) - (L_{Bv} \otimes I_r)T = L_{Bv - v} \otimes \Delta^T.$$

If $B \neq I$, this gives $\operatorname{rank} T \geq r2^{d-2}$. If $B = I$ but $\Delta \neq I$, then $\operatorname{rank} T \geq 2^d$. Removing degrees zero and one and quotienting by I loses at most $r(d+3)$. The common lower bound $2^{d-2} - r(d+3)$ therefore suffices. Since $|G_{d,r}| < q^{d^2 + rd + r^2 + 1}$, the sum of these fixed proportions is at most

$$F_{d,r}^{-1} q^{d^2 + rd + r^2 + 1 - 2^{d-2} + r(d+3)}. \tag{106}$$

It remains to control pure shears, where $B = I$, $\Delta = I$, and $\alpha \neq 0$. Let $k_0 = \operatorname{rank} \alpha \leq r$. For counting, enlarge $\mathcal{P}^r_d$ to $(\mathfrak{m}^2 R)^r$, adding $2r$ dimensions. Invertible linear changes of ring variables and parameter components put the displacement map into the form

$$(x_1, \ldots, x_{k_0}, y) \longmapsto (x_1 + p_1, \ldots, x_{k_0} + p_{k_0}, y).$$

These changes need not preserve I; it becomes a two-dimensional space I'. Set $S = k[y_1, \ldots, y_m]/(y_i^2)$, $m = d - k_0$, and expand

$$p_a = \sum_{T \subseteq \{1, \ldots, k_0\}} x_T b_{a,T}, \qquad a_a = b_{a,\varnothing} \in \mathfrak{m}_S^2.$$

A necessary condition for a fixed parameter is obtained by setting all active variables x_i to zero:

$$\sum_{T \neq \varnothing} \left(\prod_{i \in T} a_i \right) b_{a,T} \in \mathrm{pr}_S(I') \qquad (1 \leq a \leq r). \tag{107}$$

This is an exact specialization; it retains every product involving different parameters. The target space has dimension at most two.

Condition on all a_a. If some a_i, $1 \leq i \leq k_0$, has a nonzero quadratic coefficient, multiplication by a_i on S has rank at least 2^{m-2}: restrict inputs to variables outside a nonzero

quadratic monomial and project outputs to that monomial's full coefficient. The resulting map multiplies by a unit of the complementary ring. Restricting inputs to $b_{a,\{i\}} \in \mathfrak{m}_S$ loses at most one rank. Fixing all other coefficients in one equation of (107) therefore imposes codimension at least $2^{m-2} - 3$. The chosen a_i is held fixed independently of this coefficient block, so this is an affine-linear count with no independence assumption between the different fixed equations.

If all of $a_1, \dots, a_{k_0}$ have zero quadratic part, the enlarged parameter space has codimension $k_0\binom{m}{2}$. For $d \geq r+2$, use $m \geq d-r$ and sum over fewer than q^{rd} nonzero matrices α. The total pure-shear fixed proportion is at most

$$F_{d,r}^{-1} q^{rd+2r} \left(q^{3-2^{d-r-2}} + q^{-\binom{d-r}{2}} \right). \tag{108}$$

For fixed q, r, both (106) and (108) tend to zero. They bound the sum of all nonidentity fixed proportions, not only a probability of nonfree points. Burnside's lemma gives

$$|\mathcal{Y}_{d,r}^{\mathrm{fr}}/G_{d,r}| = \frac{q^{rD_{\mathrm{par}}} F_{d,r}\, q^{f_{d,r}}}{|G_{d,r}|}(1 + o(1)).$$

Nonfree parameter and orbit proportions also tend to zero, by the same stabilizer-sum argument as in Corollary 8.13.

D.4 The whole layer and its typical automorphism order

Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}}+r(r-1)}$; multiplying by the entire fiber size costs only $q^{O_r(d^2)}$ beyond this exponent. The branches with $h \geq 1$ likewise contribute at most $q^{(r-1)2^d+O_{q,r}(d^2)}$, by Proposition C.3. Both are negligible compared with the displayed orbit main term. Substituting the group order proves (43) for all objects in the layer.

An automorphism inducing the identity on U has a principal-unit module block, with q^{w_0-1} choices. Every such unit lifts, because it acts trivially on H^2. Its section blocks form a torsor under $Z^1(U, K)$, whose dimension is

$$(w_0 - 1) + (d + 1 + r) = w_0 + d + r.$$

Thus each faithful minimum-derived object satisfies

$$|\operatorname{Aut}_k(L_y)| = q^{2w_0+d+r-1} |\operatorname{Stab}_{G_{d,r}}(y)|.$$

Generic freeness and the negligible complementary strata prove (44). At $q = 4$, apply the isomorphism correspondence of Theorem 3.2.

E The critical constant and coupled derived defects

The action-space normalization uses the standard Gaussian enumeration of subspaces over finite fields; see, for example, [15, Section 2]. Ordinary extension equivalence is used in the form of a module intertwiner together with cocycles modulo coboundaries [3, Theorems 1.1–1.2]. The large principal-unit kernel follows from the usual contraction identity in the Chevalley–Eilenberg complex: if T_i is the action of e_i, then $\delta\iota_{e_i} + \iota_{e_i}\delta = T_i$. The uniform comparison here combines these tools with the intrinsic near-minimum module and a coupled one-dimensional central-defect family. In particular, the critical constant comes from comparing complete isomorphism-type counts.

E.1 Action subspaces and the lower side of the critical constant

Put $n = 2^d$, $w_0 = n-2$, $C_d = \binom{d}{2}$, and $D_{\mathrm{par}} = n-d-3$. Let B_h count the class-$d+1$ objects of total dimension $d + w_0 + r$ with derived dimension $w_0 + h$, and write $s = r - h$. All counts are unlabelled k-linear isomorphism counts, where $k = \mathbb{F}_q$ has characteristic two and $q \geq 4$.

Lemma E.1 (Normalizing the action subspace)**.** *Fix a normalized core R-module W in the h-branch, with core actions $P_0, \dots, P_{d-1}$. These actions are linearly independent. Set*

$$E = \mathrm{End}_R(W), \qquad S_0 = \mathrm{span}_k\{P_0, \dots, P_{d-1}\}.$$

The possible tuples of s additional actions, up to changes of the additional quotient basis and addition of core quotient directions, have a covering set of size at most

$$\sum_{j=0}^{\min(s,L)} \begin{bmatrix} L \\ j \end{bmatrix}_q, \qquad L = w_0 + hd + h^2 - d.$$

If $2s \leq L$, this is at most $\frac{3}{2}(s+1)q^{s(L-s)}$.

Proof. The filtration in Lemma C.2 has one factor K. A linear relation among the P_i on W would hold on that factor, contradicting the absence of a nonzero linear form in $\mathrm{Ann}_R K = I$. Thus $\dim S_0 = d$, and $\dim E/S_0 \leq L$.

For each subspace $F \subset E/S_0$, choose once and for all an ordered basis and lifts to E. Given an actual tuple $(T_1, \dots, T_s)$, let F be the span of its images and put $j = \dim F$. A change in $\mathrm{GL}_s(k)$ makes those images the chosen basis followed by zeros. Adding suitable core directions to the additional quotient generators then makes their actual actions equal to the chosen lifts followed by zeros. This is an invertible block change of quotient coordinates fixing the core. It keeps the R-module W and all core actions unchanged.

The brackets transform along with these coordinates. Once the actions have been normalized, the same core, mixed, and extra-pair cocycle covering used in Proposition C.3 applies. Some preselected lifts may give no admissible Lie algebra; including them only enlarges an upper bound. The argument chooses presentations and never divides a parameter count by a group with unknown stabilizers.

There are $\begin{bmatrix} L \\ j \end{bmatrix}_q$ subspaces of dimension j in an L-dimensional space. The usual product formula and $\prod_{i \geq 1}(1 - q^{-i}) \geq 2/3$ give $\begin{bmatrix} L \\ j \end{bmatrix}_q \leq \frac{3}{2} q^{j(L-j)}$. If $2s \leq L$, the exponent increases for $0 \leq j \leq s$, proving the last assertion. □

In particular, replacing the raw action-tuple census in Proposition C.3 by this covering gives

$$B_h \leq \tfrac{3}{2}(h+1)(s+1)q^{\,sw_0 + (2h+1)C_d + d(h^2+2sh+1) + sh^2 + r + \binom{s}{2}(h+1) - s^2}, \qquad 2s \leq w_0 + hd + h^2 - d. \tag{109}$$

The core section changes leave the actions fixed, and the later changes of additional lifts affect only the remaining brackets, as in that proposition. Thus the new normalization does not invalidate the successive ordinary-cohomology quotients used there.

Lemma E.2 (The minimum-derived count at growing surplus)**.** *Uniformly for $1 \leq r \leq D_{\mathrm{par}}/2$,*

$$\log_q B_0 = rn - \tfrac{1}{2}r^2 + O(rd + d^2). \tag{110}$$

More explicitly,

$$\log_q B_0 \geq rn - \tfrac{1}{2}r^2 - dr - C_d - \tfrac{5}{2}r - 3,$$
$$\log_q B_0 \leq rn - \tfrac{1}{2}r^2 + C_d + d - \tfrac{5}{2}r - 1 + \log_q\big(\tfrac{3}{2}(r+1)\big).$$

Proof. For $W = K$, subtracting the linear parts of the additional actions places them in the D_{par}-dimensional space $\mathcal{P}_d$. The standard Koszul elimination of redundant generators gives

$$H^2(U_0 \oplus k^r, K) \simeq H^2(U_0, K) \oplus \big(H^1(U_0, K) \otimes (k^r)^*\big) \oplus \big(\mathrm{Soc}_R(K) \otimes \Lambda^2(k^r)^*\big).$$

Indeed, write each $p_a = \sum_i t_i a_{ia}$ with $a_{ia} \in \mathfrak{m}$, and change cochain generators by $e_i^* \mapsto e_i^* + \sum_a a_{ia} v_a^*$. All mixed coefficients have degree at least two, so intrinsic generation is still exactly the nonzero core-star condition. After normalizing that coefficient to one, every action tuple has q^{f_r} cocycle classes, where

$$f_r = d + C_d + r(d+1) + \binom{r}{2} - 1.$$

Apply the action-subspace normalization directly in $\mathcal{P}_d$:

$$B_0 \le q^{f_r} \sum_{j=0}^{r} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q \le \tfrac{3}{2}(r+1) q^{f_r + r(D_{\mathrm{par}} - r)}.$$

This is the stated upper bound.

For the lower bound take every independent ordered r-frame in $\mathcal{P}_d$, together with all its normalized cocycle classes. The vector implicit-recovery equation in Theorem 5.3 makes the target frame unique for each fixed quotient matrix. There are at most $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ such matrices. For a fixed quotient matrix, any two module intertwiners differ by a unit of R/I. Principal units act trivially on ordinary cohomology, by the core contraction homotopies. Thus there are at most $q-1$ different effects on the target cocycle class. This is a fiber bound; it requires neither a free action nor an explicit full orbit classification. Consequently

$$B_0 \ge \frac{q^{f_r} \prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)}{(q-1) q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|}.$$

The frame product is at least $\frac{2}{3} q^{r D_{\mathrm{par}}}$, while the denominator is at most $q^{dr + d^2 + r^2 + 1}$. Taking logarithms and using $\log_q(2/3) \ge -1$ gives the stated lower bound. □

Proposition E.3 (Minimum-derived dominance below the constant)**.** *Put*

$$Z(d, r) = 2d^2 + 3dr + 2r + 6.$$

For $1 \le r \le D_{\mathrm{par}}/2$ and $1 \le h \le r$,

$$\frac{B_h}{B_0} \le q^{\,h[-n + \frac{1}{2}r^2 - \frac{1}{2}h^2 + Z(d,r)]}. \tag{111}$$

In particular, if $r/\sqrt{n} \to \alpha < \sqrt{2}$ as $d \to \infty$, then $B_0 / \sum_h B_h \to 1$.

Proof. The condition $r \le D_{\mathrm{par}}/2$ implies the condition on s in (109) for every h. Subtract the lower exponent in Lemma E.2 from the exponent there, and substitute $s = r - h$. After removing $h(-n + r^2/2 - h^2/2)$, the remainder is

$$d^2 h + d^2 - dh^2 + 2dhr - dh + dr + \tfrac{1}{2}hr + \tfrac{5}{2}h + r + 3.$$

For $h \ge 1$, dropping the negative terms bounds this by $h(2d^2 + 3dr + 3r/2 + 11/2)$. Also $\log_q[\frac{3}{2}(h+1)(s+1)] \le (r+1)/2$, since $\log_q(t+1) \le t/2$ for nonnegative integers t. Absorbing this into $hZ(d, r)$ proves (111).

If $a = n - r^2/2 - Z(d, r) > 0$, a geometric sum gives

$$1 - \frac{B_0}{\sum_h B_h} \le \sum_{h \ge 1} \frac{B_h}{B_0} \le \frac{q^{-a}}{1 - q^{-a}}.$$

For the stated sequence, $a/n \to 1 - \alpha^2/2 > 0$. No monotonicity assumption on the population proportions is used. □

E.2 A coupled central defect and the upper side of the constant

Retain the notation of Appendix E.1: $n = 2^d$, $w_0 = n-2$, $C_d = \binom{d}{2}$, and $D_{\mathrm{par}} = n-d-3$, with $d \geq 4$ and $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$. Here B_h counts the class-$d+1$ objects of total dimension $d + w_0 + r$ and derived dimension $w_0 + h$. Put $A = R/I$, and write $\tau = t_0 \cdots t_{d-1}$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of an abelian quotient, with its specified module action.

Proposition E.4 (One coupled derived defect)**.** *For $s = r-1$ with $2 \leq s \leq D_{\mathrm{par}}$, put*

$$f_s = d + C_d + s(d+1) + \binom{s}{2} - 1.$$

Then

$$B_1 \geq \frac{q^{f_s} \prod_{j=0}^{s-1} (q^{D_{\mathrm{par}}} - q^j)\big(q^{\binom{s}{2}} - 1\big)}{q^{ds+d+2} |\operatorname{GL}_d(q)|\,|\operatorname{GL}_s(q)|}. \tag{112}$$

Every algebra in the family used for this lower bound is indecomposable. In particular,

$$\log_q B_1 \geq sn - ds - C_d - d - 3s - 4. \tag{113}$$

For $3 \leq r \leq D_{\mathrm{par}}/2$, it follows that

$$\frac{B_0}{B_1} \leq q^{\,n - r^2/2 + dr + d^2 + r/2 + \log_q(\frac{3}{2}(r+1))}. \tag{114}$$

Proof. Apply Proposition F.1 with $h = 1$, restricting its central cocycle to $\Lambda^2 V_s$ and setting all other central coefficients to zero. A nonzero form on that subspace gives exactly the earlier coupled slice. Its complete isomorphism fibre is bounded by $(q-1)^2 q^{ds+d} |\operatorname{GL}_d(q)||\operatorname{GL}_s(q)|$, which is at most the denominator in (112). The indecomposability proof in that proposition applies to this subfamily. Extracting the frame product and the nonzero-form factor, using bounds 2/3 and 3/4, gives (113). Subtracting it from the upper bound in Lemma E.2 gives (114). The general construction retains further central cocycles; its stronger bound is used in Corollary E.7. □

Remark E.5 (Indecomposability of the binary solutions). At $q = 4$, the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy. Indeed, suppose a solution is conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its polar Lie algebra splits accordingly, so both factors are metabelian. The intrinsic derived ideal, quotient, and their scalar actions also split. In the proof of Theorem 3.2, choose the bases and normalized-section lifts separately in the two factors. Scalar-weight projection, the actions P_i, and the cocycle refinement Q then preserve these blocks. The auxiliary bracket (15) is consequently the direct sum of the two auxiliary Lie algebras, contradicting Proposition E.4. A nonzero linear factor gives a nonzero abelian auxiliary factor and is excluded by the same argument. This statement concerns direct sums of solutions. It does not assert indecomposability of the original polar Lie algebra, nor require an extension of the correspondence to arbitrary homomorphisms.

E.3 Principal orbits when the surplus grows

Fix the field $k = \mathbb{F}_q$, $q \geq 4$, of characteristic two. Retain $d, n = 2^d, D_{\mathrm{par}} = n-d-3, w_0 = n-2$, and write $P = \operatorname{rad}(R/I)^2$, a vector space of dimension D_{par}. Let B_0 count the minimum-derived stratum in dimension $d + w_0 + r$. The exact orbit model of Theorem 5.7 remains valid for every r.

Lemma E.6 (Pure shears through invariant subspaces)**.** *Let $\mathcal{F}_{d,r}$ be the ordered independent r-frames in P, and let N be the group of automorphisms of R/I whose linear part is the identity. For $\sigma \in N$, set*

$$U_\sigma = \operatorname{span}_k\{\sigma(t_i) - t_i : 0 \le i < d\}, \qquad F_\sigma = \ker(\sigma - 1 : P \to P).$$

Then the summed fixed proportion of all nonidentity pure shears is

$$\sum_{\alpha \neq 0} \frac{|\operatorname{Fix}_{\mathcal{F}_{d,r}}(\alpha)|}{|\mathcal{F}_{d,r}|} = \sum_{\substack{\sigma \in N \\ \sigma \neq 1}} \frac{|\{Q \le P : \dim Q = r,\ U_\sigma \le Q \le F_\sigma\}|}{\begin{bmatrix} D_{\mathrm{par}} \\ r \end{bmatrix}_q}. \tag{115}$$

Proof. Every choice $u_i \in P$ gives an automorphism $t_i \mapsto t_i + u_i$ of R/I. To see the lifting assertion, use the canonical representatives in $\mathfrak{m}^2 R$. Their substitution has identity linear part, hence is an automorphism of R. It preserves K; the image of $z = t_1 \cdots t_{d-1}$ lies in $kz + k\tau = I$, since terms with two positive-degree increments have degree at least $d+1$. Thus it preserves I, and the induced map is determined by the u_i's. Conversely every member of N has these increments.

For a fixed ordered frame $\mathbf{p}$, the coefficients of $u_i = \sum_a \alpha_{ai} p_a$ are unique. The pure-shear base equation is exactly $\sigma(p_a) = p_a$ for every a. Hence fixed pairs $(\alpha, \mathbf{p})$ correspond to pairs $(\sigma, \mathbf{p})$ with $U_\sigma \le \operatorname{span}(\mathbf{p}) \le F_\sigma$. The number of ordered bases is the same for every r-subspace. Dividing the pair count gives (115). □

For $2 \le j \le d-1$, put

$$a_j = \dim(P/\operatorname{rad}(R/I)^j), \qquad \rho_j = \max\{0, 2^{d-j-1} - d - 3\}.$$

If j is the least order of a nonzero increment of σ, then

$$\operatorname{rank}(\sigma - 1 : P \to P) \ge \rho_j. \tag{116}$$

Indeed lift σ to R, write $T = \sigma - 1$, and use

$$TL_{t_i} - L_{\sigma(t_i)} T = L_{\sigma(t_i) - t_i}.$$

An increment of order j has multiplication rank at least 2^{d-j}: restrict inputs to the variables complementary to one nonzero degree-j monomial and project to its full coefficient; the resulting multiplier has nonzero constant term. The commutator therefore gives $\operatorname{rank} T \ge 2^{d-j-1}$. Restricting to $\mathfrak{m}^2 R$ loses at most $d+1$ rank, and passage to the quotient by I loses at most two more. This proves (116).

If $k_0 = \dim U_\sigma$, the number of possible increment tuples of order at least j and rank k_0 is at most

$$\tfrac{3}{2} q^{k_0(D_{\mathrm{par}} - a_j + d - k_0)}.$$

For each actual tuple the Gaussian ratio in (115) is zero unless $U_\sigma \le F_\sigma$; otherwise it is at most

$$\tfrac{3}{2} q^{-k_0(D_{\mathrm{par}} - r) - (r - k_0)\rho_j}.$$

Thus the contribution from order j, rank k_0, is at most

$$\tfrac{9}{4} q^{k_0(r - a_j + d - k_0) - (r - k_0)\rho_j}. \tag{117}$$

This estimates the sum of stabilizers, rather than only the probability of a nontrivial stabilizer.

Let $j_0 = \lfloor d/2 \rfloor$, $a_0 = a_{j_0}$, and $\rho_0 = \rho_{j_0}$. Symmetry and the central-binomial estimate give $a_0 \sim n/2$, while $\rho_0 \geq \sqrt{n}/2 - d - 3$. For $r \geq 2d$, summing (117) separately over $j < j_0$ and $j \geq j_0$ gives the uniform bound

$$S_{\text{pure}} \leq \tfrac{9}{4} d^2 \left(q^{d(r+d)-(r-d)\rho_0} + q^{r+d-a_0} \right). \tag{118}$$

For the first group, drop $-k_0 a_j - k_0^2$ and use $k_0 \leq d$, $\rho_j \geq \rho_0$. For the second group use $a_j \geq a_0$, drop the nonpositive rank term, and use $k_0(r - a_0 + d - k_0) \leq r + d - a_0$ once $a_0 > r + d$. In particular (118) tends to zero uniformly on

$$2d \leq r \leq D_{\text{par}}/(4d). \tag{119}$$

For completeness the nonpure linear parts admit a bound on the same range. Put $F_{d,r} = \prod_{i<r}(1 - q^{i-D_{\text{par}}})$, which tends to one uniformly there. For $B \neq I$, retain the stronger exterior rank $r2^{d-2}$ in Appendix D. The resulting sum is at most

$$F_{d,r}^{-1} q^{d^2+rd+r^2+1-r2^{d-2}+r(d+3)}. \tag{120}$$

For $B = I$, put $t = \operatorname{rank}(\Delta - I)$. In every retained graded monomial the diagonal linear block is $\Delta^T - I$, so the total rank is $D_{\text{par}} t$. There are at most $\frac{3}{2} q^{t(2r-t)}$ such matrices Δ. Including all q^{rd} shears and summing over $t \geq 1$ gives

$$S_\Delta \leq \tfrac{3}{2} r F_{d,r}^{-1} q^{rd+2r-1-D_{\text{par}}}. \tag{121}$$

Both displays tend to zero uniformly in (119).

Proof of Theorem 6.4. Every normalized cohomology fiber has size $q^{f_{d,r}}$, so each fixed-base proportion also bounds the corresponding fixed proportion in the whole frame-fiber space. Equations (118)–(121) show that the summed nonidentity fixed proportion is $o(1)$, uniformly. Burnside therefore gives the frame count divided by $|G_{d,r}|$ with multiplicative error $1 + o(1)$, and makes nonfree orbit proportions negligible by the same stabilizer-sum argument.

It remains to cover rank-deficient action tuples. Normalizing their action subspace, while retaining all r quotient directions and all $q^{f_{d,r}}$ cocycles, bounds their class count by

$$q^{f_{d,r}} \sum_{j<r} \begin{bmatrix} D_{\text{par}} \\ j \end{bmatrix}_q \leq \tfrac{3}{2} r q^{f_{d,r}+(r-1)(D_{\text{par}}-r+1)}.$$

Its ratio to the faithful orbit main term is at most

$$\tfrac{3}{2} r F_{d,r}^{-1} q^{-D_{\text{par}}+rd+d^2+2r} = o(1),$$

uniformly in (119). This proves (50). The exact full-automorphism kernel calculation in Appendix D is valid for every faithful frame, so generic effective freeness gives the stated order. □

The independent subcritical estimate of Proposition E.3 transfers (50) to the entire dimension layer on the subrange $2d \leq r \leq \sqrt{2n} - 3d - 3$. No such whole-layer conclusion is made above the critical constant.

Corollary E.7 (A sharper upper-side localization)**.** *If $r \geq \sqrt{2n} - d + 1$ eventually, then $\pi_{d,r}(q) \to 0$. In the portion of* (119) *with $r \geq 3$, put*

$$T = \binom{d+r-1}{2} - (d-1).$$

The full-central-cocycle coupled family gives

$$\pi_{d,r}(q) \leq (1 + o(1)) \frac{(q^{D_{\text{par}}} - q^{r-1})(q-1)q^2(q^d-1)}{(q^r-1)(q^T-1)}. \tag{122}$$

Proof. Divide (50) by the independent coupled lower bound (123), with $h = 1$ and $s = r - 1$. Its full central-cocycle count is $q^{d-1}(q^T - 1)$; its effective module-block bound is $(q-1)^2 q^d$. Use $|\operatorname{GL}_m|/|\operatorname{GL}_{m-1}| = q^{m-1}(q^m - 1)$. This gives (122). The logarithm of its right-hand side is

$$n - \tfrac{1}{2}(r+d)^2 + \tfrac{1}{2}(r+d) + 2d - 3 + \log_q(q-1) + o(1).$$

The quadratic expression decreases for $r+d \geq 1$. At $r = \sqrt{2n} - d + 1$ it equals $-\sqrt{2n}/2 + 2d - 3$, which tends to minus infinity. This proves the assertion on the indicated range. On all remaining indices with $r > D_{\mathrm{par}}/(4d)$, one has $r/\sqrt{n} \to \infty$; invoke (47). □

F Coupled central defects and the postcritical population

Fix $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$, and retain $R, K, I, A = R/I$ from Appendix E. Put $d = c - 1 \geq 4$, $n = 2^d$, $w_0 = n - 2$, $C_d = \binom{d}{2}$, $D_{\mathrm{par}} = n - d - 3$, and $\tau = t_0 \cdots t_{d-1}$. Write B_h for the branch of $B_{d+1,r}(q)$ with derived dimension $w_0 + h$, and put $s = r - h$. Ordinary cohomology always means the Chevalley–Eilenberg cohomology for the specified abelian quotient action. No restricted cohomology is used.

F.1 A family with arbitrary central defect

Proposition F.1 (The coupled central-defect family)**.** *Suppose* $1 \leq s \leq D_{\mathrm{par}}$, $r = s + h$, *and*

$$T_{d,s} = \binom{d+s}{2} - (d-1), \qquad 1 \leq h \leq T_{d,s}.$$

Set

$$f_s = d + C_d + s(d+1) + \binom{s}{2} - 1.$$

There is an indecomposable family in the branch B_h *giving*

$$B_h \geq \frac{q^{f_s + h(d-1)} \prod_{j=0}^{s-1} (q^{D_{\mathrm{par}}} - q^j) \prod_{j=0}^{h-1} \left(q^{T_{d,s}} - q^j\right)}{(q-1) q^{dr} |\operatorname{GL}_d(q)| |\operatorname{GL}_s(q)| |\operatorname{GL}_h(q)|}. \tag{123}$$

Consequently the following estimate is uniform in this range:

$$\log_q B_h \geq sn + h\binom{d+s}{2} - \tfrac{1}{2}s^2 - dr - C_d - h^2 - \tfrac{5}{2}s - 3. \tag{124}$$

At $q = 4$, *the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy.*

Proof. Let $Z = k^h$, $W = K \oplus Z$, and $U = U_0 \oplus V_s$, where $U_0 = k^d$ has basis $e_0, \ldots, e_{d-1}$. Choose an ordered independent s-frame $\mathbf{p} = (p_1, \ldots, p_s)$ in $\mathcal{P}_d$. On K, the core and added quotient generators act by t_i and p_a, respectively; they kill Z. Take every K-valued ordinary two-cohomology class with normalized core-star coefficient one. The complete Koszul decomposition of Lemma D.1 shows that there are exactly q^{f_s} such classes, retaining all core, mixed, and extra-pair coefficients.

Write $\mathcal{N} \subset \Lambda^2 U$ for the span of all basis pairs except the $d-1$ star pairs $e_0 \wedge e_i$, $1 \leq i < d$. Thus $\dim \mathcal{N} = T_{d,s}$. Independently choose an alternating map

$$\gamma : \Lambda^2 U \longrightarrow Z$$

whose restriction to $\mathcal{N}$ is surjective; its star values are arbitrary. It is a Z-valued cocycle because the quotient acts trivially on Z. The combined cocycle defines an abelian Lie extension of U by W.

To check its derived ideal, the specified action already gives $[U, W] = \mathfrak{m}K \subset L'$, as a pure submodule, without first assuming that $W = L'$. Every nonstar K-component in the complete cocycle formula (105) lies in $\mathfrak{m}K$: the nonstar core terms have degree at least two, including the λ-terms of degree $d - 2 \geq 2$; mixed terms have degree at least two; extra-pair terms have degree at least three, apart from their degree-d socle term. Subtracting these pure $\mathfrak{m}K$-components from nonstar brackets gives all values of $\gamma|_{\mathcal{N}}$ as pure central vectors. Its surjectivity supplies all of Z. The central star components may then be subtracted. Their K-components q_i satisfy $q_i \equiv t_i \pmod{\mathfrak{m}K}$, and hence generate all of K by Nakayama. More generally, for any central map γ in this slice,

$$\dim_k L' = w_0 + \operatorname{rank}(\gamma|_{\mathcal{N}}).$$

Indeed, modulo $[U, W] = \mathfrak{m}K$, the star brackets have independent $K/\mathfrak{m}K$-components, whereas every nonstar K-component is zero. The bracket image therefore has dimension $d - 1 + \operatorname{rank}(\gamma|_{\mathcal{N}})$ in that quotient. Thus surjectivity on $\mathcal{N}$ is precisely the condition for $L' = W$, independently of the star values. For our surjective choice, from the third term onwards the lower central series is $\gamma_j(L) = \mathfrak{m}^{j-2}K$, so its class is $d + 1$. The commuting quotient actions square to zero, and the extension is metabelian 3-Engel. Its dimension is $d + s + w_0 + h = d + w_0 + r$.

We bound complete isomorphism fibres rather than assume a free parameter action. The action algebra on W is faithfully A, and the quotient action itself is injective: the degree-one terms detect U_0, and independence of the p_a's detects V_s. The kernel of $U \to \operatorname{rad}(A)/\operatorname{rad}(A)^2$ is intrinsically V_s. Thus an isomorphism between two members has a quotient matrix

$$\begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix}, \qquad B \in \mathrm{GL}_d(k), \quad \Delta \in \mathrm{GL}_s(k).$$

There are at most $q^{ds} |\mathrm{GL}_d(q)||\mathrm{GL}_s(q)|$ such matrices. For each fixed matrix, the vector implicit-recovery equation (38) fixes at most one target frame. It uses only the faithful algebra A and the radical degree of the added actions, so adding the trivial summand Z does not alter its conclusion.

The full module automorphism group has blocks

$$\operatorname{Aut}_R(K \oplus Z) = \left\{ \begin{pmatrix} u & \tau v \\ \lambda & C \end{pmatrix} : u \in A^\times,\ v \in k^{1\times h},\ \lambda \in \operatorname{Hom}_R(K, Z),\ C \in \mathrm{GL}_h(k) \right\}.$$

Indeed, $\operatorname{Hom}_R(Z, K)$ consists of socle-valued maps, and $\dim \operatorname{Hom}_R(K, Z) = h(d-1)$. Since $\tau \in \mathfrak{m}K$, one has $\lambda(\tau) = 0$. Every automorphism must have $u_0 \neq 0$: on $W/\mathfrak{m}W = (K/\mathfrak{m}K) \oplus Z$, its upper-right block is zero and its upper-left block is $u_0 I_{d-1}$. For a unit u, the Schur complement is just C, since $u^{-1}\tau = u_0^{-1}\tau$ and $\lambda(\tau) = 0$. This proves the displayed block description and gives

$$|\operatorname{Aut}_R(W)| = (q-1)|\mathrm{GL}_h(q)|q^{w_0-1+hd}.$$

The subgroup $\operatorname{diag}(1 + a, I_h)$, $a \in \operatorname{rad}(A)$, has order q^{w_0-1} and acts trivially on all of $H^2(U, W)$. To see this, lift $a = \sum_i t_i b_i$ in $\mathfrak{m}R$ and use the ordinary contraction identity

$$\delta\Big(\sum_i b_i \iota_{e_i}\Big) + \Big(\sum_i b_i \iota_{e_i}\Big)\delta = a.$$

On W, positive-degree multiplication is the map $\operatorname{diag}(a, 0)$. Hence each displayed principal unit differs from the identity by a null-homotopic map. The effective module effects on cohomology number at most $(q-1)|\mathrm{GL}_h(q)|q^{hd}$.

For a fixed quotient matrix and target frame, all intertwiners differ by an element of $\mathrm{Aut}_R(W)$, and section changes are two-coboundaries. Each effective module effect gives at most one target cohomology class. The selected K-valued classes and Z-valued surjections inject into $H^2(U,K) \oplus H^2(U,Z)$, so each such class represents at most one member of the chosen family. An effect need not preserve our slice: an image outside it contributes no target. This bounds every complete isomorphism fibre by the denominator in (123). The number of surjections $k^{T_{d,s}} \to k^h$ is the second product in its numerator; the first product counts independent action frames. The arbitrary star values contribute $q^{h(d-1)}$. This proves the exact lower bound.

Every family member is indecomposable. The endomorphism ring $\mathrm{End}_R K = A$ is local, so K is indecomposable and is not isomorphic to the trivial module k. If $L = L_1 \oplus L_2$, then $W = L_1' \oplus L_2'$ as R-modules. By finite-length Krull–Schmidt, at most one derived summand contains the indecomposable factor K; any other nonzero derived summand is a sum of trivial modules. All core and polynomial extra actions kill that summand. Its Lie factor also kills the other derived summand by the direct-sum condition. Faithfulness of L/W then puts the whole factor inside the abelian ideal W, contradicting its nonzero derived ideal. If one derived summand is zero, its Lie factor kills W; faithfulness puts it inside W, which lies in the other factor, so it is zero. At $q = 4$, suppose the corresponding quadratic solution were conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its derived ideal, quotient, and their intrinsic scalar actions split accordingly. Choose the bases and normalized-section lifts block by block in Theorem 3.2. The scalar-weight projections, operators P_i, and quadratic refinement Q preserve these blocks, so the resulting auxiliary Lie algebra is the direct sum of their two nonzero auxiliary algebras. This contradicts the Lie indecomposability just proved. A nonzero linear factor likewise gives a nonzero abelian auxiliary factor and is excluded. This proves indecomposability of the quadratic solution; it makes no assertion about arbitrary decompositions of its original polar Lie algebra.

Finally, both products after extracting their leading powers of q are at least $\prod_{j\ge1}(1 - q^{-j}) \ge 2/3$. Use $(q-1) < q$ and $|\mathrm{GL}_j(q)| \le q^{j^2}$. Substituting $D_{\mathrm{par}} = n - d - 3$ and f_s gives (124), with the constant -3 covering the extracted product factors. For $h = 1$, $s = r - 1$, the resulting bound is $\log_q B_1 \ge sn - 3s - d - 4$. Relative to the extra-pair-only family, the larger central cocycle space contributes the factor q^{ds+C_d} at leading order. □

F.2 Uniform branch comparison

For real $\rho \ge 0$ and $0 \le \beta \le \rho$, define

$$U(\rho,\beta) = \rho + \left(\frac{\rho^2}{2} - 1\right)\beta - \frac{\beta^3}{2} = (\rho-\beta)\left(1 + \beta^2 + \frac{\beta(\rho-\beta)}{2}\right),$$
$$L(\rho,\beta) = \rho - \beta + \frac{\beta}{2}(\rho-\beta)^2.$$

These are coarse upper and lower branch comparisons. Appendix G sharpens the upper covering and determines the exact leading count of the whole population.

Lemma F.2 (Uniform normalized comparison exponents)**.** *Fix $M > 0$. Uniformly for integers $0 \le h \le r \le M\sqrt{n}$, putting $\rho = r/\sqrt{n}$, $\beta = h/\sqrt{n}$, one has*

$$\log_q B_h \le n^{3/2} U(\rho,\beta) + O_M(dn + d^2\sqrt{n} + n). \tag{125}$$

If $r/\sqrt{n} \to \alpha$, and $0 < \beta_0 < \alpha$ is fixed, then for $h = \lfloor \beta_0\sqrt{n} \rfloor$,

$$\liminf_{d\to\infty} n^{-3/2} \log_q B_h \ge L(\alpha,\beta_0). \tag{126}$$

Proof. For the upper bound the condition on the action-subspace census in (109) holds eventually, uniformly in these branches, because $2s = O_M(\sqrt{n})$ whereas its right-hand dimension is at least $n - 2 - d$. The leading terms in that exponent are

$$sn + sh^2 + \tfrac{1}{2}hs^2 = n^{3/2}U(\rho, \beta).$$

All remaining terms are uniformly $O_M(dn+d^2\sqrt{n}+n)$, including the logarithm of its prefactor. This error is $o(n^{3/2})$.

For the lower bound, $s = r - h \sim (\alpha - \beta_0)\sqrt{n}$. Thus eventually $s \le D_{\mathrm{par}}$ and $h \le \binom{s}{2}$, so Proposition F.1 applies. In (124), the leading terms are $sn + hs^2/2$. Its other terms are $O_M(dn+d^2\sqrt{n}+n)$; rounding h changes the normalized leading expression by $o(1)$. Continuity of L gives the liminf assertion. No rate of convergence for $r/\sqrt{n}$ is required. □

F.3 Proof of the typical surplus order

Proof of Proposition 6.6. Put $\delta = \alpha^2/2 - 1 > 0$, and choose

$$\beta_0 = \frac{\delta}{2\alpha}, \qquad G = \frac{\delta^2}{4\alpha}.$$

Here $0 < \beta_0 < \alpha/4$, and

$$L(\alpha, \beta_0) - \alpha = \delta\beta_0 - \alpha\beta_0^2 + \tfrac{1}{2}\beta_0^3 \ge G > 0.$$

The lower branch $h = \lfloor \beta_0\sqrt{n} \rfloor$ therefore shows that the total population has size at least $q^{(\alpha+G+o(1))n^{3/2}}$.

Set $\epsilon_h = \delta/(16\alpha)$. For $0 \le \beta \le \epsilon_h$,

$$U(\alpha, \beta) \le \alpha + \delta\epsilon_h = \alpha + G/4.$$

By the uniform upper estimate, continuity, and $\log_q(r + 1) = O(d)$, the combined count of all branches with $h \le \epsilon_h\sqrt{n}$ is at most $q^{(\alpha+G/4+o(1))n^{3/2}}$. Their proportion is at most $q^{-(3G/4-o(1))n^{3/2}}$.

For the other endpoint put $\epsilon_s = \alpha/[4(1 + 6\alpha^2)]$. Eventually $\rho = r/\sqrt{n} \le 2\alpha$. If $s \le \epsilon_s\sqrt{n}$, the factorized form of U, with $\beta \le 2\alpha$, gives

$$U(\rho, \beta) \le \epsilon_s(1 + 6\alpha^2) = \alpha/4.$$

The combined proportion of these branches is consequently at most $q^{-(3\alpha/4+G-o(1))n^{3/2}}$. Adding the two exceptional proportions and absorbing their factor two proves the theorem with $\eta = \min\{G/4, \alpha/4\}$. The constants depend on the fixed supercritical α, and no statement at a converging critical ratio is used in the proof. □

G Socle-visible actions and the exact leading population law

The residual action entropy in the branch upper bound can be coupled to a loss of ordinary two-cohomology. This coupling closes the leading-order comparison of Appendix F for all bounded values of $r/2^{d/2}$, including the supercritical regime. Fix $k = \mathbb{F}_q$ of characteristic two, $q \ge 4$, and retain $R, K, d, n = 2^d, w_0 = n - 2$ and $C_j = \binom{j}{2}$. Write B_h for the branch with derived dimension $w_0 + h$ and quotient dimension $d + s$, where $h + s = r$. All cohomology in this appendix is ordinary Chevalley–Eilenberg cohomology of the indicated abelian quotient. The one-direction cohomology calculation is the classical codimension-one exact sequence of [7, Proposition 1].

G.1 Restriction to the core socle

Lemma G.1 (The socle-restriction kernel)**.** *Let W be any normalized bounded-defect core module supplied by Lemma C.2. Put*

$$S = \mathrm{Soc}_R(W), \qquad m = \dim_k S, \qquad t = h + 1 - m.$$

Then $1 \le m \le h+1$, and the restriction map $\mathrm{End}_R(W) \to \mathrm{End}_k(S)$ has kernel of dimension at most

$$e_{h,m} = w_0 + h(d-1) + tm. \tag{127}$$

Proof. Use the two exact sequences in (96), written here as

$$0 \longrightarrow F \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N \longrightarrow 0, \qquad \dim F + \dim N = h.$$

Let $a = \dim(S \cap F)$, $b = \dim(S \cap W_0)$, and $\epsilon = b - a$. The image of $S \cap W_0$ in K is annihilated by the maximal ideal of R, hence lies in $\mathrm{Soc}_R K = k\tau$. Thus $\epsilon \in \{0, 1\}$. The quotient by S has the more precise filtration

$$0 \longrightarrow F/(S \cap F) \longrightarrow W_0/(S \cap W_0) \longrightarrow K/k^{\epsilon} \longrightarrow 0, \qquad W/(W_0 + S) \simeq N/\operatorname{im}(S \to N).$$

Here k^{ϵ} is either zero or the socle $k\tau$ of K. The two small factors have combined length

$$(\dim F - a) + (\dim N - m + b) = h - m + \epsilon = t - 1 + \epsilon.$$

Refine them by composition series. Subadditivity of $\mathrm{Hom}_R(-, W)$, and precomposition with the quotient map $K \longrightarrow K/k^{\epsilon}$, give

$$\dim \mathrm{Hom}_R(W/S, W) \le \dim \mathrm{Hom}_R(K, W) + (t - 1 + \epsilon)m.$$

Target subadditivity for the original two extensions gives

$$\dim \mathrm{Hom}_R(K, W) \le \dim \mathrm{End}_R(K) + h \dim \mathrm{Hom}_R(K, k) = w_0 + h(d-1).$$

The kernel of restriction is precisely $\mathrm{Hom}_R(W/S, W)$. Since $\epsilon \le 1$, the result follows. This argument uses K as a subquotient; it does not assume that K embeds in W. □

Lemma G.2 (The action–cohomology loss)**.** *Let the extra quotient $V = k^s$ act on W through commuting square-zero R-endomorphisms $T_1, \dots, T_s$. Denote their restrictions to S by Q_a, and put*

$$\rho = \max_{v \in k^s} \operatorname{rank}\Big(\sum_a v_a Q_a\Big), \qquad A_{d,h,s} = d + C_d + s(d+1) + C_s + hC_{d+s}.$$

Then $0 \le \rho \le \lfloor m/2 \rfloor$, and

$$\dim_k H^2(U_0 \oplus V, W) \le A_{d,h,s} - (t + \rho)C_s. \tag{128}$$

Proof. Take core cohomology first in the ordinary CE double complex. Its total degree-two term bounds the resulting two-cohomology by

$$\dim H^0(V, H^2(U_0, W)) + \dim H^1(V, H^1(U_0, W)) + \dim H^2(V, S).$$

Lemma C.2 bounds the first two terms by $d + C_d + hC_d$ and $s(d + 1 + hd)$, respectively.

If $s \ge 1$, choose an actual k-rational quotient direction attaining rank ρ on S. On this one-dimensional abelian quotient both degree-zero and degree-one cohomology have dimension $m - \rho$. Taking its cohomology first, then that of a complementary $(s-1)$-space, gives

$$\dim H^2(V, S) \le \big(C_{s-1} + s - 1\big)(m - \rho) = C_s(m - \rho).$$

For $s = 0$ this term is zero directly. Summing gives (128). Neither spectral sequence is assumed to collapse. All linear combinations of the Q_a's are square zero, so the stated range of ρ follows. Its definition uses the finite base field, not an algebraic closure. □

G.2 Rank-sensitive action coverings

Lemma G.3 (A common-flag rank covering)**.** *The number of ordered commuting square-zero s-tuples of m-square matrices whose span has maximum rank ρ over k is at most*

$$K_{m,s,\rho} := [m]_q!\, 2^{ms} q^{s\rho(m-\rho)}, \qquad [m]_q! = \prod_{j=1}^{m} \frac{q^j - 1}{q - 1}. \tag{129}$$

For $\rho = 0$ one may replace $K_{m,s,0}$ by one. Moreover

$$\log_q K_{m,s,\rho} \le s\rho(m-\rho) + \tfrac{1}{2}m(m-1) + ms \log_q 2 + m \log_q (1 - q^{-1})^{-1}. \tag{130}$$

Proof. Let $a_{m,j}$ count strictly upper triangular square-zero matrices of rank j. The classical rank recurrence is

$$a_{m,j} = q^j a_{m-1,j} + (q^{m-j} - q^{j-1}) a_{m-1,j-1}.$$

Indeed a last column v appended to A must lie in $\ker A$. If $\operatorname{rank} A = j$, the rank remains j for the q^j choices in $\operatorname{im} A$; if $\operatorname{rank} A = j-1$, it increases for the $q^{m-j} - q^{j-1}$ choices outside $\operatorname{im} A$. Induction gives $a_{m,j} \le \binom{m}{j} q^{j(m-j)}$, hence

$$\sum_{j=0}^{\rho} a_{m,j} \le 2^m q^{\rho(m-\rho)}.$$

Commuting nilpotent operators have a common complete flag over k: their common kernel contains a rational line, after which one argues inductively on the quotient. They are strictly triangular relative to that flag. There are $[m]_q!$ complete flags. For each flag, ignoring mutual commutation and counting each generator independently gives (129). This is a covering argument and assumes no free flag action. The estimate $[m]_q! \le q^{m(m-1)/2}(1 - q^{-1})^{-m}$ gives (130). □

Classical input. The triangular rank recurrence is Kirillov's, as recorded in [9, p. 1, equation *(Sasha)*]. The common-flag covering and the displayed inequality are elementary consequences. The population bound below couples this classical matrix count to the socle-restriction kernel and the rank-sensitive ordinary-cohomology loss.

Proposition G.4 (A full branch covering)**.** *Put*

$$S_e(s) = \sum_{j=0}^{\min(s,e)} \begin{bmatrix} e \\ j \end{bmatrix}_q, \qquad M_h(d) = (h+1) q^{hC_d + dh^2 + h}.$$

For every $h, s \ge 0$, the full branch satisfies

$$B_h \le M_h(d) q^{A_{d,h,s}} \sum_{m=1}^{h+1} S_{e_{h,m}}(s) \sum_{\rho=0}^{\lfloor m/2 \rfloor} K_{m,s,\rho}\, q^{-(h+1-m+\rho)C_s}. \tag{131}$$

Terms $\rho > 0$ may be omitted when $s = 0$.

Proof. The module census in Lemma C.2 supplies at most $M_h(d)$ representatives. Fix one of them, of socle dimension m, and let E_0 be the kernel of $E = \operatorname{End}_R(W) \to \operatorname{End}_k(S)$. Choose a vector-space section of $E \to E/E_0$. Each action then has a kernel coordinate and a visible socle coordinate.

Normalize the kernel-coordinate map $k^s \to E_0$ by $\mathrm{GL}_s(k)$: for its image choose a prescribed basis, followed by zeros. This supplies $S_{\dim E_0}(s)$ choices, at most $S_{e_{h,m}}(s)$ by Lemma G.1.

The same quotient-coordinate change transforms the visible tuple; allowing all such tuples still covers every algebra. For a fixed socle tuple the visible coordinate has a unique lift to the chosen complement of E_0, so no additional factor is required. Lemma G.3 supplies its count.

For each action tuple, ordinary two-cohomology covers every abelian Lie extension up to section changes. Lemma G.2 bounds the number of these classes. The class and derived-generation conditions impose further restrictions and may be dropped in an upper covering. Every unlabelled Lie type admits a normalized presentation, so multiplying these coverings, then summing over m and ρ, proves the result. No orbit size is divided out and no module splitting is assumed. □

G.3 The branch envelope

Define the continuous function

$$\Psi(\beta,\gamma)=\begin{cases}\gamma+\frac{1}{2}\beta\gamma^2, & \beta\le\gamma/2,\\ \gamma+\frac{1}{3}\beta^2\gamma+\frac{1}{6}\beta\gamma^2+\frac{1}{12}\gamma^3, & \beta\ge\gamma/2.\end{cases}\tag{132}$$

Proposition G.5 (Uniform leading branch bound)**.** *For each fixed $A_*>0$, uniformly for integers $h,s\ge 0$ with $h+s\le A_*\sqrt{n}$,*

$$\log_q B_h\le n^{3/2}\Psi(h/\sqrt{n},s/\sqrt{n})+O_{q,A_*}(dn).\tag{133}$$

Proof. In (131), put $\beta=h/\sqrt{n}$, $\gamma=s/\sqrt{n}$, $\zeta=m/\sqrt{n}$, and $\xi=\rho/\sqrt{n}$. The module census costs $O_{A_*}(dn)$. Moreover

$$e_{h,m}=n+(h+1-m)m+O_{A_*}(d\sqrt{n}),$$

so eventually $2s\le e_{h,m}$ uniformly. The Gaussian estimate of Appendix E gives $S_e(s)\le\frac{3}{2}(s+1)q^{s(e-s)}$. The flag and triangular prefactors in (130) cost $O_{q,A_*}(n)$. All remaining terms involving d, the difference $C_s-s^2/2$, and the number of summands contribute $O_{q,A_*}(dn)$. The resulting leading expression is

$$F(\beta,\gamma,\zeta,\xi)=\gamma+\tfrac{1}{2}\beta\gamma^2+\gamma(\beta-\zeta)\zeta-\tfrac{1}{2}(\beta-\zeta+\xi)\gamma^2+\gamma\xi(\zeta-\xi).$$

The true range $0\le\zeta\le\beta+n^{-1/2}$ can be replaced by $0\le\zeta\le\beta$ at a cost $O_{A_*}(n)$, since F has uniformly bounded first derivatives on the relevant compact set. The range of ξ is $0\le\xi\le\zeta/2$.

For $\gamma>0$, complete squares in

$$F/\gamma=1+\zeta(\beta+\gamma/2)-\zeta^2+\xi(\zeta-\gamma/2)-\xi^2.$$

The maximizing ξ is $(\zeta-\gamma/2)_+/2$. If $\beta\le\gamma/2$, the maximizing ζ is β; if $\beta\ge\gamma/2$, it is $2\beta/3+\gamma/6$. The resulting values are exactly (132). For $\gamma=0$ all leading expressions are zero. This proves the uniform estimate. □

G.4 The exact leading law and the typical allocation

Put

$$\varphi(\alpha)=\begin{cases}\alpha, & 0\le\alpha\le\sqrt{2},\\ \dfrac{\alpha^3+9\alpha+(\alpha^2+6)^{3/2}}{27}, & \alpha\ge\sqrt{2},\end{cases}\tag{134}$$

and

$$\beta_*(\alpha)=\begin{cases}0, & 0\le\alpha\le\sqrt{2},\\ \dfrac{2\alpha-\sqrt{\alpha^2+6}}{3}, & \alpha>\sqrt{2}.\end{cases}\tag{135}$$

Theorem G.6 (The full leading population law)**.** *For every fixed $A_* > 0$, uniformly for integers $1 \le r \le A_*\sqrt{n}$,*

$$\log_q B_{d+1,r}(q) = n^{3/2}\varphi(r/\sqrt{n}) + O_{q,A_*}(dn). \tag{136}$$

In particular, if $r/\sqrt{n} \to \alpha > 0$, then

$$\frac{\log_q B_{d+1,r}(q)}{n^{3/2}} \longrightarrow \varphi(\alpha).$$

For each $\epsilon > 0$, there is $\delta > 0$ such that along this sequence the proportion of types satisfying

$$\left|\frac{\dim_k L' - w_0}{\sqrt{n}} - \beta_*(\alpha)\right| \ge \epsilon$$

is at most $q^{-\delta n^{3/2}}$ for all sufficiently large d. Thus the allocation of surplus to the derived ideal has a unique limiting value. This assertion concerns scaled defects and makes no claim of exact minimum-derived dominance in a shrinking window.

Proof. Fix $\alpha \ge 0$ and write $\gamma = \alpha - \beta$. On the second branch of Ψ, equivalently $0 \le \gamma \le 2\alpha/3$, its value is

$$E_\alpha(\gamma) = \gamma + \alpha^2\gamma/3 - \alpha\gamma^2/2 + \gamma^3/4.$$

It is strictly increasing because

$$E'_\alpha(\gamma) = 1 + \tfrac{3}{4}(\gamma - 2\alpha/3)^2 > 0.$$

Hence its maximum occurs at the boundary with the first branch. On the first branch the value is

$$L_\alpha(\gamma) = \gamma + \tfrac{1}{2}\alpha\gamma^2 - \tfrac{1}{2}\gamma^3, \qquad 2\alpha/3 \le \gamma \le \alpha.$$

Its unique maximum is at $\gamma = \alpha$ when $\alpha \le \sqrt{2}$, and otherwise at $\gamma_* = (\alpha + \sqrt{\alpha^2 + 6})/3$. For $\alpha > \sqrt{2}$ this point is strictly greater than $2\alpha/3$. Substitution gives $\varphi(\alpha)$, and the unique complementary value is $\beta_*(\alpha)$.

Apply Proposition G.5 and sum the $r + 1$ branches. This gives the upper bound in (136) with its stated uniform error.

For the lower bound put $\alpha = r/\sqrt{n}$ and choose $h = \lfloor \beta_*(\alpha)\sqrt{n} \rfloor$. Then $0 \le h \le r$. If $h = 0$, Lemma E.2 gives $\log_q B_0 \ge rn - O_{A_*}(dn)$. When this rounding occurs with $\alpha > \sqrt{2}$, replacing the real optimizer by zero changes the leading expression by at most $O_{A_*}(n)$, by the uniform Lipschitz bound for L_α on $0 \le \alpha \le A_*$.

If $h \ge 1$, put $s = r - h$. Uniformly in this case, $s \le A_*\sqrt{n}$, and the supercritical optimizer has $s \ge 2r/3 - O(1)$. For all sufficiently large d, $1 \le s \le D_{\mathrm{par}}$ and $h \le C_{d+s} - (d-1)$. Thus Proposition F.1 applies and gives

$$\log_q B_h \ge sn + \tfrac{1}{2}hs^2 - O_{q,A_*}(dn).$$

Changing the maximizing real h by at most one costs $O_{A_*}(n)$. This is the required lower bound, uniformly through $\alpha = \sqrt{2}$.

Finally, the maximizer of the continuous compact-domain envelope is unique. For fixed $\alpha > 0$ and $\epsilon > 0$, its value on the closed subset $|\beta - \beta_*(\alpha)| \ge \epsilon$ has a strictly positive gap from $\varphi(\alpha)$, unless that subset is empty. Continuity preserves a smaller fixed gap when $r/\sqrt{n}$ is sufficiently close to α. The uniform branch upper, the total lower, and $\log_q(r+1) = o(n^{3/2})$ then prove the exponential concentration estimate. □

What the argument establishes. For $\alpha > \sqrt{2}$, the typical derived surplus is $\beta_*(\alpha)\sqrt{n} + o(\sqrt{n})$; its leading value is positive. The matching lower is furnished by the full-central family, whereas the upper covers every bounded-defect derived module, including nonsplit modules. Equality of the leading exponents does not assert that the full-central family has density one, or that all typical algebras are indecomposable. These finer properties require information below the $n^{3/2}$ scale.

H The first positive derived defect

H.1 The modules and actions at one derived defect

Put $n = 2^d$, $w_0 = n - 2$, $C_d = \binom{d}{2}$, $D_{\mathrm{par}} = n - d - 3$, and $s = r - 1$, with $d \geq 4$. All fields below have characteristic two and fixed order $q \geq 4$. Use R, K, I from (91) and put $A = R/I$, $J = \mathfrak{m}R$, and $\tau = t_0 \cdots t_{d-1}$.

The third module needed below is denoted E. As a vector space it is $K \oplus kz$. On the monomial basis y_S, $S \cap \{1, \ldots, d-1\} \neq \varnothing$, its action agrees with multiplication in K, except that

$$t_1 y_{\{1\}} = z, \qquad \mathfrak{m}z = 0.$$

In particular, the positive generators $y_i = y_{\{i\}}$, $i > 0$, satisfy $t_1 y_1 = z$, $t_i y_i = 0$ for $i > 1$, and $t_i y_j + t_j y_i = 0$ for distinct positive indices. The t_0-direction is free on the K-part.

Lemma H.1 (Three module types at one defect)**.** *Every normalized core module of derived defect one is isomorphic, after a linear change of normalized core coordinates, to exactly one of*

$$K \oplus k, \qquad J, \qquad E.$$

For the core quotient $U_0 = k^d$, the first and third modules have

$$\dim H^0 = 2, \qquad \dim H^1 = 2d + 1, \qquad \dim H^2 = d + 2C_d.$$

For J, these dimensions are respectively $1, d + 1, d + C_d$.

Proof. The two extensions in (96), with $h = 1$, give two cases. If the core derived module is K, the full module is an extension of k by K. Since $\mathrm{Ext}^1_R(k, K)$ has dimension one, it is either split or the unique nonsplit type. The sequence $0 \to K \to J \to k \to 0$ is nonsplit: its socle is $k\tau \subset K$, so no trivial summand lifts the quotient. This gives the first two modules.

Otherwise the core derived module is the whole module, and there is an extension $0 \to kz \to W \to K \to 0$. The minimal presentation of K, obtained by extending the presentation of the augmentation ideal in $k[t_1, \ldots, t_{d-1}]/(t_i^2)$, identifies its extension class with coefficients q_i, q_{ij} in the relations

$$t_i y_i = q_i z, \qquad t_i y_j + t_j y_i = q_{ij} z \quad (1 \leq i < j < d).$$

There are $(d-1) + \binom{d-1}{2} = C_d$ coefficients; the t_0-direction introduces no further relation. The ordinary-cohomology connecting map sends the normalized core-star cocycle to $\sum_{i<j} q_{ij} z\, e_0^* \wedge e_i^* \wedge e_j^*$. All the remaining core cohomology representatives have coefficient degree at least two and contribute zero to this map. Since the star coefficient is nonzero, a core cocycle can lift only if every $q_{ij} = 0$. The diagonal vector (q_i) is either zero, giving $K \oplus k$, or can be normalized to $(1, 0, \ldots, 0)$ by $\mathrm{GL}_{d-1}(k)$ and rescaling z. Here the field is perfect, so the Frobenius twist causes no obstruction. The latter module is E. These types are distinct, as their top/socle dimensions are respectively $(d, 2), (d, 1), (d-1, 2)$.

For E, the connecting maps on $H^0(U_0, K)$, $H^1(U_0, K)$, and $H^2(U_0, K)$ all vanish. The top socle representative lifts to a socle vector. The displayed $H^1(K)$ representatives can be taken to be

$$\zeta_*(e_0) = 0, \quad \zeta_*(e_i) = t_0 t_i \ (i > 0), \qquad \zeta_j(e_i) = \mathbf{1}_{i=j}\tau/t_j \ (0 \le j < d).$$

The first is the connecting image of the socle of R/K, and the last d classes map to the standard Koszul basis for $H^1(U_0, R)$. This also makes the dimension $d+1$ and the coefficient-degree bound explicit. All these coefficients have degree at least two; the only additional action in E occurs when t_1 multiplies the degree-one vector y_1. In the basis of Lemma B.3, the remaining $H^2(K)$ representatives are β_κ, with coefficient degree two, and β_λ, with coefficient degree $d-2 \ge 2$. Consequently the only potentially affected representative is the star, whose connecting value is zero because $q_{ij} = 0$. The long exact sequence therefore gives the three claimed dimensions, also immediate for $K \oplus k$. For J, use $0 \to J \to R \to k \to 0$. The core Koszul complex of R has $\dim H^i = \binom{d}{i}$; its representatives have coefficient degree $d-i$. Thus augmentation induces zero on H^i for $i = 0, 1, 2 < d$, and the same long exact sequence gives the dimensions for J. □

For $K \oplus k$, let N_i send the top generator t_i of K to z, killing $\mathfrak{m}K$ and z, and let $\Theta(z) = \tau$, $\Theta(K) = 0$. For E, define $N_i(y_j) = \mathbf{1}_{i=j} z$, $N_i(\mathfrak{m}E) = 0$, and

$$\Theta(y_1) = \tau/t_1, \qquad \Theta(z) = \tau, \qquad \Theta(y_S) = 0 \quad (S \ne \{1\}).$$

These are R-module maps.

Lemma H.2 (Admissible extra actions)**.** *On $K \oplus k$ or E, every additional action compatible with a generating normalized core cocycle has, after subtracting its core linear part, the form*

$$T_a = p_a + v_a\Theta, \qquad p_a \in \mathcal{P}_d, \quad v_a \in k.$$

On J, every additional action is polynomial, in a space of dimension $D_{\mathrm{par}}+1$, after subtracting its core linear part.

Proof. For the split module the usual block description gives, after nilpotency removes the two constant diagonal blocks, $T = p + \sum_i a_i N_i + v\Theta$. For E, one has

$$\mathrm{End}_R(E) = A \oplus \mathrm{Span}_k\{N_i : 1 \le i < d\} \oplus k\Theta.$$

For completeness its dimension is $w_0 + d$. In $0 \to k \to E \to K \to 0$, the map $\mathrm{End}(K) \to \mathrm{Ext}^1(K, k)$ sends multiplication by u to $u_0 q$. Consequently $\dim \mathrm{Hom}(K, E) = w_0 + d - 2$. Restriction $\mathrm{End}(E) \to \mathrm{Soc}(E)$ has image dimension two, realized by the identity and Θ. This proves the dimension; the displayed independent maps give all endomorphisms. Its residue scalar is p_0, which is zero for a square-zero action. Subtracting core linear multiplication leaves $p \in \mathcal{P}_d$.

The core/core/extra Jacobi equations require $T_*[\beta_0] = 0$ in $H^2(U_0, W)$. Positive-degree polynomial multiplication induces zero on cohomology. The map Θ also induces zero on core H^2. Indeed its effect on the lifted star for E is $(\tau/t_1)e_0^* \wedge e_1^*$, which is the boundary of the one-cochain having value $\tau/(t_0 t_1)$ on e_1. It kills all higher K-coefficient representatives. Its effect on a central two-form is a τ-valued form, also a boundary: for $i \ne j$, the cochain with value τ/t_i on e_j has precisely that boundary. These observations also cover the split module.

By contrast, N_i sends the normalized star to the central class $z e_0^* \wedge e_i^*$. These classes are independent in $H^2(U_0, W)$: the inclusion from $H^2(U_0, k)$ is injective because the preceding $H^1(K)$ connecting map is zero. All other normalized core representatives are killed by N_i. Jacobi therefore forces every $a_i = 0$. For J, finite duality gives $\mathrm{End}_R(J) = R/(\tau)$; nilpotency removes its constant term, and core linear subtraction leaves dimension $D_{\mathrm{par}} + 1$. □

Lemma H.3 (The cohomology cost of a nonpolynomial action)**.** *For $W = K \oplus k$ or E, polynomial actions in s extra directions give*

$$\dim H^2(U_0 \oplus k^s, W) = d + 2C_d + s(2d+1) + 2\binom{s}{2}.$$

If at least one coefficient v_a is nonzero, the dimension is instead

$$d + 2C_d + s(2d+1) + \binom{s}{2}.$$

Thus the exact loss is $\binom{s}{2}$, not a loss in the first one-direction H^2 calculation.

Proof. Eliminate polynomial actions by the ordinary R-linear Koszul change of cochain generators. Since Θ is R-linear, this leaves the residual actions $v_a\Theta$. If nonzero, a quotient basis change makes them $\Theta, 0, \dots, 0$. On core cohomology, Θ_* has rank one on H^0 and rank zero on H^1, H^2. The H^2 assertion was proved above. On H^1, lifted K-representatives have degree at least two and are killed; central representatives map to τe_i^*, the boundary of τ/t_i. On H^0, it sends z to τ. The classical codimension-one exact sequence [7, Proposition 1], or its mapping cone, consequently gives

$$H^0 : 1, \qquad H^1 : (2d+1) + 1, \qquad H^2 : (d + 2C_d) + (2d+1)$$

as its dimensions. Tensoring with the exterior cochains in the remaining $s-1$ zero directions gives the second formula; the all-zero residual action gives the first. □

H.2 Polynomial base orbits and the one-defect count

For E, every identity-linear substitution $\sigma(t_i) = t_i + u_i$, $u_i \in \mathfrak{m}^2 R$, preserving K, I, admits a semilinear lift to E. Choose polynomial lifts $\widetilde{a} \in E$ for $a \in K$. For $a, b \in K$, the defining relations give

$$a\widetilde{b} + b\widetilde{a} = 0, \qquad a\widetilde{a} = q(a_{[1]})z,$$

where q is the diagonal Frobenius-linear form. The map $y_i \mapsto \widetilde{\sigma(t_i)}$, $z \mapsto z$, extended σ-semilinearly, respects all presentation relations because σ has identity linear part. It induces $\sigma|_K$ on the quotient and the identity on the kernel, so is invertible. The split module has the analogous lift. Both modules have annihilator precisely I: the quotient K forces the annihilator to lie in I, and the explicit action shows that I kills E (and also $K \oplus k$). Thus all of these statements depend only on the classes of the polynomials modulo I, as required by the base action.

It follows that the base group

$$G_0 = k^{ds} \rtimes \mathrm{GL}_s(k)$$

acts by isomorphisms between the corresponding polynomial-action presentation families. A genuine action on a chosen cohomology slice is unnecessary for the following upper covering: choose one base in each base orbit, transport all presentations to it, then normalize their core-star coefficient and central star values. The latter $d-1$ values are uniquely removed by $1 + \sum_i a_i N_i$, which commutes with polynomial actions. Each resulting fiber has at most

$$q^{F_s}, \qquad F_s = 2C_d + s(2d+1) + 2\binom{s}{2}$$

representatives. No orbit-size division on that fiber is used. The uniform base stabilizer-sum bounds proved for Theorem 6.4 apply to the subgroup G_0. Hence independent polynomial frames have

$$(1+o(1))\frac{\prod_{j<s}(q^{D_{\mathrm{par}}}-q^j)}{q^{ds}|\operatorname{GL}_s(q)|}$$

base orbits, uniformly in $2d \le s \le D_{\mathrm{par}}/(4d)$. The two polynomial module types therefore contribute at most

$$q^{sn-3s+2C_d+O_q(1)} \tag{137}$$

classes in that range.

The other presentations have smaller covering bounds. Action-subspace normalization in the $(D_{\mathrm{par}}+1)$-dimensional space $\mathcal{P}_d + k\Theta$, combined with Lemma H.3 and the same star normalizations, gives at most

$$\tfrac{3}{2}(s+1)q^{sn+ds-s^2/2-3s/2+2C_d}$$

classes with a nonpolynomial action, for each of the two module types. Here the star normalization is valid even though it need not fix the individual action tuple. If $N=\sum a_iN_i$, then $N^2=0$, $N\Theta=0$, and $\Theta N=\sum a_iM_{\tau/t_i}$. Conjugation by $1+N$ therefore sends

$$p+v\Theta \longmapsto p+v\sum_i a_iM_{\tau/t_i}+v\Theta.$$

It preserves the polynomial-plus-Θ space and the coefficient v, because $\tau/t_i\in\mathcal{P}_d$. One first normalizes the core star and its central star values, and then chooses a basis for the resulting action subspace. This covers every type by a normalized fiber of dimension $2C_d+s(2d+1)+\binom{s}{2}$; no stabilizer assumption is used here. For J, using all core and extra cohomology without a generation restriction gives the upper bound

$$\tfrac{3}{2}(s+1)q^{sn-s^2/2-3s/2+d+C_d}.$$

Rank-deficient polynomial tuples are bounded by $q^{F_s}\sum_{j<s}\begin{bmatrix}D_{\mathrm{par}}\\ j\end{bmatrix}_q$; relative to the faithful polynomial main covering, their ratio is $q^{-D_{\mathrm{par}}+ds+2s+O_q(\log s+d^2)}$. Indeed every tuple of rank $j<s$ can first be put, by an extra quotient basis change, into a tuple consisting of a basis of its j-dimensional polynomial action space and $s-j$ zero directions. All s quotient directions and the complete fiber are retained; a choice of a Gaussian action subspace covers every such presentation. Thus the independent-frame count and this rank-deficient covering together exhaust the polynomial cases.

Theorem H.4 (One-defect enumeration at the transition scale)**.** *Fix $0<a<b<\infty$. Uniformly for integers $a\sqrt{n}\le r\le b\sqrt{n}$, as $d\to\infty$, with $s=r-1$,*

$$\log_q B_1 = sn-3s+O_{q,a,b}(d^2), \qquad \log_q(B_1/B_0) = -n+\binom{r+d}{2}+O_{q,a,b}(d^2).$$

The nonpolynomial, J-module, and rank-deficient presentations have total type proportion $O_{q,a,b}(q^{-\kappa n})$, for some $\kappa=\kappa(a,b)>0$. Almost all one-defect types consequently admit a faithful polynomial presentation with module $K\oplus k$ or E.

Proof. The full-central-cocycle family in Proposition F.1, with $h=1$, gives $\log_q B_1 \ge sn-3s-d-4$. For all sufficiently large d, the entire indicated range lies in $2d\le s\le D_{\mathrm{par}}/(4d)$. Thus (137) supplies a uniform upper bound $sn-3s+2C_d+O_q(1)$ for the faithful polynomial

cases. Relative to the displayed lower bound, the logarithm of the nonpolynomial covering is at most

$$-\tfrac{1}{2}s^2 + ds + \tfrac{3}{2}s + 2C_d + d + 4 + \log_q\big(\tfrac{3}{2}(s+1)\big),$$

and the J covering has logarithmic ratio at most

$$-\tfrac{1}{2}s^2 + \tfrac{3}{2}s + 2d + C_d + 4 + \log_q\big(\tfrac{3}{2}(s+1)\big).$$

Since $s \geq a\sqrt{n}/2$ eventually, each is at most $-\kappa n$ for a fixed positive κ, uniformly in the range. The rank-deficient covering has logarithmic ratio $-D_{\mathrm{par}} + ds + 2s + O_q(d^2 + \log s) \leq -n/2$ eventually, also uniformly. Summing these coverings proves the first formula and the assertion about the exceptional presentations.

Finally Theorem 6.4, including the Gaussian and general-linear product estimates, gives uniformly

$$\log_q B_0 = rn - dr - \tfrac{1}{2}r^2 - \tfrac{5}{2}r + C_d - (d-1)^2 - \log_q(q-1) + O_q(1).$$

Subtract this from the first formula, use $s = r - 1$, and absorb the remaining terms of order d^2. The result is the stated formula for B_1/B_0. □

In particular, the comparison changes sign within $O_{q,a,b}(d^2/\sqrt{n})$ of $r = \sqrt{2n} - d + \frac{1}{2}$. For every fixed $\varepsilon > 0$, the ratio B_1/B_0 tends to zero on the side $r \leq \sqrt{2n} - d + \frac{1}{2} - \varepsilon$, and tends to infinity on the side $r \geq \sqrt{2n} - d + \frac{1}{2} + \varepsilon$, within the range of the theorem. This compares the first positive-defect branch with the minimum-derived branch. It does not by itself control defects $h \geq 2$ or establish a whole-layer lower-side threshold.

I The discrete critical transition

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. Retain R, K, I from (91), and put

$$n = 2^d, \quad w_0 = n - 2, \quad D_{\mathrm{par}} = n - d - 3, \quad C_j = \binom{j}{2}, \quad A = R/I, \quad \tau = t_0 \cdots t_{d-1}, \quad d \geq 4.$$

The branch B_h has derived dimension $w_0 + h$, quotient dimension $d + s$, and $h + s = r$. All cohomology is ordinary Chevalley–Eilenberg cohomology. Write

$$T = \sqrt{2n}, \quad R_d^* = T - d + \tfrac{1}{2}, \quad x_{d,r} = r - R_d^*, \quad E_{d,r} = -n + C_{d+r}, \quad z'(L) = \dim_k(Z(L) \cap L').$$

I.1 Maximum core socle and polynomial rigidity

Proposition I.1 (The modules at maximum core socle)**.** *Every normalized core module W with $\dim W = w_0 + h$ and $\dim \mathrm{Soc}_R W = h + 1$ is, after a linear core-coordinate change, a module $E_Q = K \oplus Z$, $Z = k^h$, as follows. The core kills Z and acts by usual monomial multiplication on K, except that*

$$t_i y_i = q_i \in Z \quad (1 \leq i < d), \qquad t_i y_j + t_j y_i = 0 \quad (i \neq j),$$

where y_i lifts t_i. The map $Q : k^{d-1} \to Z$, $Q(e_i) = q_i$, is classified by its rank $0 \leq u \leq \min(h, d-1)$. Moreover $\mathrm{Ann}_R E_Q = I$.

Proof. In the two extensions (96), $0 \to F \to W_0 \to K \to 0$ and $0 \to W_0 \to W \to N \to 0$, one has

$$\dim \mathrm{Soc}\, W \leq \dim \mathrm{Soc}\, W_0 + \dim \mathrm{Soc}\, N \leq \dim \mathrm{Soc}\, F + 1 + \dim \mathrm{Soc}\, N \leq h + 1.$$

Equality forces F, N to be trivial and $\operatorname{Soc} W \to N$ to be surjective. A section into the socle splits the second extension, leaving $0 \to Z \to W \to K \to 0$. The minimal presentation of K has relations $t_i y_i$ and $t_i y_j + t_j y_i$; the t_0-factor is free. Thus this extension has diagonal values q_i and mixed values q_{ij}. The core-star connecting class is $\sum_{i<j} q_{ij} e_0^* \wedge e_i^* \wedge e_j^*$; all other core representatives have coefficient degree at least two and map to zero. The nonzero generating star therefore forces $q_{ij} = 0$. A positive core change M transforms the diagonal map by $M^{[2]}$; Frobenius is bijective over k, so row and column changes leave only its rank. Distinct ranks have distinct top dimensions $d - 1 + h - u$. Finally $E_Q/Z = K$ forces its annihilator into I, and the displayed monomial action shows that I kills E_Q. □

Lemma I.2 (Low core cohomology)**.** *The connecting maps from $H^j(U_0, K)$ to $H^{j+1}(U_0, Z)$ vanish for $j = 0, 1, 2$. Consequently*

$$\dim H^0(U_0, E_Q) = h + 1, \quad \dim H^1(U_0, E_Q) = d + 1 + hd, \quad \dim H^2(U_0, E_Q) = d + C_d + hC_d,$$

and $H^2(U_0, Z) \to H^2(U_0, E_Q)$ is injective.

Proof. The socle vector τ lifts. Representatives for $H^1(U_0, K)$ are $\zeta_*(e_i) = t_0 t_i$ for $i > 0$, $\zeta_*(e_0) = 0$, and $\zeta_j(e_i) = \mathbf{1}_{i=j}\tau/t_j$. Their coefficients have degree at least two, as do the κ- and λ-representatives in Lemma B.3. These lift unchanged, while the star lifts because the mixed relations vanish. The long exact sequence proves the assertions; no higher-degree connecting map is being used. □

Proposition I.3 (Polynomial actions and the normalized fiber)**.** *An R-endomorphism of E_Q that kills its socle and satisfies $T_*[\beta] = 0$ for a generating core cocycle is multiplication by a unique element of* $\operatorname{rad} A$*. Subtracting core linear parts therefore puts every extra action in $\mathcal{P}_d = \operatorname{rad}(A)^2$, of dimension D_{par}. For s such actions, put*

$$A_{d,h,s} = d + C_d + s(d+1) + C_s + hC_{d+s}, \qquad F_{d,h,s} = A_{d,h,s} - 1 - h(d-1).$$

The full ordinary two-cohomology has dimension $A_{d,h,s}$, and normalized generating presentations are covered, up to module isomorphism, by

$$q^{F_{d,h,s}} \tag{138}$$

fiber representatives.

Proof. Since T kills Z, it induces multiplication by $p \in A$ on K; killing τ forces $p_0 = 0$. Subtract that actual polynomial endomorphism. The difference is $N_\Lambda(y_i) = \Lambda_i \in Z$, killing Z and nonsingleton monomials. Positive multiplication is null-homotopic on the core complex, whereas

$$(N_\Lambda)_*[\beta] = \sum_i \Lambda_i e_0^* \wedge e_i^*.$$

These classes are independent by Lemma I.2, so compatibility forces $N_\Lambda = 0$. Faithfulness on the quotient K gives uniqueness.

Write $p_a = \sum_i t_i a_{ia}$, $a_{ia} \in \mathfrak{m}R$. The triangular R-linear change of exterior cochain generators eliminates the polynomial actions, giving

$$H^2(U_0 \oplus k^s, E_Q) \simeq H^2(U_0, E_Q) \oplus \big(H^1(U_0, E_Q) \otimes (k^s)^*\big) \oplus \big(\operatorname{Soc}_R(E_Q) \otimes \Lambda^2(k^s)^*\big).$$

This gives $A_{d,h,s}$. Scalar multiplication normalizes the nonzero star coefficient to one. The commuting automorphisms $1 + N_\Lambda$ then translate its $h(d-1)$ central star coordinates independently. Their action on this value-one fiber is free, since restriction to core cohomology detects every nonzero Λ. Its orbit representatives have the cardinality (138). No further stabilizer is divided out. □

I.2 Transport over polynomial base orbits

Lemma I.4 (The linear core stabilizer and base covering)**.** *For a fixed rank-u representative Q, put*

$$H_Q = \{M \in \mathrm{GL}_{d-1}(k) : M^{[2]} \ker Q = \ker Q\}.$$

Uniformly for $2d \le s \le D_{\mathrm{par}}/(4d)$, its faithful polynomial presentations have at most

$$(1+o(1))\frac{q^{F_{d,h,s}} \prod_{j<s}(q^{D_{\mathrm{par}}} - q^j)}{q^{ds} |\mathrm{GL}_s(k)||H_Q|} \tag{139}$$

types, with error independent of h, Q. Moreover

$$|H_Q| = q^{u(d-1-u)} |\mathrm{GL}_u(k)||\mathrm{GL}_{d-1-u}(k)|, \quad \log_q |H_Q| = (d-1)^2 - u(d-1-u) + O_q(1). \tag{140}$$

Writing $V_h = sn - 3s + (h-1)C_{d+s}$, the logarithm of (139) is at most

$$V_h + (2-h)(d-1) + u(d-1-u) + O_q(1). \tag{141}$$

After summing the module ranks,

$$\log_q B_h^{\mathrm{pol}} \le V_h + 2(d-1) + \log_q(\min(h, d-1)+1) + O_q(1). \tag{142}$$

The same bound includes rank-deficient polynomial tuples when $h \le d+O(1)$ and $r = T+O(d)$.

Proof. An identity-linear substitution $\sigma(t_i) = t_i + u_i$, $u_i \in \mathfrak{m}^2 R$, preserves K, I: for $z = t_1 \cdots t_{d-1}$, its only surviving increment terms give $\sigma(z) = z + a\tau$, $\sigma(\tau) = \tau$. For polynomial lifts in E_Q, the defining relations give $a\widetilde{b} + b\widetilde{a} = 0$ and $a\widetilde{a} = Q(a_{[1]}^{[2]})$. Thus $y_i \mapsto \widetilde{\sigma(t_i)}$, fixing Z, is a semilinear automorphism. For $M \in H_Q$, an invertible $C : Z \to Z$ with $CQ = QM^{[2]}$ exists: define it first on $\operatorname{im} Q$, then extend over a complement. The maps $y_i \mapsto \sum_j M_{ji} y_j$, $Z \mapsto CZ$ lift the linear substitution $\mathrm{diag}(1, M)$.

Hence the genuine polynomial base action restricts to $G_Q = k^{ds} \rtimes (H_Q \times \mathrm{GL}_s(k))$. Transport to one base per orbit and apply (138). This needs neither canonical semilinear lifts nor an action on a chosen fiber slice. The whole-base stabilizer sum in Theorem 6.4 bounds the subgroup sum term by term, uniformly in h, Q. Burnside gives (139). Frobenius identifies H_Q with the parabolic fixing a kernel of dimension $d-1-u$, proving (140). Taking logarithms gives (141); use $(2-h)(d-1) + u(d-1-u) \le 2(d-1)$ and sum the ranks.

For deficient frames use the full fiber times $\sum_{j<s} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q$. Relative to the faithful covering, including its linear core denominator, this is at most

$$q^{-D_{\mathrm{par}} + ds + 2s + (d-1)^2 + O_q(\log(s+1))} = q^{-n+o(n)}.$$

This holds uniformly in the stated critical ranges and proves the last assertion without an orbit-size assumption on deficient frames. □

The principal product formula will be used in the form

$$\log_q B_0 = rn - 3r - C_{d+r} + d - 1 - \log_q(q-1) + O_q(1). \tag{143}$$

Corollary I.5 (The refined first-defect comparison)**.** *Uniformly for $r/\sqrt{n}$ in a fixed compact positive interval, with $s = r-1$,*

$$\log_q B_1 = sn - 3s + O_q(d), \tag{144}$$

$$\log_q(B_1/B_0) = E_{d,r} + O_q(d). \tag{145}$$

More precisely, $E_{d,r} - 2d + O_q(1) \le \log_q(B_1/B_0) \le E_{d,r} + d + O_q(1)$. The compact interval may affect the sufficiently large lower bound on d, but not these bounded-error constants.

Proof. For $h = 1$, the two module ranks in (141) give constants $d - 1$ and $2d - 3$. The exceptional one-defect coverings in Appendix H.2 are exponentially smaller on this scale. The coupled lower bound and the polynomial upper therefore give $sn - 3s - d - 4 \le \log_q B_1 \le sn - 3s + 2d - 3 + O_q(1)$. Subtract (143) to obtain all assertions. □

I.3 Uniform control of the other module and action strata

For a normalized core write $m = \dim \mathrm{Soc}_R W$, $t = h + 1 - m$, and let ρ be the maximum rank of its visible extra action on that socle. Since

$$Z(L) \cap L' = \mathrm{Soc}_R W \cap \bigcap_a \ker Q_a,$$

the condition $z'(L) < h + 1$ is equivalent to $t > 0$ or $\rho > 0$, independently of the chosen normalized core. Denote these types by B_h^{bad}. Put

$$U = hC_d + dh^2 + h + A_{d,h,s} + s\big(w_0 + h(d-1) - s\big).$$

For $2s \le w_0$, $h \le s/4$, and $s \ge 40$, the full branch covering (131) implies

$$\log_q B_h \le U + O_q(\log(r+2)), \tag{146}$$

$$\log_q B_h^{\mathrm{bad}} \le U - s^2/16 + O_q(\log(r+2)). \tag{147}$$

Indeed use $S_e(s) \le \frac{3}{2}(s+1)q^{s(e-s)}$ with the upper ambient dimension $e = e_{h,m} \ge w_0$. Relative to $m = h + 1, \rho = 0$, the exponent changes by

$$t(sm - C_s) + \rho\big(s(m-\rho) - C_s\big).$$

For $t \ge 1$, the first term is at most $-t(s^2/4 - 3s/2)$. For $\rho = 0$ there is no flag factor. For $\rho \ge 1$, the second term is at most $-s^2/4 + s/2$, and the common-flag prefactor has logarithm at most

$$\tfrac{1}{2}m(m-1) + \tfrac{1}{2}ms + \tfrac{1}{4}m \le 5s^2/32 + 11s/16 + 1/4.$$

Here $\log_q 2 \le 1/2$ and $\log_q(1 - q^{-1})^{-1} \le 1/4$. The combined loss is at least $s^2/16$ for $s \ge 40$. All census and summation prefactors cost only $O_q(\log(r+2))$, proving both estimates.

Subtracting (143) gives

$$\log_q(B_h/B_0) \le \Delta_{d,n}(r,h) + O_q(\log(r+2)), \tag{148}$$

where the exact polynomial is

$$\begin{aligned}\Delta_{d,n}(r,h) = {} & h^3/2 - (r+d-1)h^2 \\ & + h\big[-n + r^2/2 + (2d - 1/2)r + d^2 - 2d + 5/2\big] \\ & + r(2d+1) + d^2 - d + 1.\end{aligned}$$

Lemma I.6 (Uniform critical cutoffs)**.** *Fix $A_0 \ge 1/2$ and restrict to $r = T - d + a$, $|a| \le A_0$. There are positive constants κ, depending only on the fixed parameters, such that*

$$\sum_{h \le r/5} B_h^{\mathrm{bad}}/B_0 \le q^{-\kappa n}, \tag{149}$$

$$\sum_{h \ge r/5} B_h/B_0 \le q^{-\kappa n^{3/2}}, \tag{150}$$

$$\sum_{h \ge d + \lceil A_0 + 3/2 \rceil} B_h/B_0 \le q^{-\kappa T}. \tag{151}$$

On the one-sided range $-A_0 \le a \le 1/2$, the last cutoff can be replaced by $d + 2$.

Proof. For $h \le r/5$, one has $s \ge 4r/5$, and $h^3/2 - rh^2 \le -9rh^2/10$. Completing a square in the remaining $O_{A_0}(dr)h + O_{A_0}(dr)$ terms gives $\Delta \le O_{A_0}(d^2T) = o(n)$. Thus (147) proves the first estimate. For $h \ge r/5$, the scaled defect stays a fixed distance from zero. The unique envelope maximizer at $r/\sqrt{n} = \sqrt{2}$, its uniform $O(dn)$ error, and the principal lower bound prove the second estimate. This use of the continuous envelope is confined to a fixed macroscopic gap.

For the third estimate substitute $r = T - d + a$ in Δ. Its coefficient of T is $-h^2 + (d + a - 1/2)h + 2d + 1$. At $h = d + k$, this equals $(a + 3/2 - k)d - k^2 + (a - 1/2)k + 1$. Take $k = \lceil A_0 + 3/2 \rceil$. This is negative eventually; if its coefficient of d vanishes, its constant is $1 - 2k \le -3$. The remaining terms are $O_{A_0}(d^3) = o(T)$. The derivative of Δ in h is negative uniformly for $d + k \le h \le r/5$: its terms are $-2Th + 3h^2/2 + (d + O_{A_0}(1))T + O_{A_0}(dh + d^2)$, and $h \le T/4$ eventually. Hence this entire interval has $\Delta \le -\Omega_{A_0}(T)$. Summing and adding the macroscopic tail proves the claim. For $a \le 1/2$, take $k = 2$; the coefficient becomes $(a - 1/2)d + 2a - 4 \le -3$. □

I.4 The discrete critical profile

Define $f_x(h) = hx - h(h-1)$ for integers $h \ge 0$, and $M(x) = \max_{h \ge 0} f_x(h)$.

Lemma I.7 (Fixed-defect logarithms)**.** *For fixed $h \ge 1$ and $A_* > 0$, uniformly for $|x_{d,r}| \le A_*$,*

$$\log_q B_h = V_h + O_{q,h,A_*}(d). \tag{152}$$

Moreover

$$\log_q(B_h/B_0) = \mathcal{D}_h(r) + O_{q,h,A_*}(d), \tag{153}$$

$$= T f_{x_{d,r}}(h) + O_{q,h,A_*}(d), \tag{154}$$

where $\mathcal{D}_h(r) = hE_{d,r} - h(h-1)(r + d - (h+1)/2) + 3h$. For $h = 0$, the branch comparison is identically zero.

Proof. The coupled lower (124) simplifies exactly to

$$\log_q B_h \ge V_h - dh - h^2 - 3. \tag{155}$$

For fixed h its frame and central-surjectivity conditions hold eventually, since $s = T + O(d)$ and $C_{d+s} - (d-1) \gg h$. The polynomial upper (142) differs from V_h by $O_q(d + \log(h+1))$. The bad count is $q^{-\Omega(n)}B_0$, smaller than this fixed-defect lower family. This proves (152). Use $C_{d+r-h} = C_{d+r} - h(d+r) + h(h+1)/2$ and (143) to obtain $\mathcal{D}_h$. Finally $r + d = T + 1/2 + x$ and $E_{d,r} = Tx + x^2/2 - 1/8$, giving (154). □

Theorem I.8 (The critical logarithmic profile)**.** *For each fixed $A_* > 0$, uniformly for integer r with $|x_{d,r}| \le A_*$,*

$$\log_q(B_{d+1,r}(q)/B_0) = TM(x_{d,r}) + O_{q,A_*}(d). \tag{156}$$

The profile is $M(x) = 0$ for $x \le 0$, and $M(x) = jx - j(j-1)$ on $2j - 2 \le x \le 2j$, $j \ge 1$.

Proof. Lemma I.6 discards $h \ge H_d = d + \lceil A_* + 2 \rceil$ and the bad derived-center stratum. For the remaining polynomial families, (142) gives uniformly

$$\log_q(B_h^{\mathrm{pol}}/B_0) \le T f_x(h) + O_{q,A_*}(h^3 + d + \log(h+1)). \tag{157}$$

Choose a fixed integer $J_{A_*} \ge 2A_* + 4$. For $h \ge J_{A_*}$, $f_x(h) \le -h^2/2$. Since $H_d = O_{A_*}(d)$, the last upper bound is at most $-Th^2/4$ on $J_{A_*} \le h < H_d$, eventually. Thus

$$\sum_{h \ge J_{A_*}} B_h/B_0 \le q^{-\kappa_{q,A_*}T}. \tag{158}$$

Every maximizing integer lies below J_{A_*}. For these finitely many branches apply Lemma I.7, then take the maximum and sum. The difference $f_x(h+1) - f_x(h) = x - 2h$ gives the stated piecewise-linear profile. □

Theorem I.9 (The derived-defect staircase)**.** *Let an integer sequence satisfy $x_{d,r} \to x \in \mathbb{R}$. For a uniformly chosen type in the complete dimension layer:*

(i) *if $x < 0$, then $h = 0$ except for proportion $q^{-\kappa T}$;*

(ii) *if $2j - 2 < x < 2j$, $j \geq 1$, then $h = j$ except for proportion $q^{-\kappa T}$;*

(iii) *if $x = 2j$, $j \geq 0$, then $h \in \{j, j+1\}$ except for proportion $q^{-\kappa T}$.*

Here $\kappa > 0$ may depend on the fixed limit and field. At a tie limit,

$$\log_q(B_{j+1}/B_j) = T(x_{d,r} - 2j) + O_{q,j}(d). \tag{159}$$

Consequently $(x_{d,r} - 2j)T/d \to -\infty$ selects $h = j$, and convergence to $+\infty$ selects $h = j+1$, each with probability tending to one. The weights inside the remaining $O_{q,j}(d/T)$ tie neighborhoods are not determined.

Proof. Choose $A_* > |x| + 1$ and use the finite cutoff J_{A_*} above. At a nontie limit, one integer maximizer has a fixed positive score gap over all other indices below J_{A_*}. At $x = 2j$, exactly $j, j+1$ maximize, and every other score is lower by at least two. Continuity, the fixed-defect estimates, $d = o(T)$, and (158) turn these gaps into the stated exponential proportions. The maximizing score is nonnegative, so the tail estimate relative to B_0 remains negligible relative to the full count. Subtracting the two adjacent fixed-defect logarithms proves (159); its sign under the variable-distance conditions proves the final assertions. □

I.5 Shrinking localization of minimum-derived dominance

Theorem I.10 (Two-branch support through the lower splice)**.** *Uniformly for integers*

$$T - 3d - 3 \leq r \leq R_d^* + 1/4, \tag{160}$$

there is $\kappa_q > 0$ such that

$$\sum_{h \geq 2} B_h/B_0 \leq q^{-\kappa_q T}. \tag{161}$$

In particular, for $|r - R_d^*| \leq 1/4$, $\pi_{d,r} = (1 + B_1/B_0)^{-1}(1 + O_q(q^{-\kappa_q T}))$. The proportion with $z'(L) < h + 1$ is at most $q^{-\kappa_q n}$.

Proof. For the bounded upper end of this strip, the preceding estimates already discard the bad stratum, the macroscopic tail, and $h \geq d + 3$. To extend them to its lower end, use the exact upper polynomials rather than any monotonicity of π:

$$\partial_r \Delta = h(r - h + 2d - 1/2) + 2d + 1 > 0, \qquad \partial_r \mathcal{D}_h = h(r + d + 1/2 - h) > 0 \quad (1 \leq h \leq r).$$

For $h \leq r/5$, evaluate Δ at $\bar{r} = T - d + 3/4$; its raw maximum is $O(d^2 T)$, whereas the bad penalty is $\Omega(n)$. At $h \geq d + 3$, the same evaluation gives the raw cutoff $-\Omega(T)$. For $h \geq r/5$ the uniform macroscopic envelope gap applies throughout the strip. Finally for $2 \leq h \leq d+2$, evaluate $\mathcal{D}_h$ at $\bar{r}$ in the strong polynomial upper. Its value is $Th(5/4 - h) + O(h^3 + d + \log(h + 1)) \leq -Th^2/4$, eventually. All uses of base enumeration have $s = T + O(d)$, uniformly within $2d \leq s \leq D_{\mathrm{par}}/(4d)$. Summing proves both assertions, and the formula for π follows from the exact branch decomposition. □

Corollary I.11 (A shrinking outer window)**.** *For every integer sequence $r = r(d) \geq 1$,*

$$r \leq R_d^* - 4d/T \text{ eventually } \implies \pi_{d,r} \to 1, \qquad r \geq R_d^* + 4d/T \text{ eventually } \implies \pi_{d,r} \to 0.$$

The outer window has width $8d/T = o(1)$. Its boundary density is not specified by these bounds.

Proof. For $\delta = r - R_d^*$, $E_{d,r} = T\delta + \delta^2/2 - 1/8$. The precise bounds in Corollary I.5 give, at the two stated endpoints, respectively $\log_q(B_1/B_0) \leq -3d + O_q(1)$ and $\log_q(B_1/B_0) \geq 2d + O_q(1)$. On left-side indices with $r \leq T - 3d - 3$, use the old outer localization. On the remaining left indices use Theorem I.10 and the increasing polynomial $E_{d,r}$ to obtain $B_1/B_0 \to 0$. On right-side indices with $r \geq T - d + 1$, use the old upper localization. On the remaining right indices, $r = T - d + O(1)$; the same increasing polynomial and $\pi \leq B_0/B_1$ suffice. This is a splice of uniform estimates, with no monotonicity assumption on π. □

Corollary I.12 (The odd-d integer lattice)**.** *Along odd d, put $r = T - d + \ell$ for a fixed integer ℓ. Almost every type has $h = \max\{0, \lceil \ell/2 \rceil\}$, with exceptional proportion $q^{-\kappa_\ell T}$. For arbitrary integer sequences on this subsequence, $r \leq T - d$ implies $\pi_{d,r} \to 1$, while $r \geq T - d + 1$ implies $\pi_{d,r} \to 0$.*

Proof. Here $T = 2^{(d+1)/2}$ is integral and $x_{d,r} = \ell - 1/2$, which never equals a tie value. Apply Theorem I.9. For the final assertions every integer lies at least $1/2$ from R_d^*, whereas $d/T \to 0$; apply Corollary I.11. □

J Exact calculations and reproducibility

The general dimension, orbit-recovery and population bounds are proved in the text and preceding appendices. Exact finite controls accompany the equality classifications and coordinate actions. In particular the class-five noncompatible orbit calculation reconstructs the action matrices and exhausts all $4^9 = 262144$ parameter vectors, giving eleven orbits; compatible low-class orbit and mass calculations are checked separately. The package also reconstructs the universal modules, ordinary CE complexes, nonlinear coordinate recovery, automorphism kernels and fixed-surplus bookkeeping.

The groups `critical_constant/`, `critical_window/`, `action_cohomology/` and `critical_refinement/` under `verification/` give finite controls of the corresponding normalization and allocation interfaces. They do not substitute for the uniform estimates or asymptotic arguments. Each stored report states its model range and whether the calculation is exhaustive.

From this project directory run

```
python verification/run_replay.py --output-dir replay_core
```

The runner copies the supplied inputs into an isolated directory, collects failures and preserves the original reports. The `--no-write` option discards temporary output; `--extended` runs the additional listed controls. The selected entry point uses the Python standard library and SymPy 1.14.0 for the symbolic staircase identities. Exact arithmetic controls are distinct from proof-assistant formalization.

References

[1] Y. Bahturin and A. Olshanskii, Nilpotent algebras, implicit function theorem, and polynomial quasigroups, *Journal of Algebra* **632** (2023), 154–193. doi:10.1016/j.jalgebra.2023.05.024. Author manuscript: `https://arxiv.org/abs/2208.09527`.

[2] Y. Bahturin and A. Olshanskii, *On generic properties of nilpotent algebras*, preprint, 2024. https://arxiv.org/abs/2406.15631.

[3] V. G. Bardakov and M. Singh, Extensions and automorphisms of Lie algebras, *Journal of Algebra and its Applications* **16** (2017), 1750162. Author manuscript: https://arxiv.org/abs/1508.01850.

[4] W. Bruns, On the Koszul algebra of a local ring, *Illinois Journal of Mathematics* **37** (1993), no. 2, 278–283. Author-hosted original.

[5] P. M. Cohn, A non-nilpotent Lie ring satisfying the Engel condition and a non-nilpotent Engel group, *Proceedings of the Cambridge Philosophical Society* **51** (1955), no. 3, 401–405. doi:10.1017/S0305004100030395.

[6] E. C. Dade, Endo-permutation modules over p-groups, II, *Annals of Mathematics* (2) **108** (1978), 317–346. doi:10.2307/1971169.

[7] J. Dixmier, Cohomologie des algèbres de Lie nilpotentes, *Acta Scientiarum Mathematicarum (Szeged)* **16** (1955), 246–250.

[8] V. Drensky, Weak polynomial identities and their applications, *Communications in Mathematics* **29** (2021), 291–324. doi:10.2478/cm-2021-0022.

[9] S. B. Ekhad and D. Zeilberger, An explicit formula for the number of solutions of $X^2 = 0$ in triangular matrices over a finite field, *Electronic Journal of Combinatorics* **3**(1) (1996), Research Paper 2. doi:10.37236/1226.

[10] J. Feldvoss, S. Siciliano, and T. Weigel, Split strongly abelian p-chief factors and first degree restricted cohomology, *Journal of Lie Theory* **24** (2014), 29–39. https://www.heldermann-verlag.de/jlt/jlt24/feldla2e.pdf.

[11] L. G. Kovács and R. Stöhr, Free centre-by-metabelian Lie algebras in characteristic 2, *Bulletin of the London Mathematical Society* **46** (2014), no. 3, 491–502. doi:10.1112/blms/bdu011. Author manuscript: https://archives.maths.anu.edu.au/people/Kovacs/K114.pdf.

[12] B. Li, *Sharp ideal growth of metabelian restricted Lie algebras*, companion preprint, 2026. Earlier deposited version: doi:10.5281/zenodo.23852662.

[13] J. Pakianathan and E. Yalçın, Quadratic maps and Bockstein closed group extensions, *Transactions of the American Mathematical Society* **359** (2007), 6079–6110. doi:10.1090/S0002-9947-07-04346-2.

[14] Ş. Papadima and A. I. Suciu, Chen Lie algebras, *International Mathematics Research Notices* **2004** (2004), no. 21, 1057–1086. doi:10.1155/S1073792804132017.

[15] Y. Qiao, Enumerating alternating matrix spaces over finite fields with explicit coordinates, *Discrete Mathematics* **344** (2021), no. 11, 112580. doi:10.1016/j.disc.2021.112580. Author manuscript: https://arxiv.org/abs/2007.05108.

[16] G. Traustason, Engel Lie-algebras, *The Quarterly Journal of Mathematics* **44** (1993), no. 3, 355–384. doi:10.1093/qmath/44.3.355.

[17] B. Li, *Quadratic orbit maps and three-Engel Lie algebras in characteristic two*, companion manuscript, 2026.

[18] B. Li, *Central-depth stability and optimal central realizations of three-Engel Lie algebras*, companion manuscript, 2026.

[19] B. Li, *Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: nilpotence, Lie reconstruction, and growth*, unified research manuscript, 2026. Earlier public version: doi:10.5281/zenodo.23852642; expanded local revision, 2 November 2026.